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Weinberg et al.

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(54) **CONTINUOUS DIALOG WITH A DIGITAL ASSISTANT**

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G10L 15/22 (2006.01)
G06F 3/01 (2006.01)
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(52) **U.S. Cl.**
CPC **G10L 25/78** (2013.01); **G06F 3/013** (2013.01); **G10L 15/083** (2013.01);
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CPC G10L 15/22; G10L 15/00; G10L 15/06; G10L 15/065; G10L 15/07; G10L 15/08;
(Continued)

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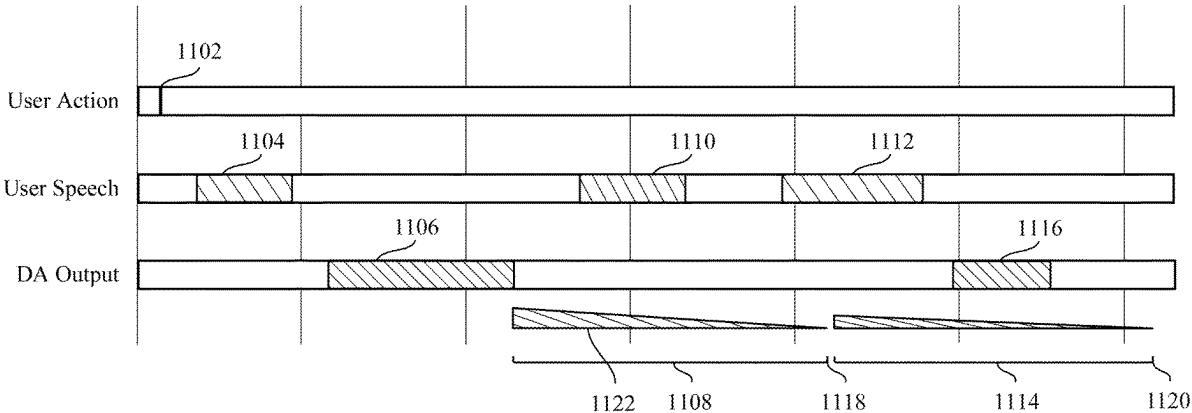
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(57) **ABSTRACT**

Systems and processes for operating an intelligent automated assistant are provided. For example, a first speech input directed to a digital assistant is received from a user. A first response is provided based on the first speech input. A session window is initiated, wherein the session window is associated with a variable speech threshold. A second speech input is received during the session window. In accordance with a determination that the second speech input includes speech directed to the digital assistant, a duration associated with the session window is increased. In accordance with a determination that the variable speech threshold does not exceed a predetermined speech threshold, the session window is ended.

36 Claims, 24 Drawing Sheets

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- (60) Provisional application No. 63/159,557, filed on Mar. 11, 2021.

(51) **Int. Cl.**

G10L 15/08 (2006.01)

G10L 15/18 (2013.01)

G10L 25/78 (2013.01)

(52) **U.S. Cl.**

CPC **G10L 15/1807** (2013.01); **G10L 15/22** (2013.01); **G10L 2015/088** (2013.01); **G10L 2015/221** (2013.01); **G10L 2015/223** (2013.01); **G10L 2025/786** (2013.01)

(58) **Field of Classification Search**

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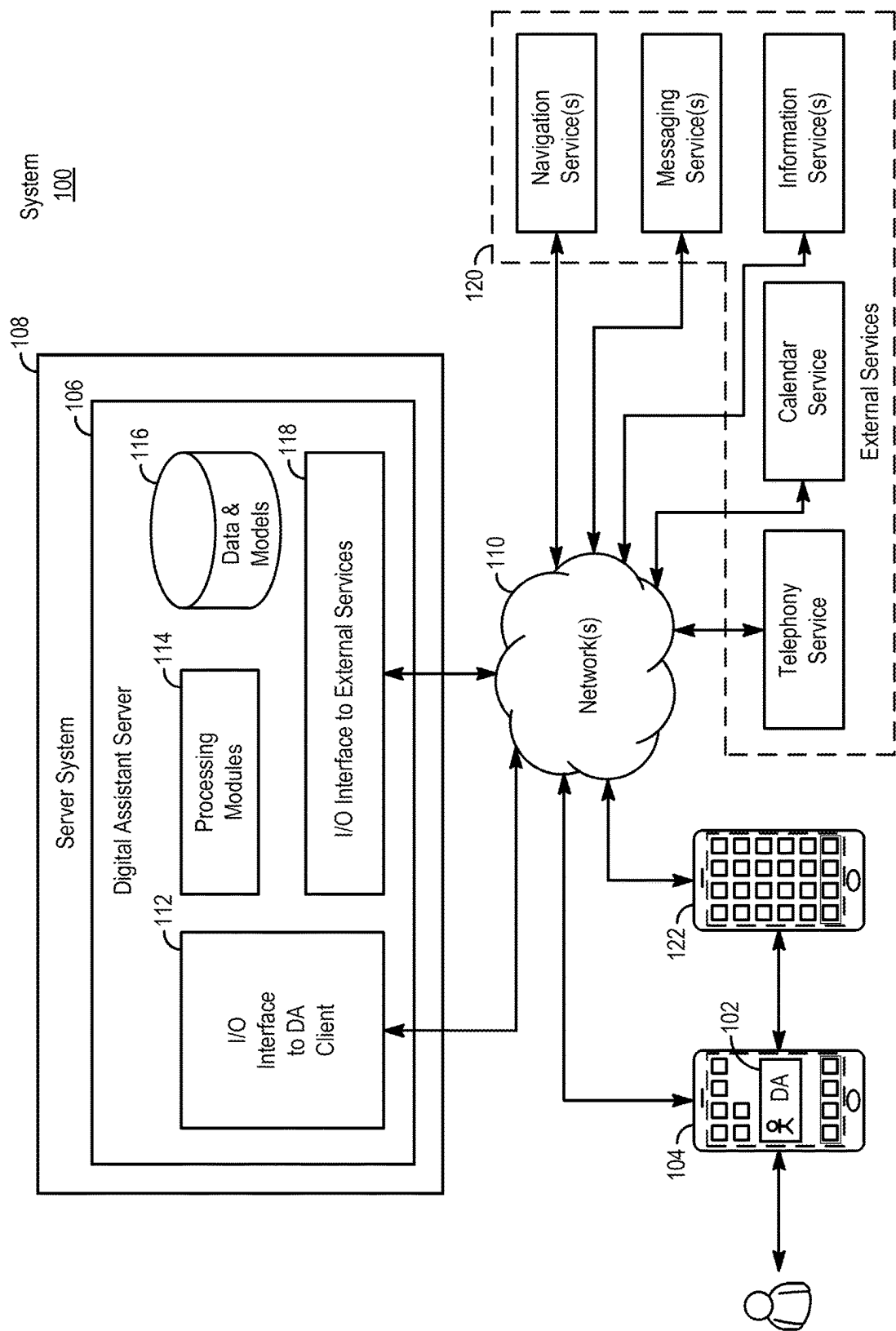


FIG. 1

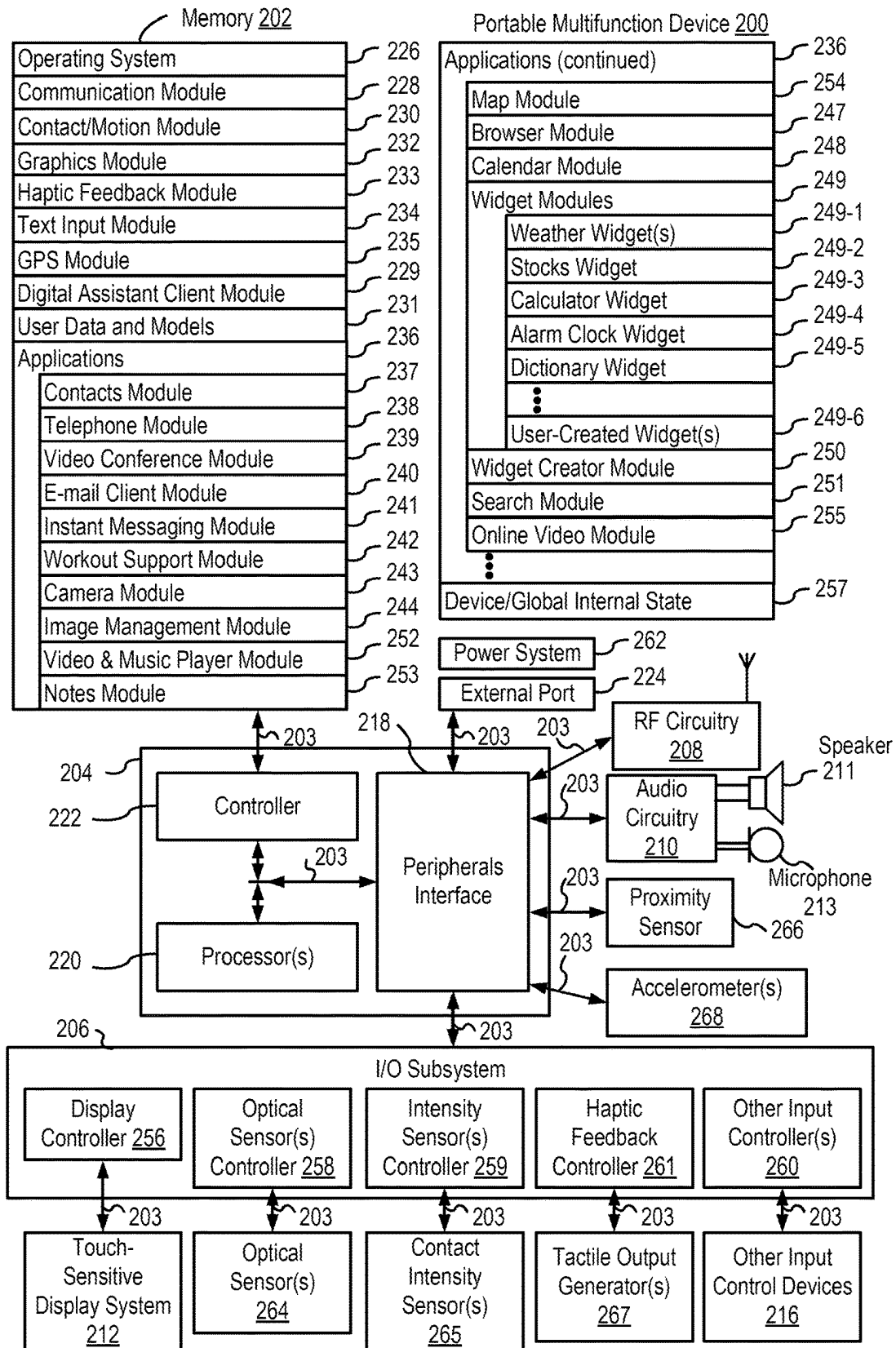
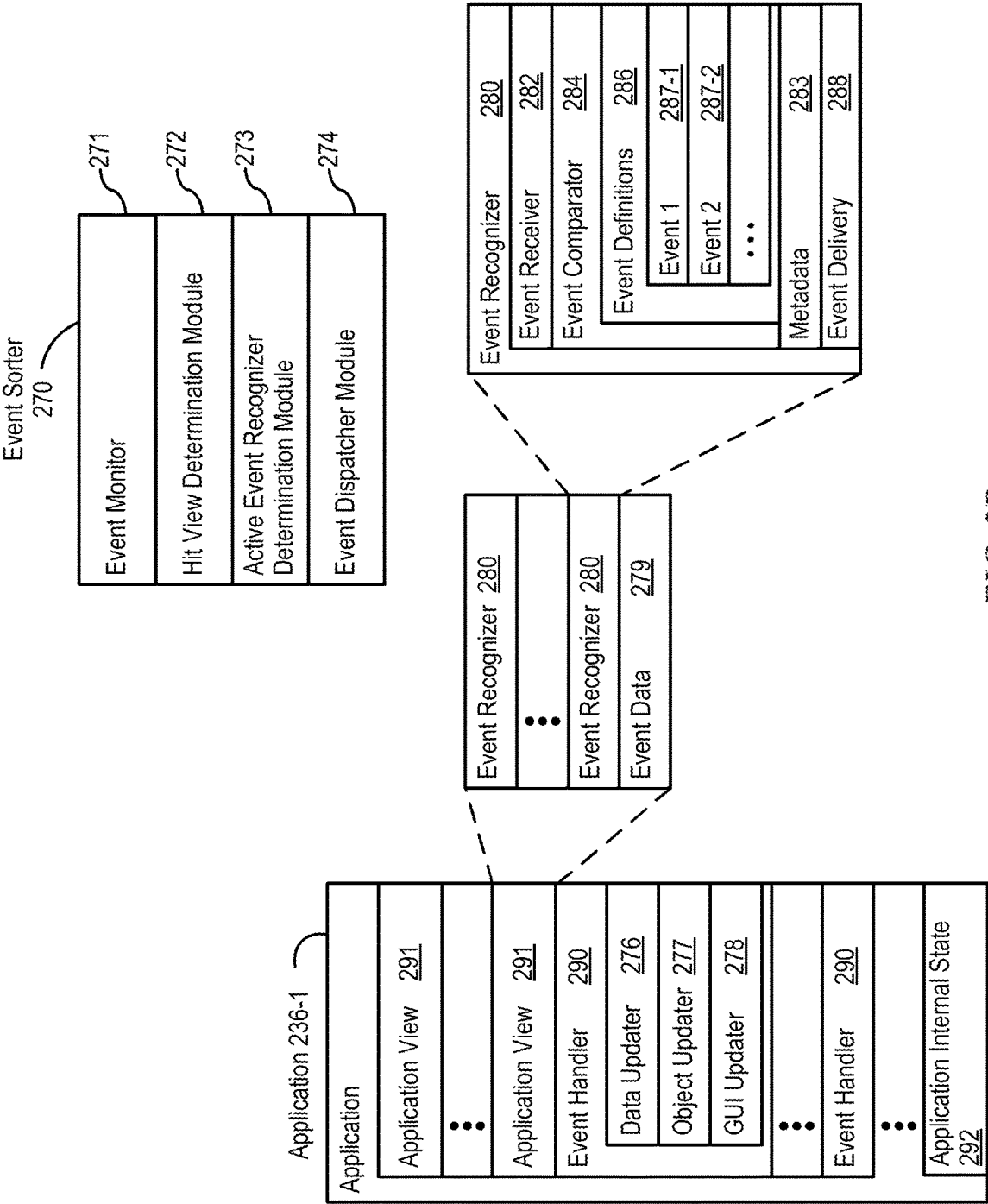


FIG. 2A



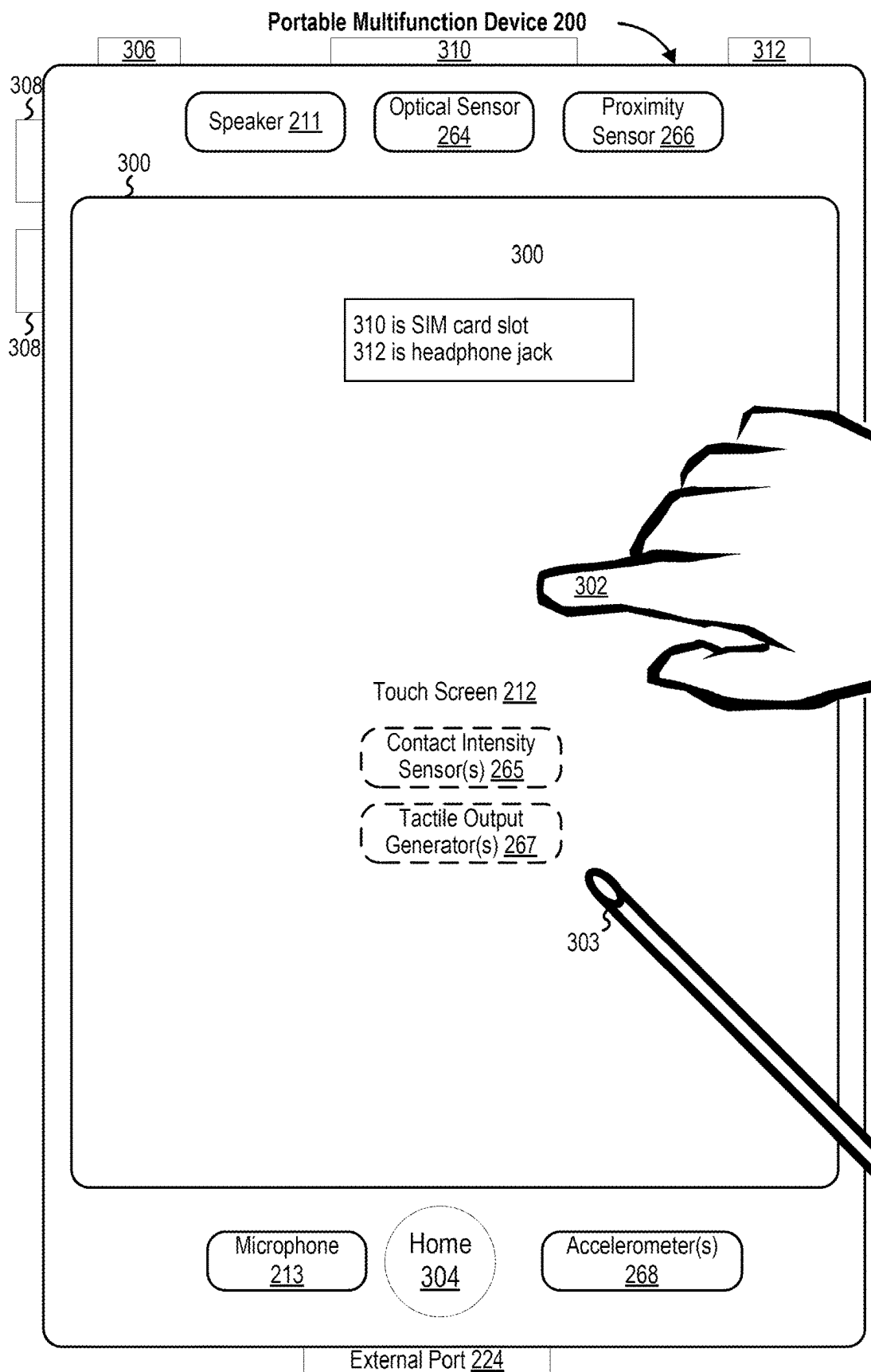


FIG. 3

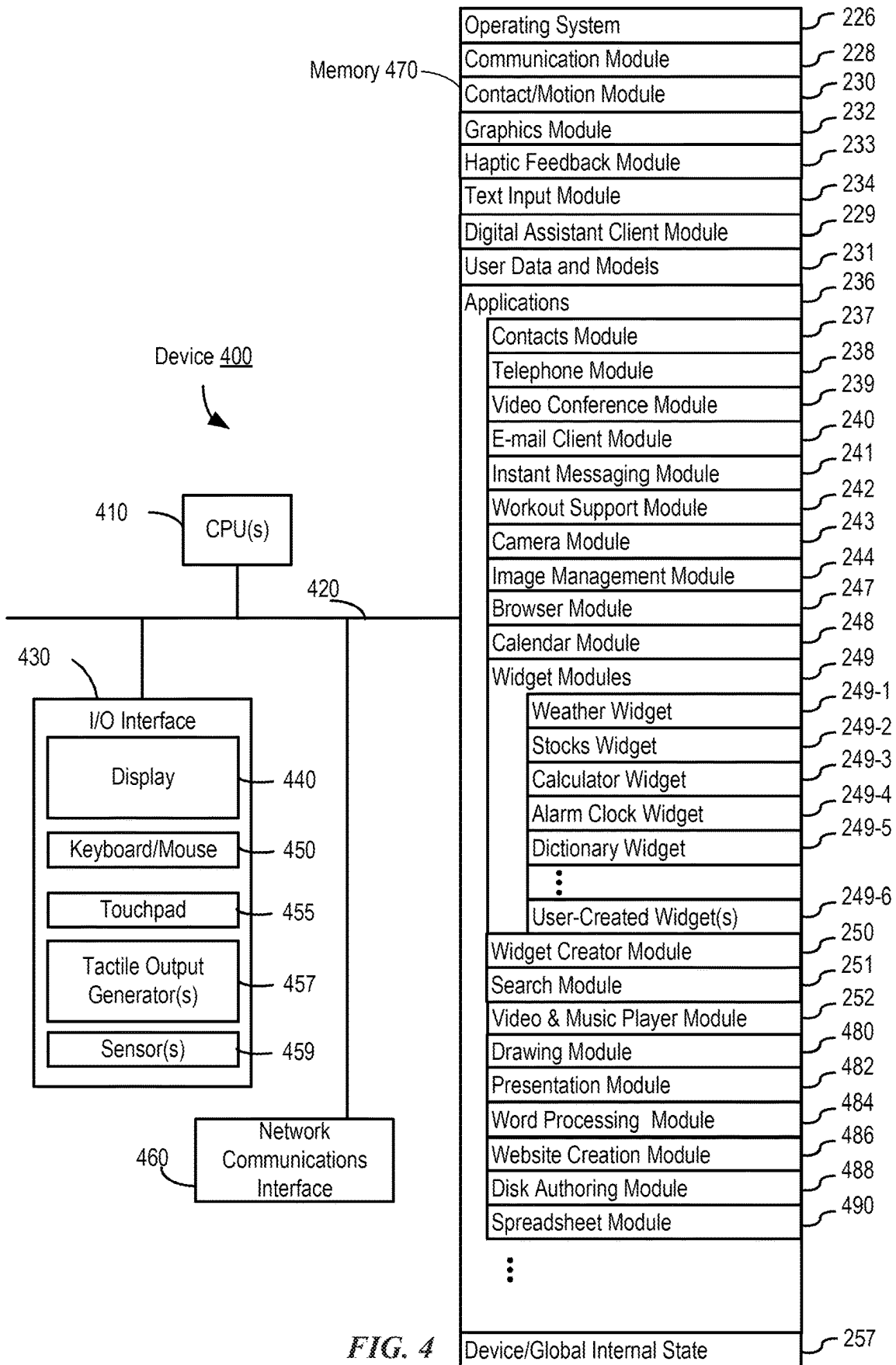


FIG. 4

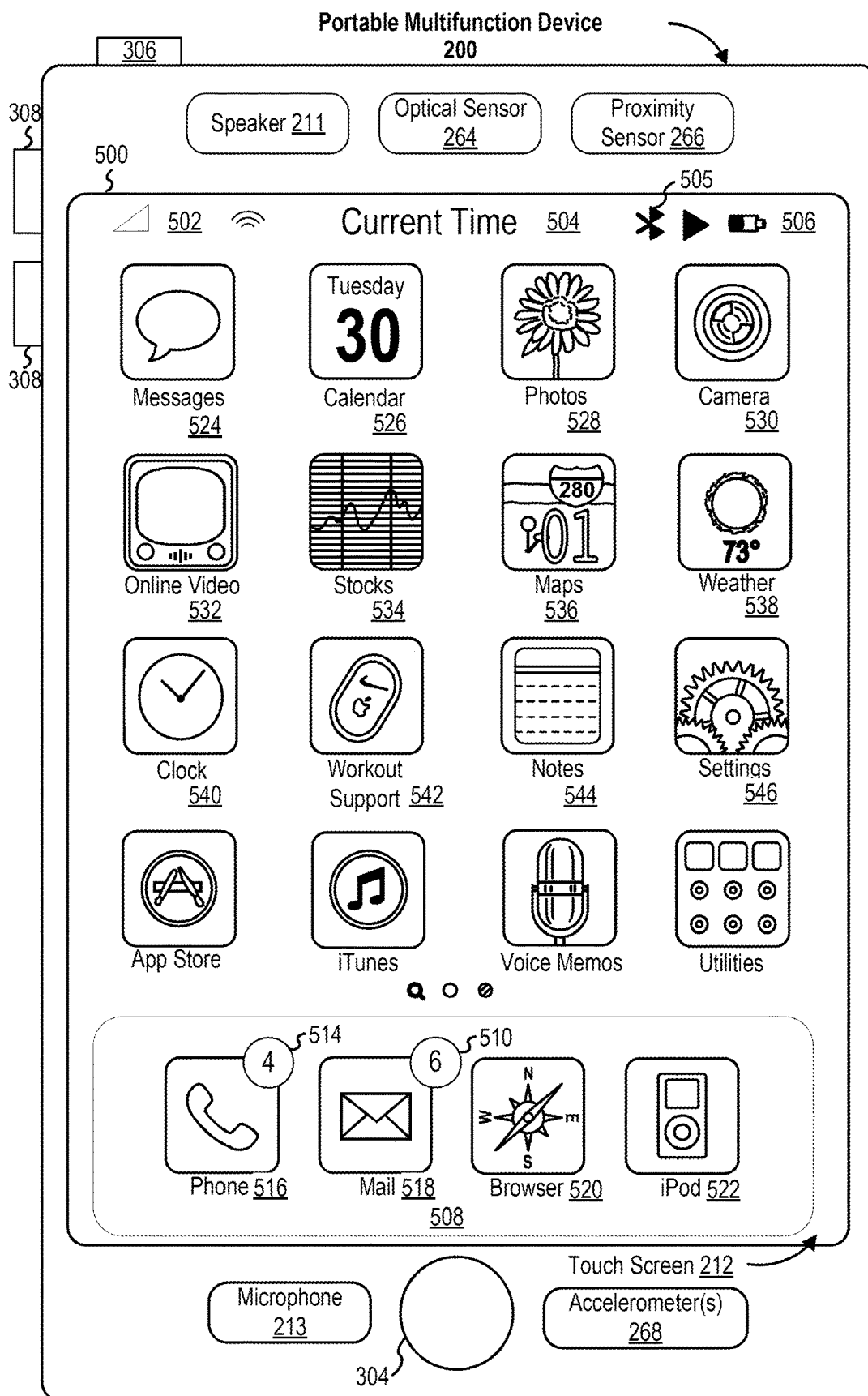


FIG. 5A

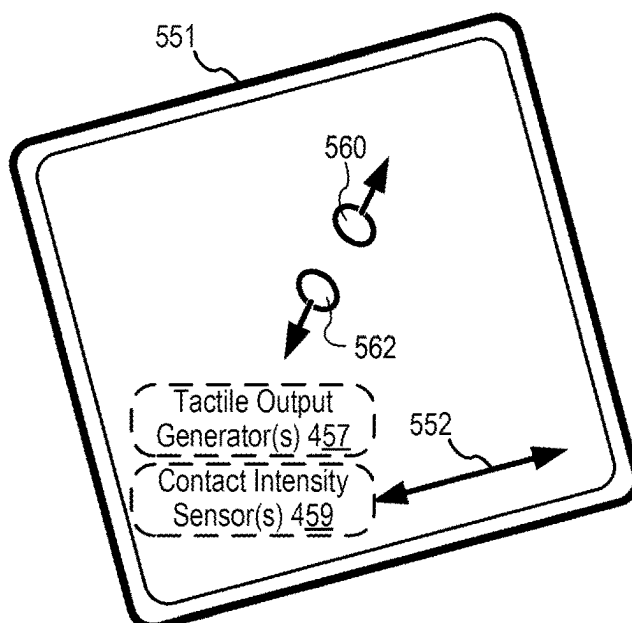
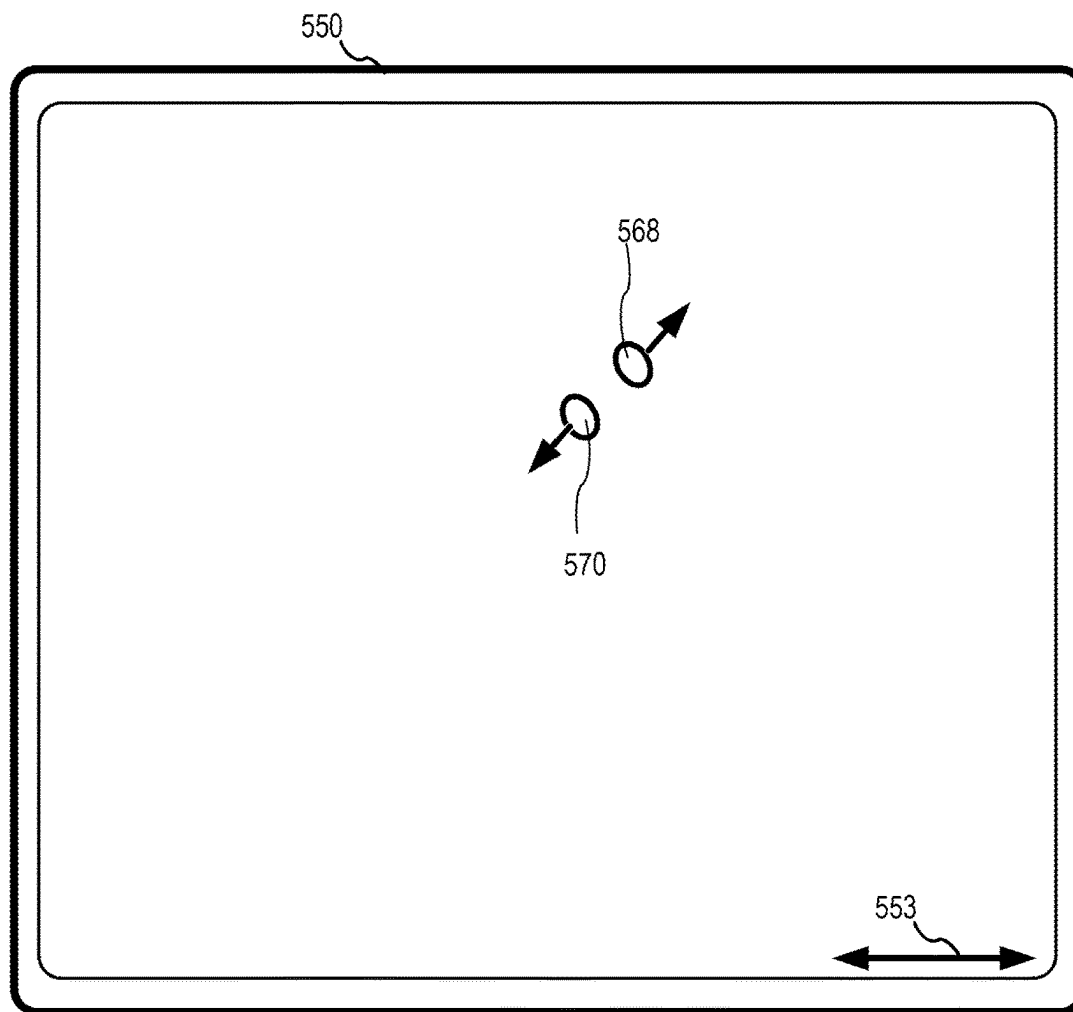
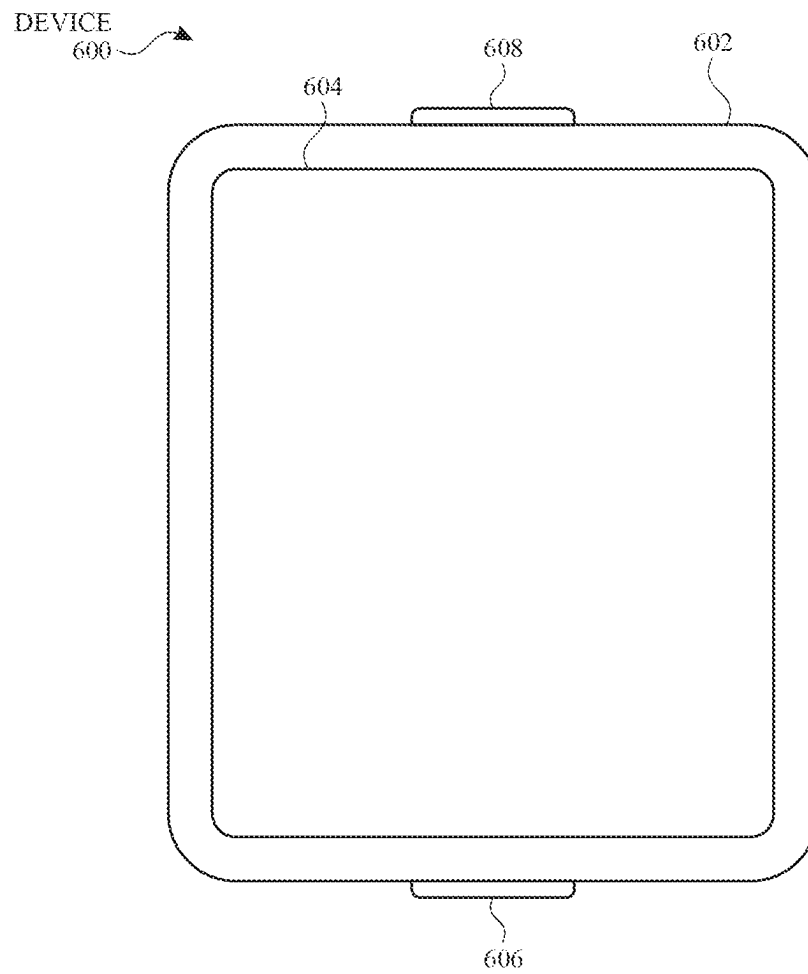


FIG. 5B

**FIG. 6A**

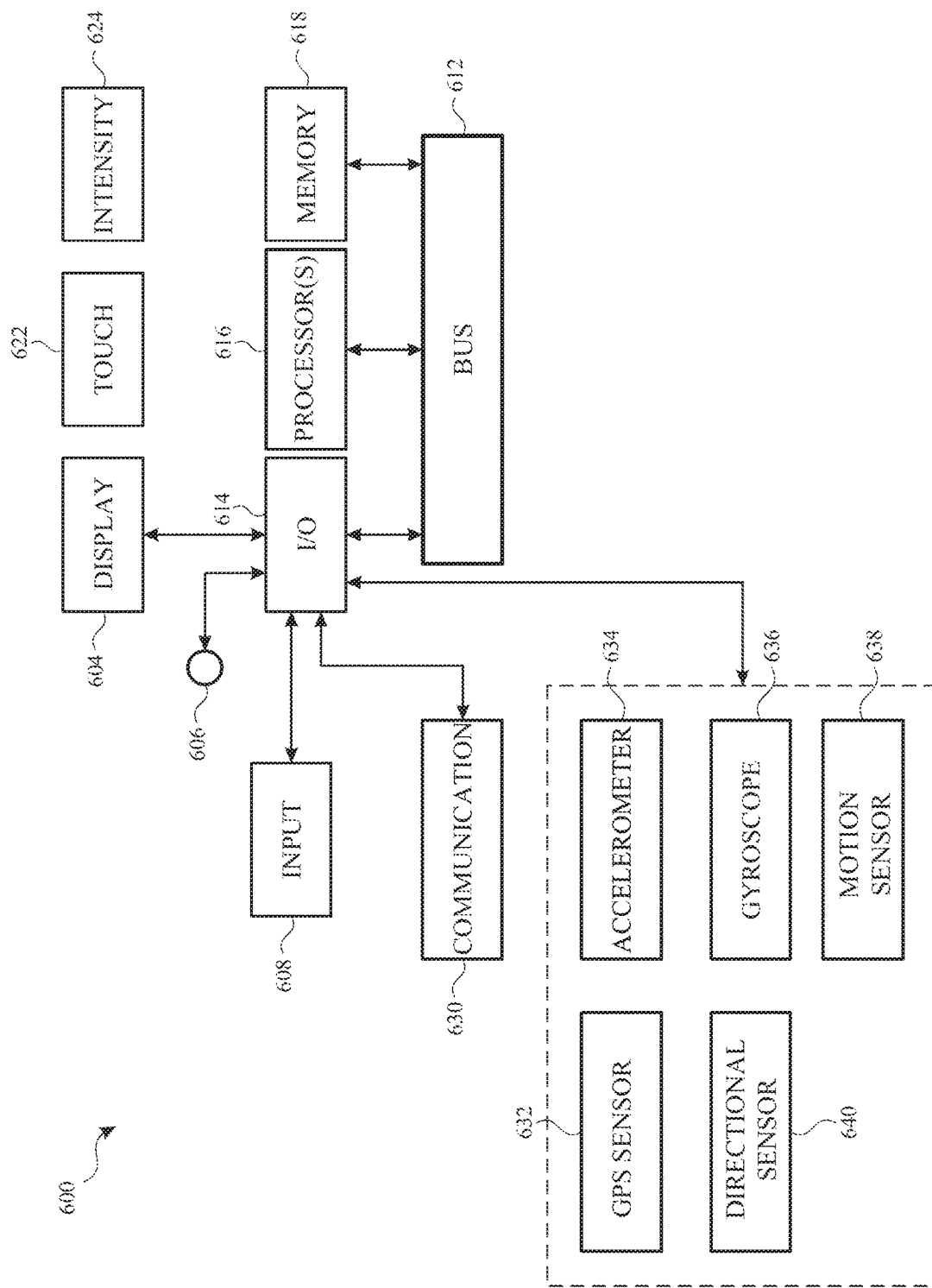


FIG. 6B

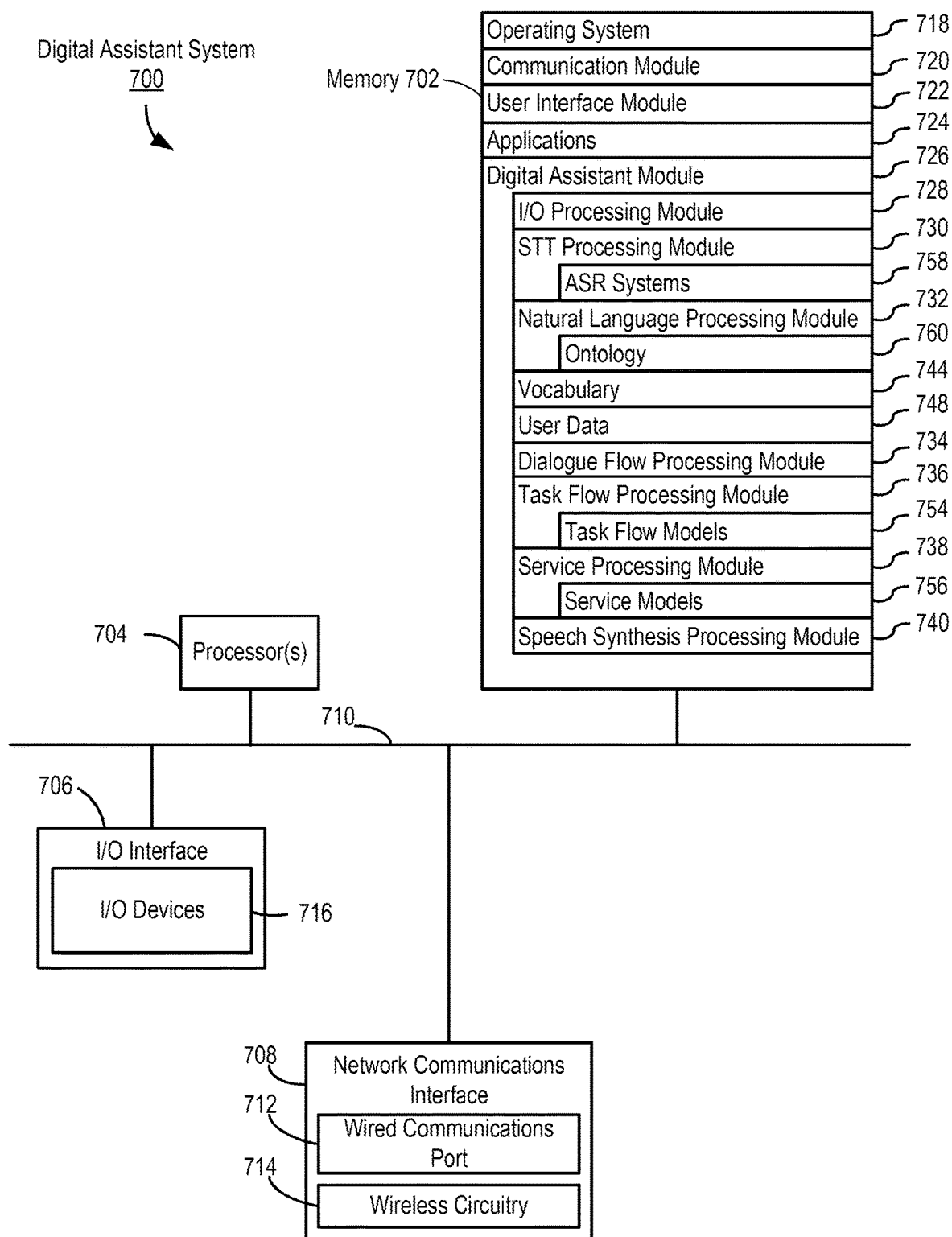


FIG. 7A

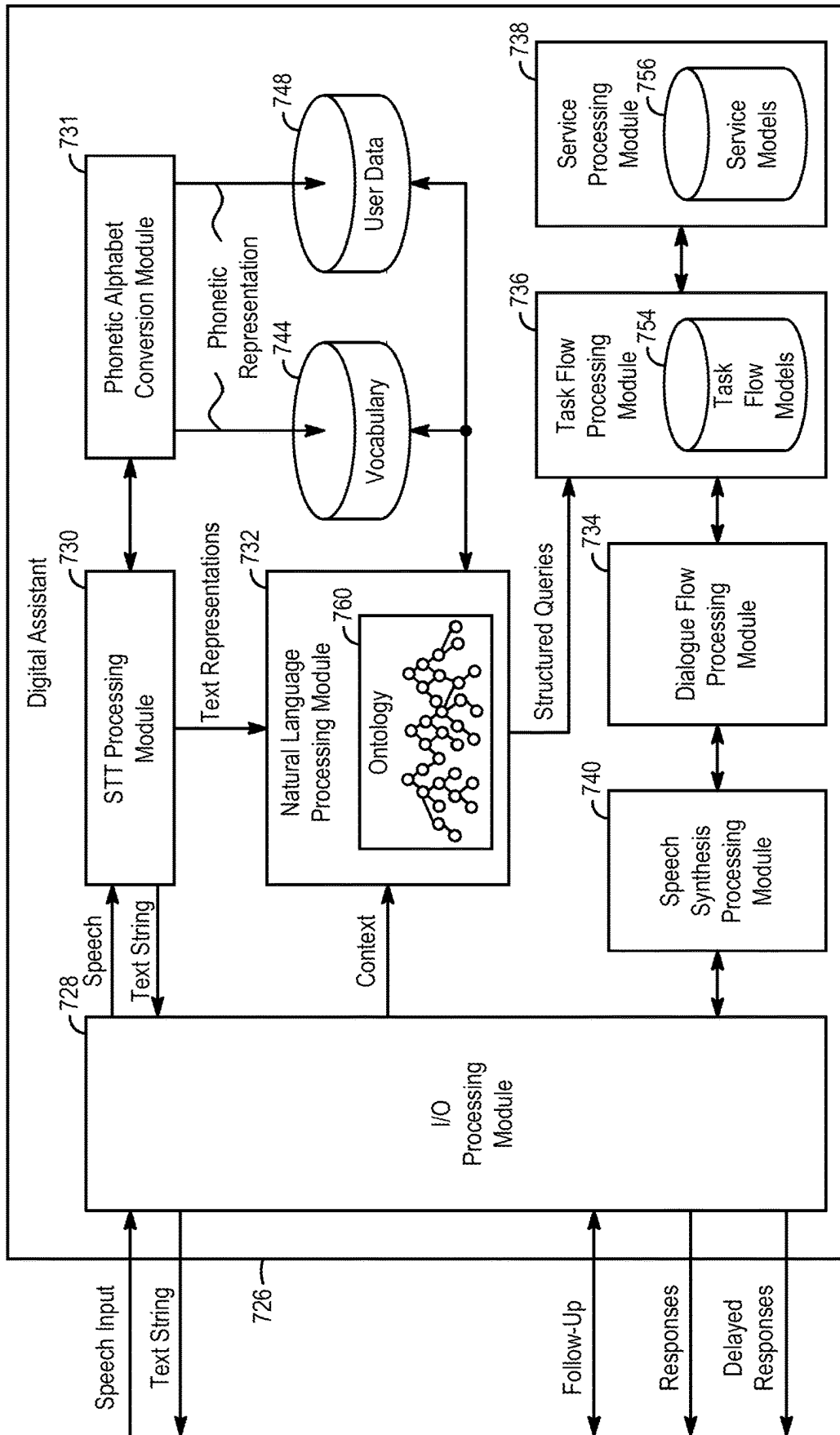


FIG. 7B

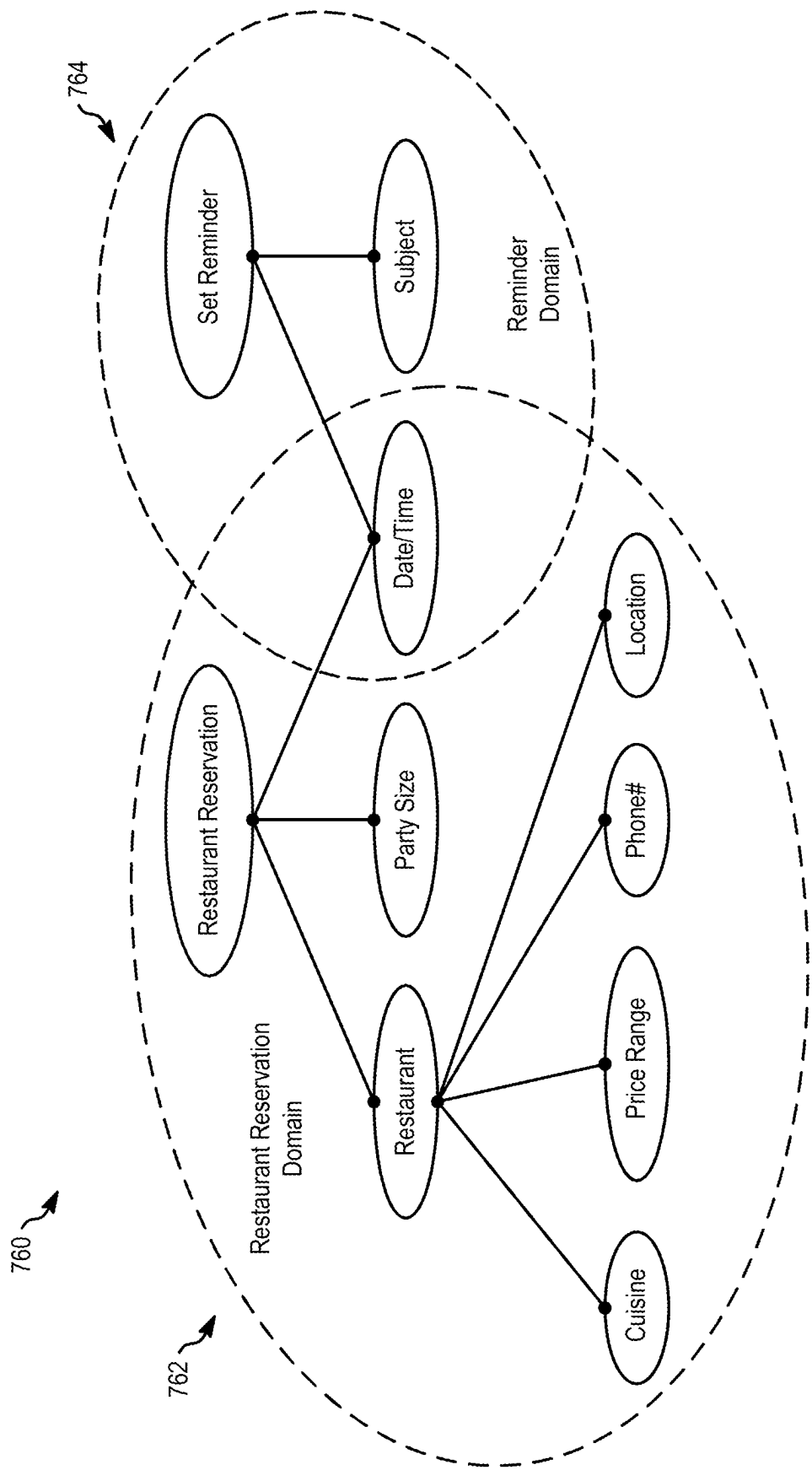
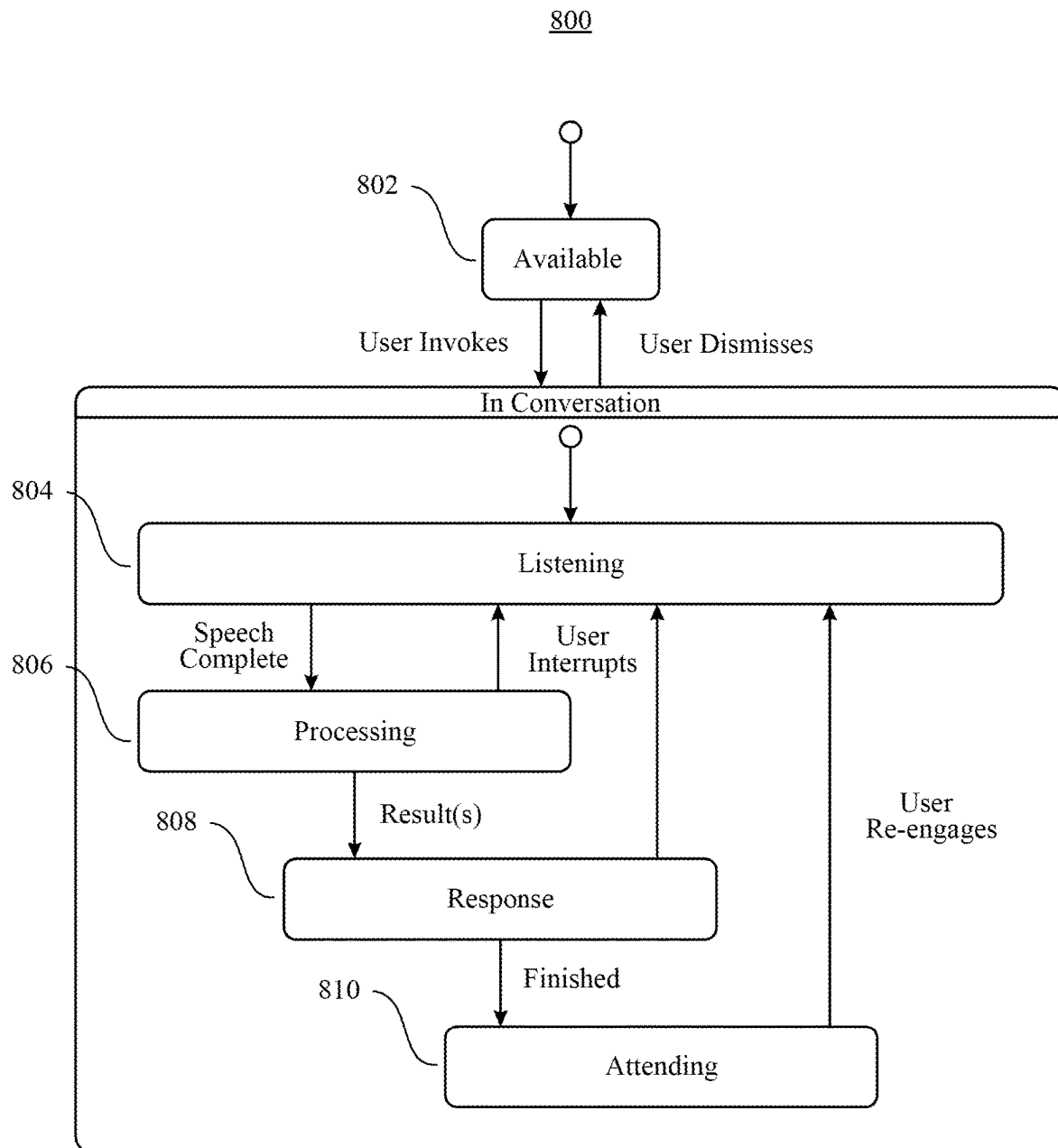


FIG. 7C

**FIG. 8**

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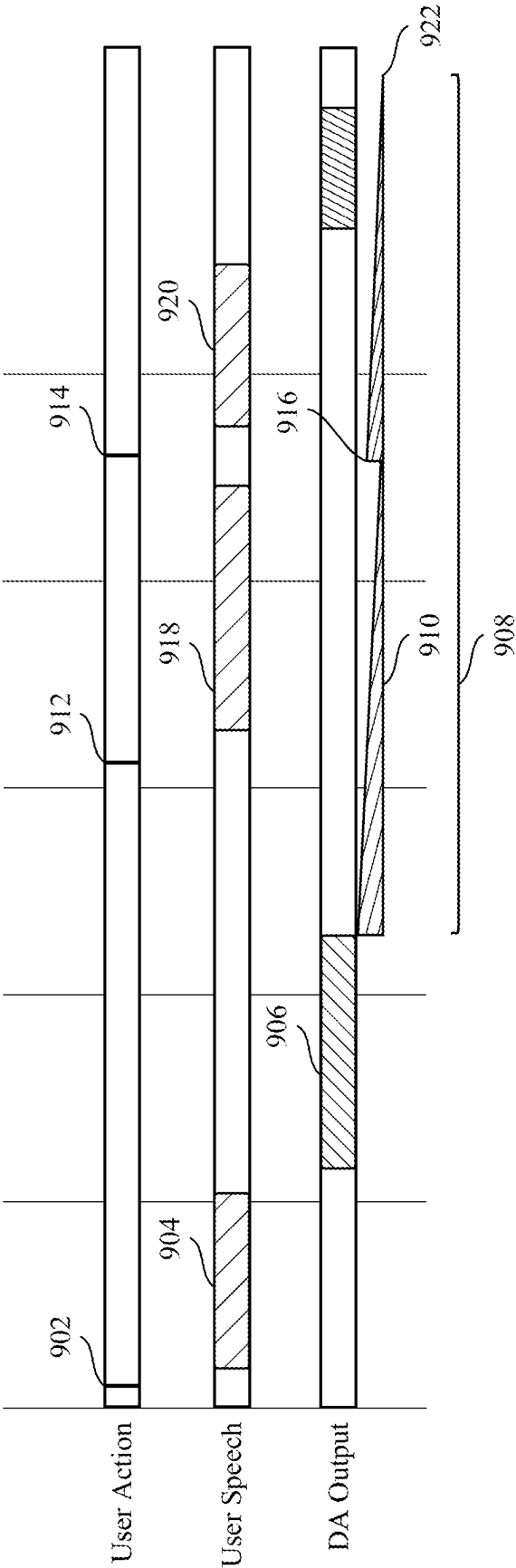


FIG. 9

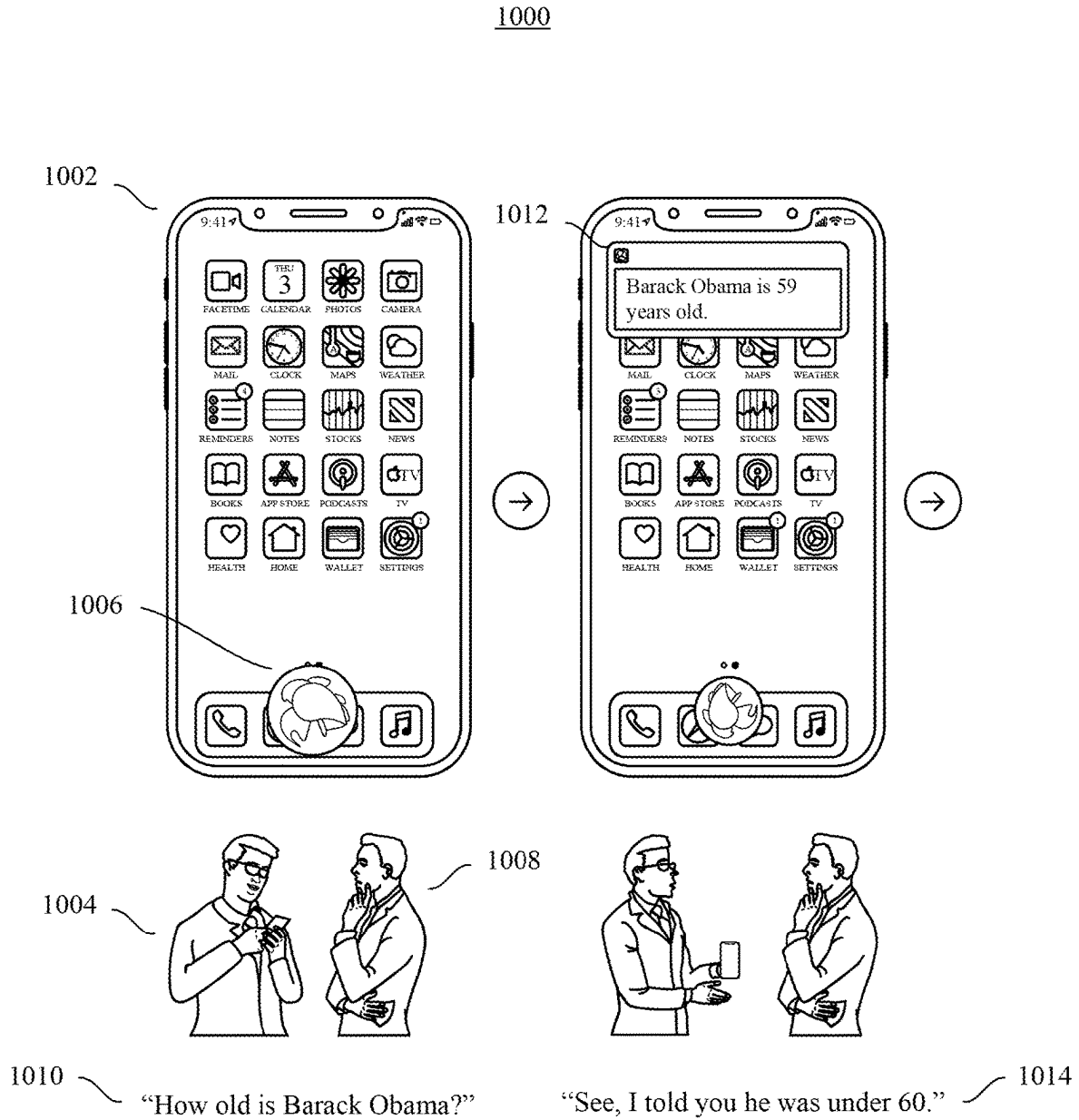


FIG. 10A

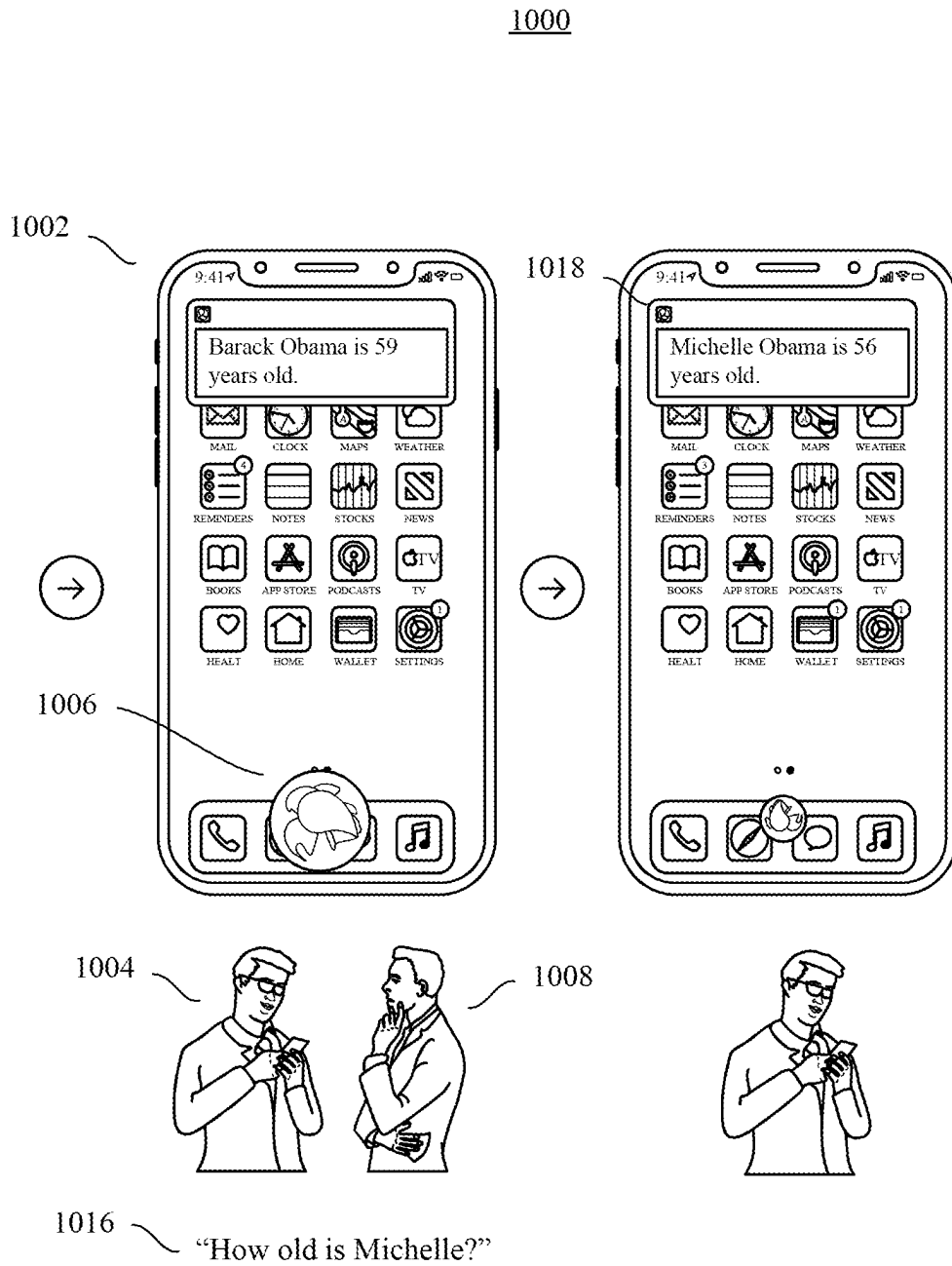


FIG. 10B

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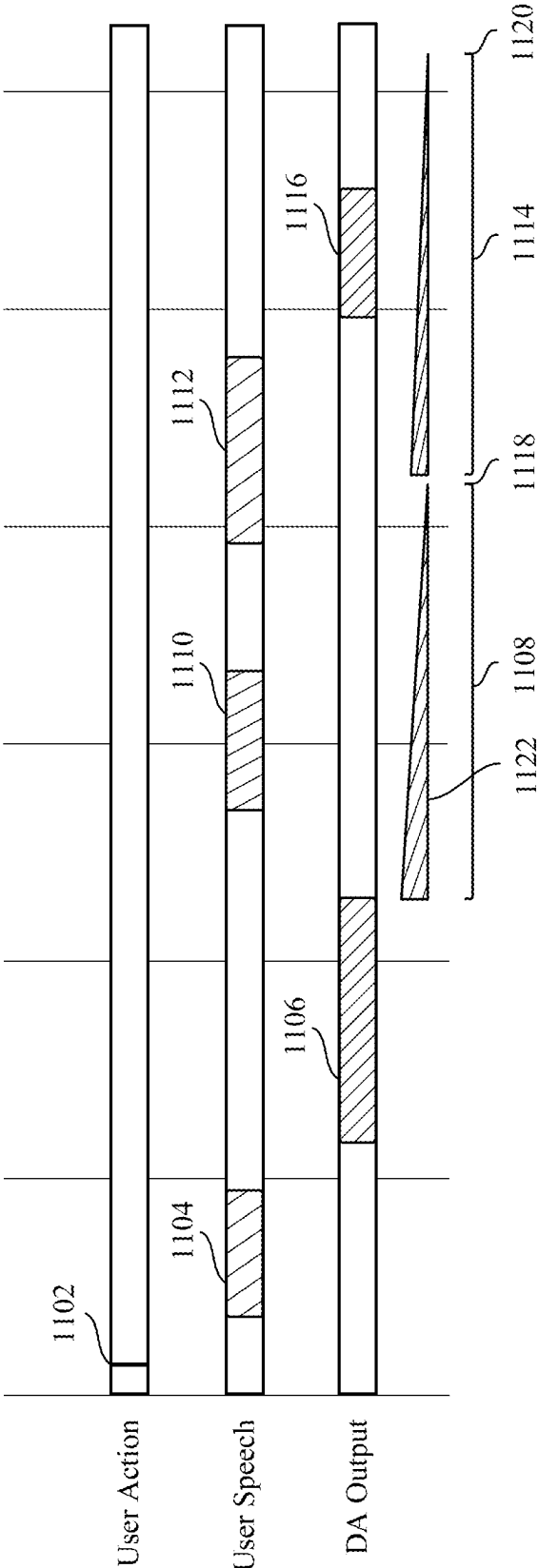
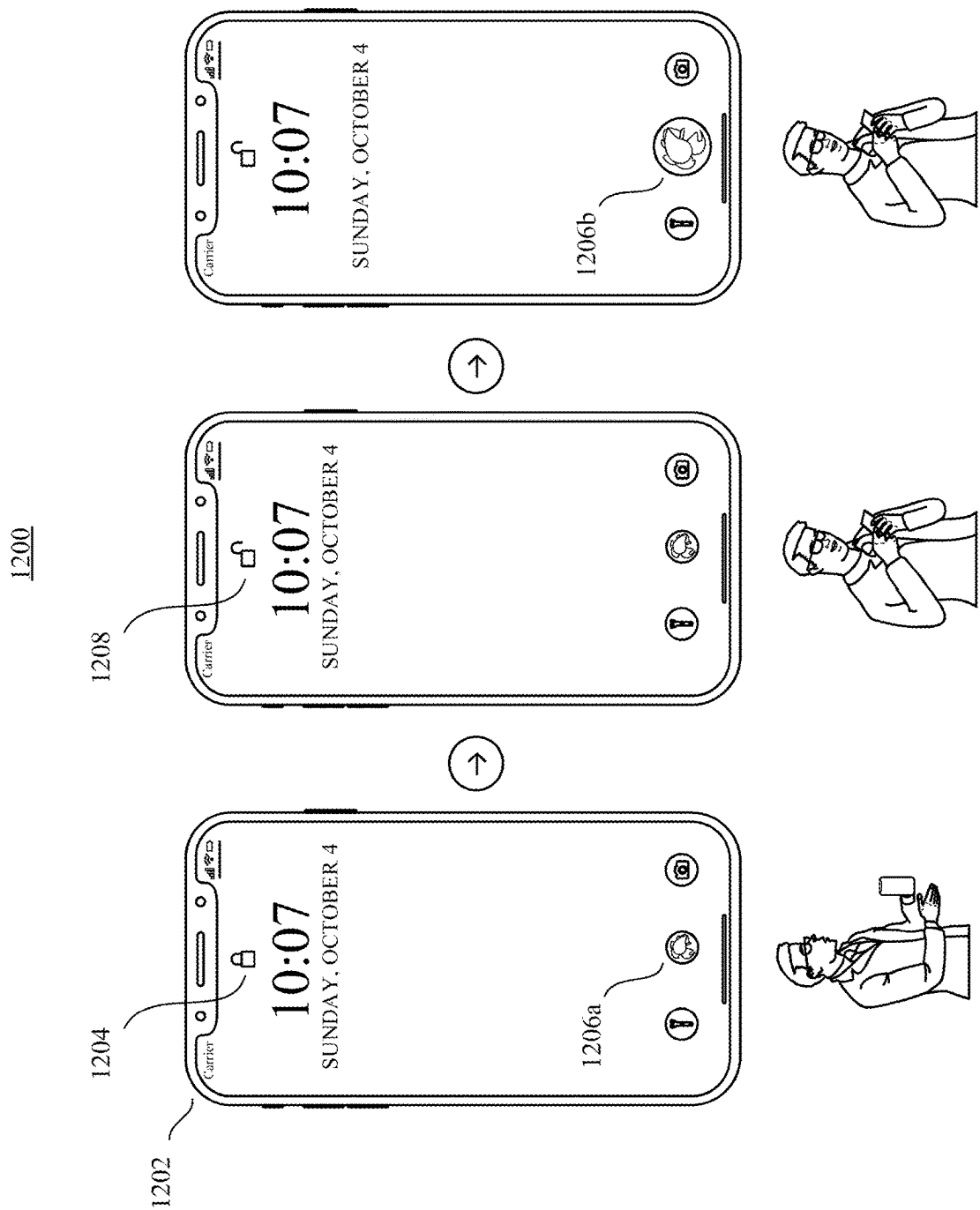


FIG. 11



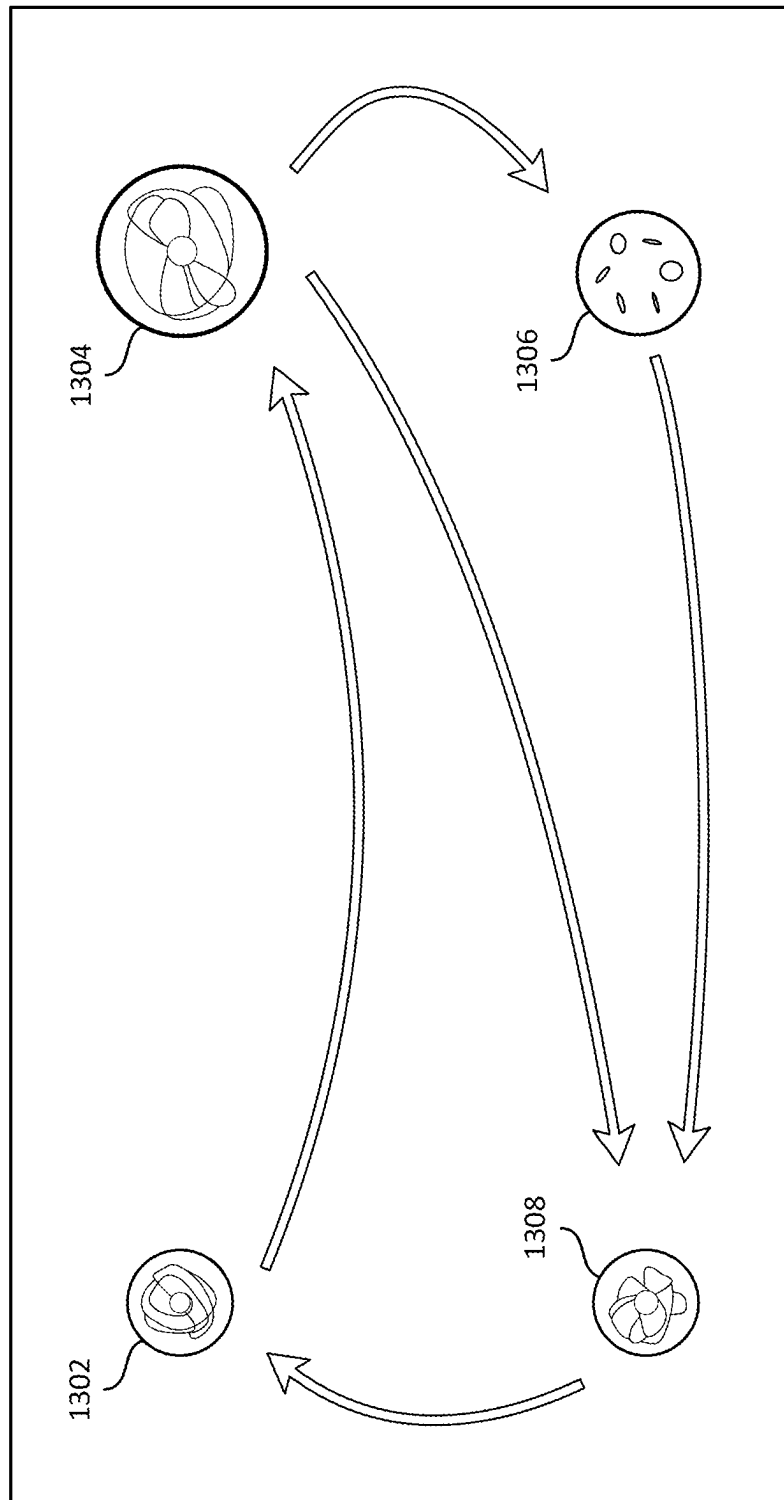
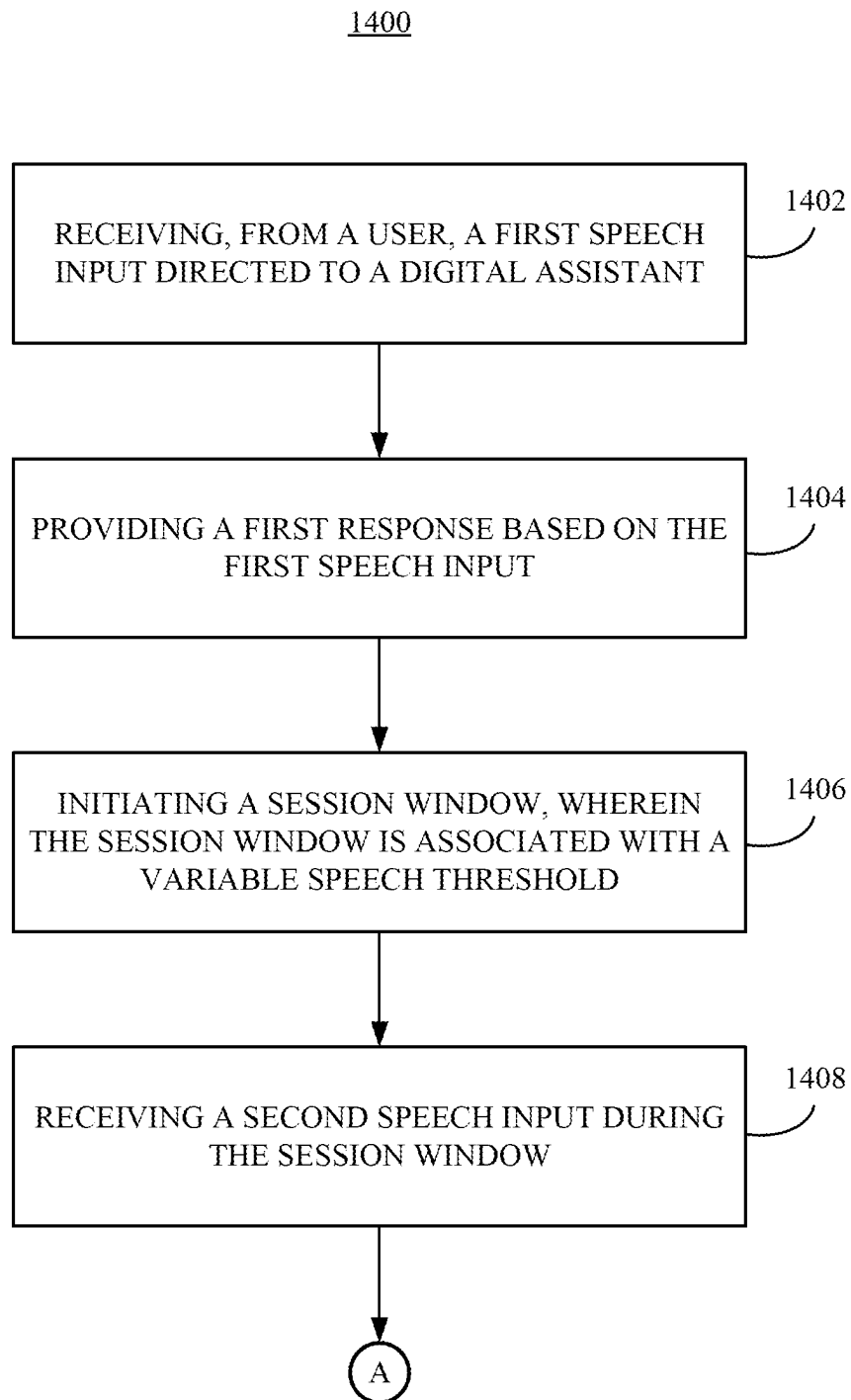
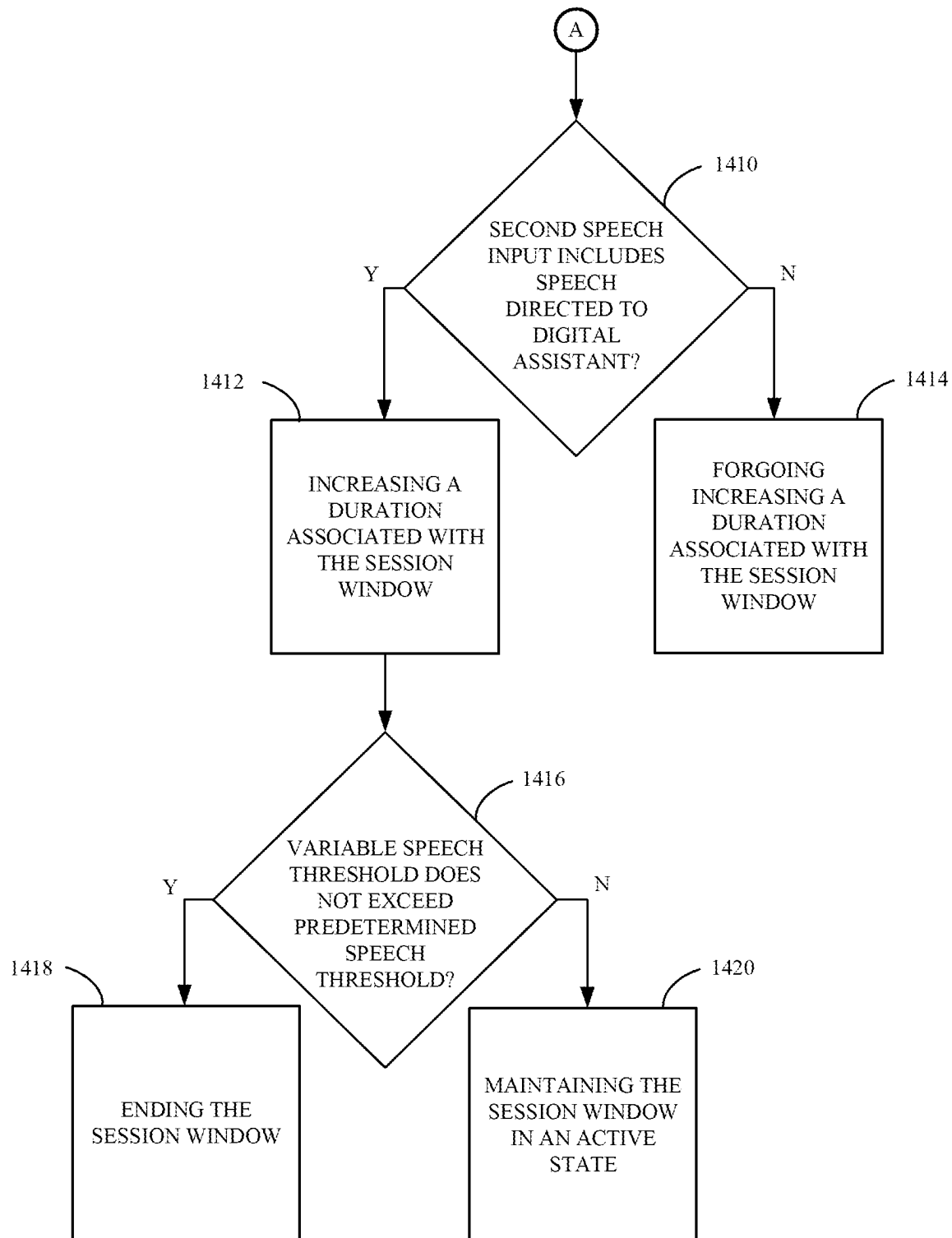
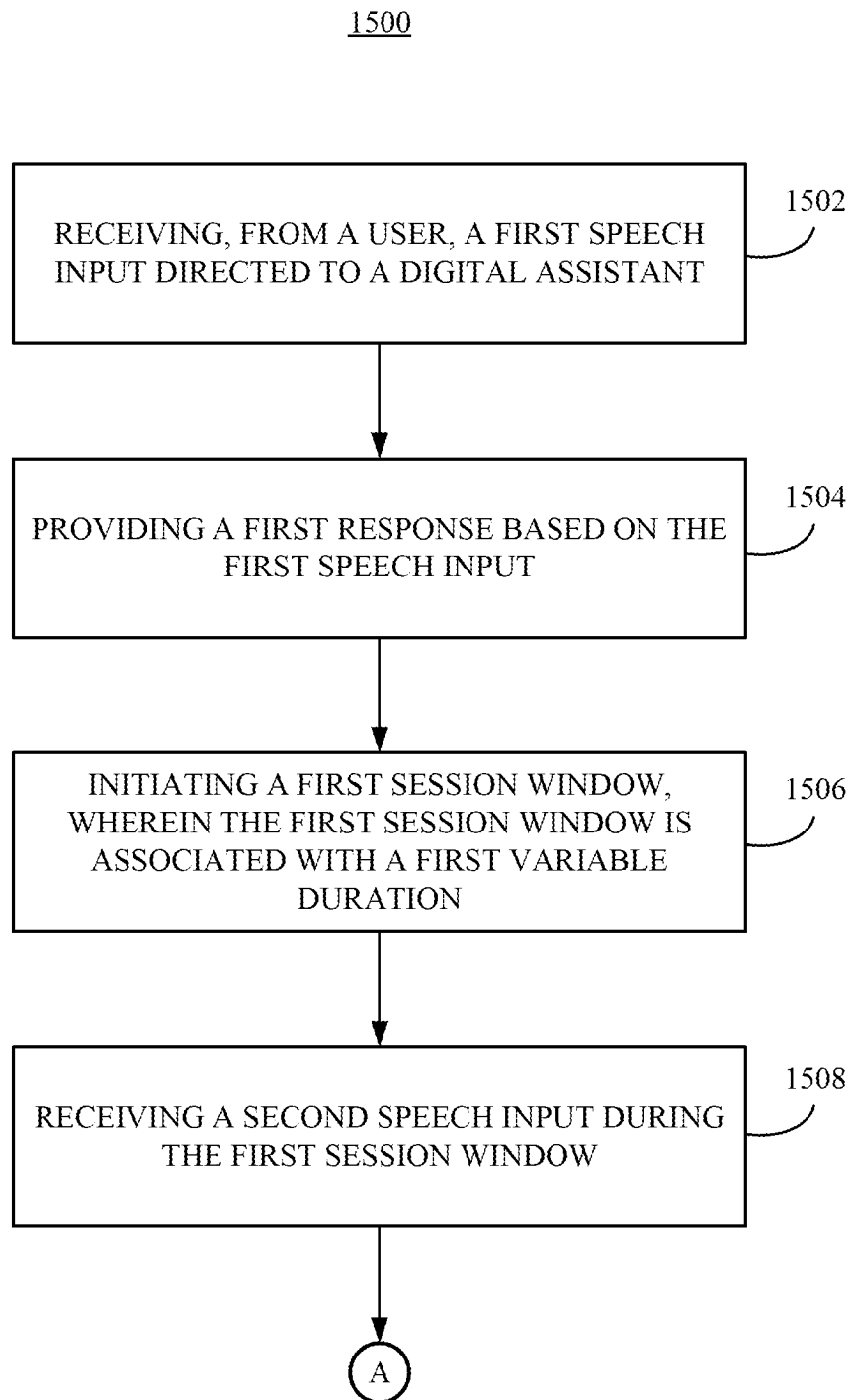
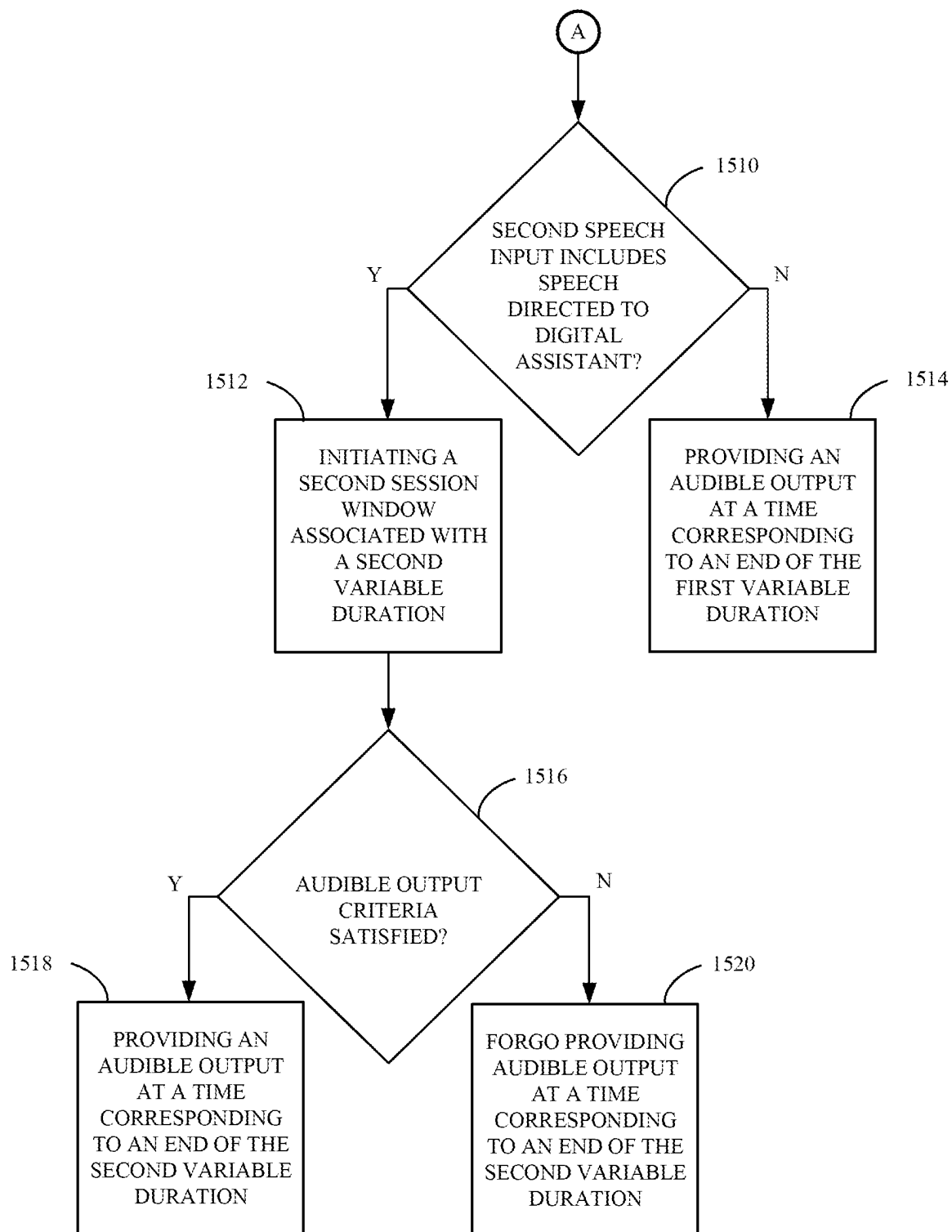


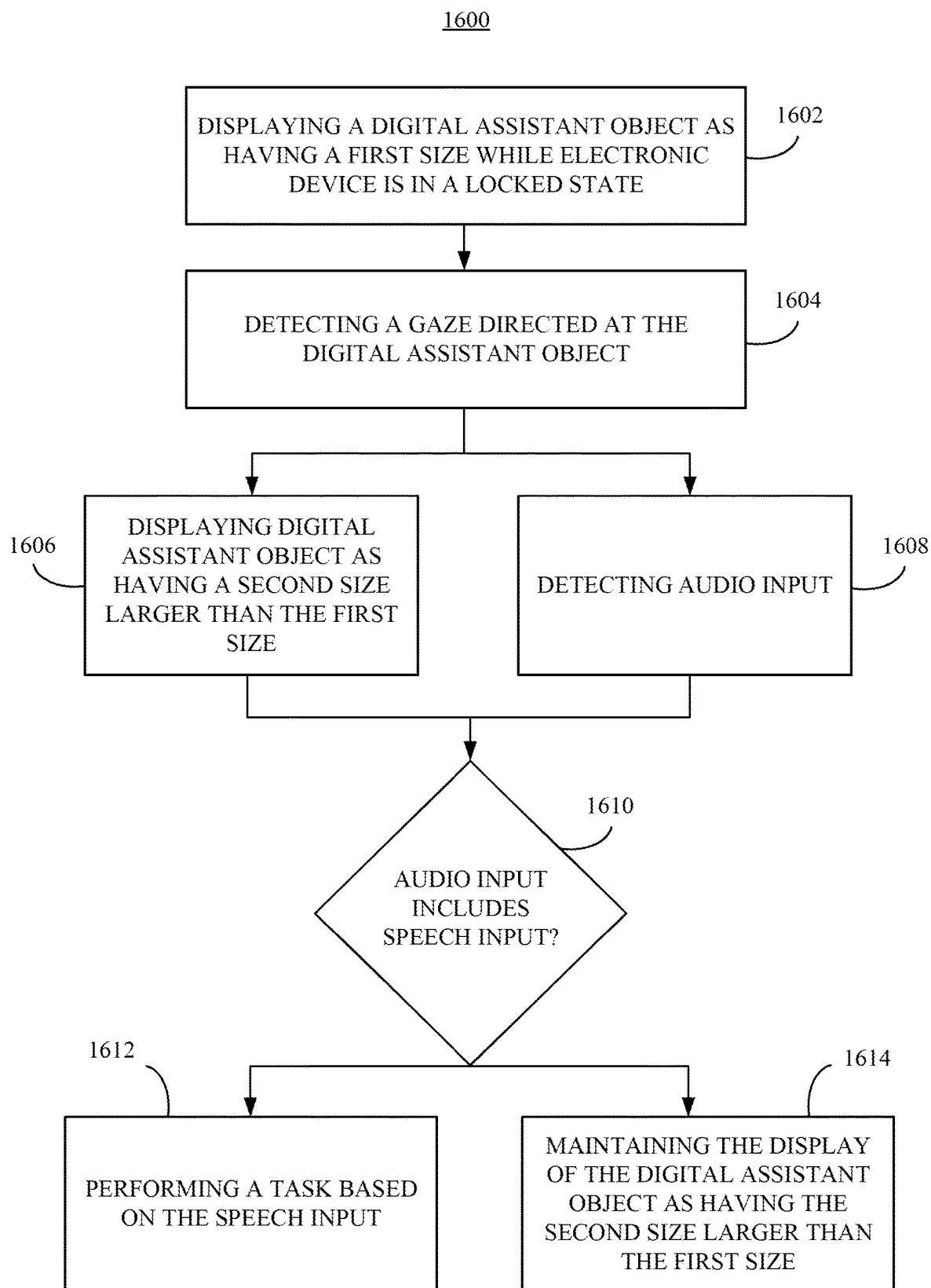
FIG. 13

**FIG. 14A**

**FIG. 14B**

**FIG. 15A**

**FIG. 15B**

**FIG. 16**

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CONTINUOUS DIALOG WITH A DIGITAL ASSISTANT

CROSS-REFERENCE TO RELATED APPLICATIONS

This application is a continuation of U.S. patent application Ser. No. 17/330,145, filed on May 25, 2021, which claims priority to U.S. Provisional Patent Application Ser. No. 63/159,557, filed on Mar. 11, 2021, titled “CONTINUOUS DIALOG WITH A DIGITAL ASSISTANT,” the contents of which are hereby incorporated by reference in their entirety.

FIELD

This relates generally to intelligent automated assistants and, more specifically, to systems for continuous dialog with a digital assistant.

BACKGROUND

Intelligent automated assistants (or digital assistants) can provide a beneficial interface between human users and electronic devices. Such assistants can allow users to interact with devices or systems using natural language in spoken and/or text forms. For example, a user can provide a speech input containing a user request to a digital assistant operating on an electronic device. The digital assistant can interpret the user's intent from the speech input and operationalize the user's intent into tasks. The tasks can then be performed by executing one or more services of the electronic device, and a relevant output responsive to the user request can be returned to the user.

While a user interacts with the device, various queries and responses may be exchanged between the user and a digital assistant. The interaction flow between the user and digital assistant can further enhance user experience when the interaction is similar to interactions between two or more humans. However, traditional digital assistant systems typically do not include capabilities to facilitate robust interactions between a user and a digital assistant. For example, conventional systems may require a “trigger” or other input that must precede user requests in order to submit such requests to the digital assistant. These systems also do not allow for the user to dynamically interrupt or correct the digital assistant if needed. Thus, an improved digital assistant system having continuous dialog capabilities is desired.

SUMMARY

Systems and processes for operating an intelligent automated assistant are provided. For example, a first speech input directed to a digital assistant is received from a user. A first response is provided based on the first speech input. A session window is initiated, wherein the session window is associated with a variable speech threshold. A second speech input is received during the session window. In accordance with a determination that the second speech input includes speech directed to the digital assistant, a duration associated with the session window is increased. In accordance with a determination that the variable speech threshold does not exceed a predetermined speech threshold, the session window is ended.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a block diagram illustrating a system and environment for implementing a digital assistant, according to various examples.

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FIG. 2A is a block diagram illustrating a portable multifunction device implementing the client-side portion of a digital assistant, according to various examples.

FIG. 2B is a block diagram illustrating exemplary components for event handling, according to various examples.

FIG. 3 illustrates a portable multifunction device implementing the client-side portion of a digital assistant, according to various examples.

FIG. 4 is a block diagram of an exemplary multifunction device with a display and a touch-sensitive surface, according to various examples.

FIG. 5A illustrates an exemplary user interface for a menu of applications on a portable multifunction device, according to various examples.

FIG. 5B illustrates an exemplary user interface for a multifunction device with a touch-sensitive surface that is separate from the display, according to various examples.

FIG. 6A illustrates a personal electronic device, according to various examples.

FIG. 6B is a block diagram illustrating a personal electronic device, according to various examples.

FIG. 7A is a block diagram illustrating a digital assistant system or a server portion thereof, according to various examples.

FIG. 7B illustrates the functions of the digital assistant shown in FIG. 7A, according to various examples.

FIG. 7C illustrates a portion of an ontology, according to various examples.

FIG. 8 illustrates a process for continuous dialog with a digital assistant, according to various examples.

FIG. 9 illustrates a process for continuous dialog with a digital assistant, according to various examples.

FIGS. 10A-10B illustrate a process for continuous dialog with a digital assistant, according to various examples.

FIG. 11 illustrates a process for continuous dialog with a digital assistant, according to various examples.

FIG. 12 illustrates a process for digital assistant interaction on a lock screen, according to various examples.

FIG. 13 illustrates a process for continuous dialog with a digital assistant, according to various examples.

FIGS. 14A-14B illustrate a process for continuous dialog with a digital assistant, according to various examples.

FIGS. 15A-15B illustrate a process for continuous dialog with a digital assistant, according to various examples.

FIG. 16 illustrates a process for continuous dialog with a digital assistant, according to various examples.

DETAILED DESCRIPTION

In the following description of examples, reference is made to the accompanying drawings in which are shown by way of illustration specific examples that can be practiced. It is to be understood that other examples can be used and structural changes can be made without departing from the scope of the various examples.

Conventional techniques for continuous digital assistant interactions are generally lacking in effectiveness. In particular, traditional systems typically require one or more triggering inputs in order to begin a digital assistant session. Such triggering inputs may be required if the user wishes to engage in follow-up or additional interactions, resulting in a cumbersome and repetitive exchange between the user and device. For example, the user may provide a triggering input, such as a spoken trigger or button press. Then, the user may provide a speech input including a user request. To the extent the response provided by the digital assistant is unsatisfactory or the user otherwise wishes to continue the

dialog, the triggering input is typically required for continuance of the dialog (e.g., submitting further queries and receiving responses). These conventional systems thus do not offer an effective and seamless means by which a user can interact with a device.

Although the following description uses terms “first,” “second,” etc. to describe various elements, these elements should not be limited by the terms. These terms are only used to distinguish one element from another. For example, a first input could be termed a second input, and, similarly, a second input could be termed a first input, without departing from the scope of the various described examples. The first input and the second input are both inputs and, in some cases, are separate and different inputs.

The terminology used in the description of the various described examples herein is for the purpose of describing particular examples only and is not intended to be limiting. As used in the description of the various described examples and the appended claims, the singular forms “a,” “an,” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. It will also be understood that the term “and/or” as used herein refers to and encompasses any and all possible combinations of one or more of the associated listed items. It will be further understood that the terms “includes,” “including,” “comprises,” and/or “comprising,” when used in this specification, specify the presence of stated features, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, and/or groups thereof.

The term “if” may be construed to mean “when” or “upon” or “in response to determining” or “in response to detecting,” depending on the context. Similarly, the phrase “if it is determined” or “if [a stated condition or event] is detected” may be construed to mean “upon determining” or “in response to determining” or “upon detecting [the stated condition or event]” or “in response to detecting [the stated condition or event],” depending on the context.

1. System and Environment

FIG. 1 illustrates a block diagram of system 100 according to various examples. In some examples, system 100 implements a digital assistant. The terms “digital assistant,” “virtual assistant,” “intelligent automated assistant,” or “automatic digital assistant” refer to any information processing system that interprets natural language input in spoken and/or textual form to infer user intent, and performs actions based on the inferred user intent. For example, to act on an inferred user intent, the system performs one or more of the following: identifying a task flow with steps and parameters designed to accomplish the inferred user intent, inputting specific requirements from the inferred user intent into the task flow; executing the task flow by invoking programs, methods, services, APIs, or the like; and generating output responses to the user in an audible (e.g., speech) and/or visual form.

Specifically, a digital assistant is capable of accepting a user request at least partially in the form of a natural language command, request, statement, narrative, and/or inquiry. Typically, the user request seeks either an informational answer or performance of a task by the digital assistant. A satisfactory response to the user request includes a provision of the requested informational answer, a performance of the requested task, or a combination of the two. For example, a user asks the digital assistant a question, such

as “Where am I right now?” Based on the user’s current location, the digital assistant answers, “You are in Central Park near the west gate.” The user also requests the performance of a task, for example, “Please invite my friends to my girlfriend’s birthday party next week.” In response, the digital assistant can acknowledge the request by saying “Yes, right away,” and then send a suitable calendar invite on behalf of the user to each of the user’s friends listed in the user’s electronic address book. During performance of a requested task, the digital assistant sometimes interacts with the user in a continuous dialogue involving multiple exchanges of information over an extended period of time. There are numerous other ways of interacting with a digital assistant to request information or performance of various tasks. In addition to providing verbal responses and taking programmed actions, the digital assistant also provides responses in other visual or audio forms, e.g., as text, alerts, music, videos, animations, etc.

As shown in FIG. 1, in some examples, a digital assistant is implemented according to a client-server model. The digital assistant includes client-side portion 102 (hereafter “DA client 102”) executed on user device 104 and server-side portion 106 (hereafter “DA server 106”) executed on server system 108. DA client 102 communicates with DA server 106 through one or more networks 110. DA client 102 provides client-side functionalities such as user-facing input and output processing and communication with DA server 106. DA server 106 provides server-side functionalities for any number of DA clients 102 each residing on a respective user device 104.

In some examples, DA server 106 includes client-facing I/O interface 112, one or more processing modules 114, data and models 116, and I/O interface to external services 118. The client-facing I/O interface 112 facilitates the client-facing input and output processing for DA server 106. One or more processing modules 114 utilize data and models 116 to process speech input and determine the user’s intent based on natural language input. Further, one or more processing modules 114 perform task execution based on inferred user intent. In some examples, DA server 106 communicates with external services 120 through network(s) 110 for task completion or information acquisition. I/O interface to external services 118 facilitates such communications.

User device 104 can be any suitable electronic device. In some examples, user device 104 is a portable multifunctional device (e.g., device 200, described below with reference to FIG. 2A), a multifunctional device (e.g., device 400, described below with reference to FIG. 4), or a personal electronic device (e.g., device 600, described below with reference to FIGS. 6A-6B.) A portable multifunctional device is, for example, a mobile telephone that also contains other functions, such as PDA and/or music player functions. Specific examples of portable multifunction devices include the Apple Watch®, iPhone®, iPod Touch®, and iPad® devices from Apple Inc. of Cupertino, California. Other examples of portable multifunction devices include, without limitation, earphones/headphones, speakers, and laptop or tablet computers. Further, in some examples, user device 104 is a non-portable multifunctional device. In particular, user device 104 is a desktop computer, a game console, a speaker, a television, or a television set-top box. In some examples, user device 104 includes a touch-sensitive surface (e.g., touch screen displays and/or touchpads). Further, user device 104 optionally includes one or more other physical user-interface devices, such as a physical keyboard, a

mouse, and/or a joystick. Various examples of electronic devices, such as multifunctional devices, are described below in greater detail.

Examples of communication network(s) **110** include local area networks (LAN) and wide area networks (WAN), e.g., the Internet. Communication network(s) **110** is implemented using any known network protocol, including various wired or wireless protocols, such as, for example, Ethernet, Universal Serial Bus (USB), FIREWIRE, Global System for Mobile Communications (GSM), Enhanced Data GSM Environment (EDGE), code division multiple access (CDMA), time division multiple access (TDMA), Bluetooth, Wi-Fi, voice over Internet Protocol (VoIP), Wi-MAX, or any other suitable communication protocol.

Server system **108** is implemented on one or more standalone data processing apparatus or a distributed network of computers. In some examples, server system **108** also employs various virtual devices and/or services of third-party service providers (e.g., third-party cloud service providers) to provide the underlying computing resources and/or infrastructure resources of server system **108**.

In some examples, user device **104** communicates with DA server **106** via second user device **122**. Second user device **122** is similar or identical to user device **104**. For example, second user device **122** is similar to devices **200**, **400**, or **600** described below with reference to FIGS. 2A, 4, and 6A-6B. User device **104** is configured to communicatively couple to second user device **122** via a direct communication connection, such as Bluetooth, NFC, BTLE, or the like, or via a wired or wireless network, such as a local Wi-Fi network. In some examples, second user device **122** is configured to act as a proxy between user device **104** and DA server **106**. For example, DA client **102** of user device **104** is configured to transmit information (e.g., a user request received at user device **104**) to DA server **106** via second user device **122**. DA server **106** processes the information and returns relevant data (e.g., data content responsive to the user request) to user device **104** via second user device **122**.

In some examples, user device **104** is configured to communicate abbreviated requests for data to second user device **122** to reduce the amount of information transmitted from user device **104**. Second user device **122** is configured to determine supplemental information to add to the abbreviated request to generate a complete request to transmit to DA server **106**. This system architecture can advantageously allow user device **104** having limited communication capabilities and/or limited battery power (e.g., a watch or a similar compact electronic device) to access services provided by DA server **106** by using second user device **122**, having greater communication capabilities and/or battery power (e.g., a mobile phone, laptop computer, tablet computer, or the like), as a proxy to DA server **106**. While only two user devices **104** and **122** are shown in FIG. 1, it should be appreciated that system **100**, in some examples, includes any number and type of user devices configured in this proxy configuration to communicate with DA server system **106**.

Although the digital assistant shown in FIG. 1 includes both a client-side portion (e.g., DA client **102**) and a server-side portion (e.g., DA server **106**), in some examples, the functions of a digital assistant are implemented as a standalone application installed on a user device. In addition, the divisions of functionalities between the client and server portions of the digital assistant can vary in different implementations. For instance, in some examples, the DA client is a thin-client that provides only user-facing input and

output processing functions, and delegates all other functionalities of the digital assistant to a backend server.

2. Electronic Devices

Attention is now directed toward embodiments of electronic devices for implementing the client-side portion of a digital assistant. FIG. 2A is a block diagram illustrating portable multifunction device **200** with touch-sensitive display system **212** in accordance with some embodiments. Touch-sensitive display **212** is sometimes called a “touch screen” for convenience and is sometimes known as or called a “touch-sensitive display system.” Device **200** includes memory **202** (which optionally includes one or more computer-readable storage mediums), memory controller **222**, one or more processing units (CPUs) **220**, peripherals interface **218**, RF circuitry **208**, audio circuitry **210**, speaker **211**, microphone **213**, input/output (I/O) subsystem **206**, other input control devices **216**, and external port **224**. Device **200** optionally includes one or more optical sensors **264**. Device **200** optionally includes one or more contact intensity sensors **265** for detecting intensity of contacts on device **200** (e.g., a touch-sensitive surface such as touch-sensitive display system **212** of device **200**). Device **200** optionally includes one or more tactile output generators **267** for generating tactile outputs on device **200** (e.g., generating tactile outputs on a touch-sensitive surface such as touch-sensitive display system **212** of device **200** or touchpad **455** of device **400**). These components optionally communicate over one or more communication buses or signal lines **203**.

As used in the specification and claims, the term “intensity” of a contact on a touch-sensitive surface refers to the force or pressure (force per unit area) of a contact (e.g., a finger contact) on the touch-sensitive surface, or to a substitute (proxy) for the force or pressure of a contact on the touch-sensitive surface. The intensity of a contact has a range of values that includes at least four distinct values and more typically includes hundreds of distinct values (e.g., at least 256). Intensity of a contact is, optionally, determined (or measured) using various approaches and various sensors or combinations of sensors. For example, one or more force sensors underneath or adjacent to the touch-sensitive surface are, optionally, used to measure force at various points on the touch-sensitive surface. In some implementations, force measurements from multiple force sensors are combined (e.g., a weighted average) to determine an estimated force of a contact. Similarly, a pressure-sensitive tip of a stylus is, optionally, used to determine a pressure of the stylus on the touch-sensitive surface. Alternatively, the size of the contact area detected on the touch-sensitive surface and/or changes thereto, the capacitance of the touch-sensitive surface proximate to the contact and/or changes thereto, and/or the resistance of the touch-sensitive surface proximate to the contact and/or changes thereto are, optionally, used as a substitute for the force or pressure of the contact on the touch-sensitive surface. In some implementations, the substitute measurements for contact force or pressure are used directly to determine whether an intensity threshold has been exceeded (e.g., the intensity threshold is described in units corresponding to the substitute measurements). In some implementations, the substitute measurements for contact force or pressure are converted to an estimated force or pressure, and the estimated force or pressure is used to determine whether an intensity threshold has been exceeded (e.g., the intensity threshold is a pressure threshold measured in units of pressure). Using the intensity of a contact

as an attribute of a user input allows for user access to additional device functionality that may otherwise not be accessible by the user on a reduced-size device with limited real estate for displaying affordances (e.g., on a touch-sensitive display) and/or receiving user input (e.g., via a touch-sensitive display, a touch-sensitive surface, or a physical/mechanical control such as a knob or a button).

As used in the specification and claims, the term “tactile output” refers to physical displacement of a device relative to a previous position of the device, physical displacement of a component (e.g., a touch-sensitive surface) of a device relative to another component (e.g., housing) of the device, or displacement of the component relative to a center of mass of the device that will be detected by a user with the user’s sense of touch. For example, in situations where the device or the component of the device is in contact with a surface of a user that is sensitive to touch (e.g., a finger, palm, or other part of a user’s hand), the tactile output generated by the physical displacement will be interpreted by the user as a tactile sensation corresponding to a perceived change in physical characteristics of the device or the component of the device. For example, movement of a touch-sensitive surface (e.g., a touch-sensitive display or trackpad) is, optionally, interpreted by the user as a “down click” or “up click” of a physical actuator button. In some cases, a user will feel a tactile sensation such as an “down click” or “up click” even when there is no movement of a physical actuator button associated with the touch-sensitive surface that is physically pressed (e.g., displaced) by the user’s movements. As another example, movement of the touch-sensitive surface is, optionally, interpreted or sensed by the user as “roughness” of the touch-sensitive surface, even when there is no change in smoothness of the touch-sensitive surface. While such interpretations of touch by a user will be subject to the individualized sensory perceptions of the user, there are many sensory perceptions of touch that are common to a large majority of users. Thus, when a tactile output is described as corresponding to a particular sensory perception of a user (e.g., an “up click,” a “down click,” “roughness”), unless otherwise stated, the generated tactile output corresponds to physical displacement of the device or a component thereof that will generate the described sensory perception for a typical (or average) user.

It should be appreciated that device **200** is only one example of a portable multifunction device, and that device **200** optionally has more or fewer components than shown, optionally combines two or more components, or optionally has a different configuration or arrangement of the components. The various components shown in FIG. **2A** are implemented in hardware, software, or a combination of both hardware and software, including one or more signal processing and/or application-specific integrated circuits.

Memory **202** includes one or more computer-readable storage mediums. The computer-readable storage mediums are, for example, tangible and non-transitory. Memory **202** includes high-speed random access memory and also includes non-volatile memory, such as one or more magnetic disk storage devices, flash memory devices, or other non-volatile solid-state memory devices. Memory controller **222** controls access to memory **202** by other components of device **200**.

In some examples, a non-transitory computer-readable storage medium of memory **202** is used to store instructions (e.g., for performing aspects of processes described below) for use by or in connection with an instruction execution system, apparatus, or device, such as a computer-based system, processor-containing system, or other system that

can fetch the instructions from the instruction execution system, apparatus, or device and execute the instructions. In other examples, the instructions (e.g., for performing aspects of the processes described below) are stored on a non-transitory computer-readable storage medium (not shown) of the server system **108** or are divided between the non-transitory computer-readable storage medium of memory **202** and the non-transitory computer-readable storage medium of server system **108**.

Peripherals interface **218** is used to couple input and output peripherals of the device to CPU **220** and memory **202**. The one or more processors **220** run or execute various software programs and/or sets of instructions stored in memory **202** to perform various functions for device **200** and to process data. In some embodiments, peripherals interface **218**, CPU **220**, and memory controller **222** are implemented on a single chip, such as chip **204**. In some other embodiments, they are implemented on separate chips.

RF (radio frequency) circuitry **208** receives and sends RF signals, also called electromagnetic signals. RF circuitry **208** converts electrical signals to/from electromagnetic signals and communicates with communications networks and other communications devices via the electromagnetic signals. RF circuitry **208** optionally includes well-known circuitry for performing these functions, including but not limited to an antenna system, an RF transceiver, one or more amplifiers, a tuner, one or more oscillators, a digital signal processor, a CODEC chipset, a subscriber identity module (SIM) card, memory, and so forth. RF circuitry **208** optionally communicates with networks, such as the Internet, also referred to as the World Wide Web (WWW), an intranet and/or a wireless network, such as a cellular telephone network, a wireless local area network (LAN) and/or a metropolitan area network (MAN), and other devices by wireless communication. The RF circuitry **208** optionally includes well-known circuitry for detecting near field communication (NFC) fields, such as by a short-range communication radio. The wireless communication optionally uses any of a plurality of communications standards, protocols, and technologies, including but not limited to Global System for Mobile Communications (GSM), Enhanced Data GSM Environment (EDGE), high-speed downlink packet access (HSDPA), high-speed uplink packet access (HSUPA), Evolution, Data-Only (EV-DO), HSPA, HSPA+, Dual-Cell HSPA (DC-HSPDA), long term evolution (LTE), near field communication (NFC), wideband code division multiple access (W-CDMA), code division multiple access (CDMA), time division multiple access (TDMA), Bluetooth, Bluetooth Low Energy (BTLE), Wireless Fidelity (Wi-Fi) (e.g., IEEE 802.11a, IEEE 802.11b, IEEE 802.11g, IEEE 802.11n, and/or IEEE 802.11ac), voice over Internet Protocol (VoIP), Wi-MAX, a protocol for e mail (e.g., Internet message access protocol (IMAP) and/or post office protocol (POP)), instant messaging (e.g., extensible messaging and presence protocol (XMPP), Session Initiation Protocol for Instant Messaging and Presence Leveraging Extensions (SIMPLE), Instant Messaging and Presence Service (IMPS)), and/or Short Message Service (SMS), or any other suitable communication protocol, including communication protocols not yet developed as of the filing date of this document.

Audio circuitry **210**, speaker **211**, and microphone **213** provide an audio interface between a user and device **200**. Audio circuitry **210** receives audio data from peripherals interface **218**, converts the audio data to an electrical signal, and transmits the electrical signal to speaker **211**. Speaker **211** converts the electrical signal to human-audible sound waves. Audio circuitry **210** also receives electrical signals

converted by microphone **213** from sound waves. Audio circuitry **210** converts the electrical signal to audio data and transmits the audio data to peripherals interface **218** for processing. Audio data are retrieved from and/or transmitted to memory **202** and/or RF circuitry **208** by peripherals interface **218**. In some embodiments, audio circuitry **210** also includes a headset jack (e.g., **312**, FIG. 3). The headset jack provides an interface between audio circuitry **210** and removable audio input/output peripherals, such as output-only headphones or a headset with both output (e.g., a headphone for one or both ears) and input (e.g., a microphone).

I/O subsystem **206** couples input/output peripherals on device **200**, such as touch screen **212** and other input control devices **216**, to peripherals interface **218**. I/O subsystem **206** optionally includes display controller **256**, optical sensor controller **258**, intensity sensor controller **259**, haptic feedback controller **261**, and one or more input controllers **260** for other input or control devices. The one or more input controllers **260** receive/send electrical signals from/to other input control devices **216**. The other input control devices **216** optionally include physical buttons (e.g., push buttons, rocker buttons, etc.), dials, slider switches, joysticks, click wheels, and so forth. In some alternate embodiments, input controller(s) **260** are, optionally, coupled to any (or none) of the following: a keyboard, an infrared port, a USB port, and a pointer device such as a mouse. The one or more buttons (e.g., **308**, FIG. 3) optionally include an up/down button for volume control of speaker **211** and/or microphone **213**. The one or more buttons optionally include a push button (e.g., **306**, FIG. 3).

A quick press of the push button disengages a lock of touch screen **212** or begin a process that uses gestures on the touch screen to unlock the device, as described in U.S. patent application Ser. No. 11/322,549, "Unlocking a Device by Performing Gestures on an Unlock Image," filed Dec. 23, 2005, U.S. Pat. No. 7,657,849, which is hereby incorporated by reference in its entirety. A longer press of the push button (e.g., **306**) turns power to device **200** on or off. The user is able to customize a functionality of one or more of the buttons. Touch screen **212** is used to implement virtual or soft buttons and one or more soft keyboards.

Touch-sensitive display **212** provides an input interface and an output interface between the device and a user. Display controller **256** receives and/or sends electrical signals from/to touch screen **212**. Touch screen **212** displays visual output to the user. The visual output includes graphics, text, icons, video, and any combination thereof (collectively termed "graphics"). In some embodiments, some or all of the visual output correspond to user-interface objects.

Touch screen **212** has a touch-sensitive surface, sensor, or set of sensors that accepts input from the user based on haptic and/or tactile contact. Touch screen **212** and display controller **256** (along with any associated modules and/or sets of instructions in memory **202**) detect contact (and any movement or breaking of the contact) on touch screen **212** and convert the detected contact into interaction with user-interface objects (e.g., one or more soft keys, icons, web pages, or images) that are displayed on touch screen **212**. In an exemplary embodiment, a point of contact between touch screen **212** and the user corresponds to a finger of the user.

Touch screen **212** uses LCD (liquid crystal display) technology, LPD (light emitting polymer display) technology, or LED (light emitting diode) technology, although other display technologies may be used in other embodiments. Touch screen **212** and display controller **256** detect contact and any movement or breaking thereof using any of

a plurality of touch sensing technologies now known or later developed, including but not limited to capacitive, resistive, infrared, and surface acoustic wave technologies, as well as other proximity sensor arrays or other elements for determining one or more points of contact with touch screen **212**. In an exemplary embodiment, projected mutual capacitance sensing technology is used, such as that found in the iPhone® and iPod Touch® from Apple Inc. of Cupertino, California.

A touch-sensitive display in some embodiments of touch screen **212** is analogous to the multi-touch sensitive touchpads described in the following U.S. Pat. No. 6,323,846 (Westerman et al.), U.S. Pat. No. 6,570,557 (Westerman et al.), and/or U.S. Pat. No. 6,677,932 (Westerman), and/or U.S. Patent Publication 2002/0015024A1, each of which is hereby incorporated by reference in its entirety. However, touch screen **212** displays visual output from device **200**, whereas touch-sensitive touchpads do not provide visual output.

A touch-sensitive display in some embodiments of touch screen **212** is as described in the following applications: (1) U.S. patent application Ser. No. 11/381,313, "Multipoint Touch Surface Controller," filed May 2, 2006; (2) U.S. patent application Ser. No. 10/840,862, "Multipoint Touchscreen," filed May 6, 2004; (3) U.S. patent application Ser. No. 10/903,964, "Gestures For Touch Sensitive Input Devices," filed Jul. 30, 2004; (4) U.S. patent application Ser. No. 11/048,264, "Gestures For Touch Sensitive Input Devices," filed Jan. 31, 2005; (5) U.S. patent application Ser. No. 11/038,590, "Mode-Based Graphical User Interfaces For Touch Sensitive Input Devices," filed Jan. 18, 2005; (6) U.S. patent application Ser. No. 11/228,758, "Virtual Input Device Placement On A Touch Screen User Interface," filed Sep. 16, 2005; (7) U.S. patent application Ser. No. 11/228,700, "Operation Of A Computer With A Touch Screen Interface," filed Sep. 16, 2005; (8) U.S. patent application Ser. No. 11/228,737, "Activating Virtual Keys Of A Touch-Screen Virtual Keyboard," filed Sep. 16, 2005; and (9) U.S. patent application Ser. No. 11/367,749, "Multi-Functional Hand-Held Device," filed Mar. 3, 2006. All of these applications are incorporated by reference herein in their entirety.

Touch screen **212** has, for example, a video resolution in excess of 100 dpi. In some embodiments, the touch screen has a video resolution of approximately 160 dpi. The user makes contact with touch screen **212** using any suitable object or appendage, such as a stylus, a finger, and so forth. In some embodiments, the user interface is designed to work primarily with finger-based contacts and gestures, which can be less precise than stylus-based input due to the larger area of contact of a finger on the touch screen. In some embodiments, the device translates the rough finger-based input into a precise pointer/cursor position or command for performing the actions desired by the user.

In some embodiments, in addition to the touch screen, device **200** includes a touchpad (not shown) for activating or deactivating particular functions. In some embodiments, the touchpad is a touch-sensitive area of the device that, unlike the touch screen, does not display visual output. The touchpad is a touch-sensitive surface that is separate from touch screen **212** or an extension of the touch-sensitive surface formed by the touch screen.

Device **200** also includes power system **262** for powering the various components. Power system **262** includes a power management system, one or more power sources (e.g., battery, alternating current (AC)), a recharging system, a power failure detection circuit, a power converter or inverter, a power status indicator (e.g., a light-emitting diode

(LED)) and any other components associated with the generation, management and distribution of power in portable devices.

Device 200 also includes one or more optical sensors 264. FIG. 2A shows an optical sensor coupled to optical sensor controller 258 in I/O subsystem 206. Optical sensor 264 includes charge-coupled device (CCD) or complementary metal-oxide semiconductor (CMOS) phototransistors. Optical sensor 264 receives light from the environment, projected through one or more lenses, and converts the light to data representing an image. In conjunction with imaging module 243 (also called a camera module), optical sensor 264 captures still images or video. In some embodiments, an optical sensor is located on the back of device 200, opposite touch screen display 212 on the front of the device so that the touch screen display is used as a viewfinder for still and/or video image acquisition. In some embodiments, an optical sensor is located on the front of the device so that the user's image is obtained for video conferencing while the user views the other video conference participants on the touch screen display. In some embodiments, the position of optical sensor 264 can be changed by the user (e.g., by rotating the lens and the sensor in the device housing) so that a single optical sensor 264 is used along with the touch screen display for both video conferencing and still and/or video image acquisition.

Device 200 optionally also includes one or more contact intensity sensors 265. FIG. 2A shows a contact intensity sensor coupled to intensity sensor controller 259 in I/O subsystem 206. Contact intensity sensor 265 optionally includes one or more piezoresistive strain gauges, capacitive force sensors, electric force sensors, piezoelectric force sensors, optical force sensors, capacitive touch-sensitive surfaces, or other intensity sensors (e.g., sensors used to measure the force (or pressure) of a contact on a touch-sensitive surface). Contact intensity sensor 265 receives contact intensity information (e.g., pressure information or a proxy for pressure information) from the environment. In some embodiments, at least one contact intensity sensor is collocated with, or proximate to, a touch-sensitive surface (e.g., touch-sensitive display system 212). In some embodiments, at least one contact intensity sensor is located on the back of device 200, opposite touch screen display 212, which is located on the front of device 200.

Device 200 also includes one or more proximity sensors 266. FIG. 2A shows proximity sensor 266 coupled to peripherals interface 218. Alternately, proximity sensor 266 is coupled to input controller 260 in I/O subsystem 206. Proximity sensor 266 is performed as described in U.S. patents application Ser. No. 11/241,839, "Proximity Detector In Handheld Device"; Ser. No. 11/240,788, "Proximity Detector In Handheld Device"; Ser. No. 11/620,702, "Using Ambient Light Sensor To Augment Proximity Sensor Output"; Ser. No. 11/586,862, "Automated Response To And Sensing Of User Activity In Portable Devices"; and Ser. No. 11/638,251, "Methods And Systems For Automatic Configuration Of Peripherals," which are hereby incorporated by reference in their entirety. In some embodiments, the proximity sensor turns off and disables touch screen 212 when the multifunction device is placed near the user's ear (e.g., when the user is making a phone call).

Device 200 optionally also includes one or more tactile output generators 267. FIG. 2A shows a tactile output generator coupled to haptic feedback controller 261 in I/O subsystem 206. Tactile output generator 267 optionally includes one or more electroacoustic devices such as speakers or other audio components and/or electromechanical

devices that convert energy into linear motion such as a motor, solenoid, electroactive polymer, piezoelectric actuator, electrostatic actuator, or other tactile output generating component (e.g., a component that converts electrical signals into tactile outputs on the device). Contact intensity sensor 265 receives tactile feedback generation instructions from haptic feedback module 233 and generates tactile outputs on device 200 that are capable of being sensed by a user of device 200. In some embodiments, at least one tactile output generator is collocated with, or proximate to, a touch-sensitive surface (e.g., touch-sensitive display system 212) and, optionally, generates a tactile output by moving the touch-sensitive surface vertically (e.g., in/out of a surface of device 200) or laterally (e.g., back and forth in the same plane as a surface of device 200). In some embodiments, at least one tactile output generator sensor is located on the back of device 200, opposite touch screen display 212, which is located on the front of device 200.

Device 200 also includes one or more accelerometers 268. FIG. 2A shows accelerometer 268 coupled to peripherals interface 218. Alternately, accelerometer 268 is coupled to an input controller 260 in I/O subsystem 206. Accelerometer 268 performs, for example, as described in U.S. patent Publication No. 20050190059, "Acceleration-based Theft Detection System for Portable Electronic Devices," and U.S. patent Publication No. 20060017692, "Methods And Apparatuses For Operating A Portable Device Based On An Accelerometer," both of which are incorporated by reference herein in their entirety. In some embodiments, information is displayed on the touch screen display in a portrait view or a landscape view based on an analysis of data received from the one or more accelerometers. Device 200 optionally includes, in addition to accelerometer(s) 268, a magnetometer (not shown) and a GPS (or GLONASS or other global navigation system) receiver (not shown) for obtaining information concerning the location and orientation (e.g., portrait or landscape) of device 200.

In some embodiments, the software components stored in memory 202 include operating system 226, communication module (or set of instructions) 228, contact/motion module (or set of instructions) 230, graphics module (or set of instructions) 232, text input module (or set of instructions) 234, Global Positioning System (GPS) module (or set of instructions) 235, Digital Assistant Client Module 229, and applications (or sets of instructions) 236. Further, memory 202 stores data and models, such as user data and models 231. Furthermore, in some embodiments, memory 202 (FIG. 2A) or 470 (FIG. 4) stores device/global internal state 257, as shown in FIGS. 2A and 4. Device/global internal state 257 includes one or more of: active application state, indicating which applications, if any, are currently active; display state, indicating what applications, views or other information occupy various regions of touch screen display 212; sensor state, including information obtained from the device's various sensors and input control devices 216; and location information concerning the device's location and/or attitude.

Operating system 226 (e.g., Darwin, RTXC, LINUX, UNIX, OS X, iOS, WINDOWS, or an embedded operating system such as VxWorks) includes various software components and/or drivers for controlling and managing general system tasks (e.g., memory management, storage device control, power management, etc.) and facilitates communication between various hardware and software components.

Communication module 228 facilitates communication with other devices over one or more external ports 224 and also includes various software components for handling data received by RF circuitry 208 and/or external port 224.

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External port **224** (e.g., Universal Serial Bus (USB), FIRE-WIRE, etc.) is adapted for coupling directly to other devices or indirectly over a network (e.g., the Internet, wireless LAN, etc.). In some embodiments, the external port is a multi-pin (e.g., 30-pin) connector that is the same as, or similar to and/or compatible with, the 30-pin connector used on iPod® (trademark of Apple Inc.) devices.

Contact/motion module **230** optionally detects contact with touch screen **212** (in conjunction with display controller **256**) and other touch-sensitive devices (e.g., a touchpad or physical click wheel). Contact/motion module **230** includes various software components for performing various operations related to detection of contact, such as determining if contact has occurred (e.g., detecting a finger-down event), determining an intensity of the contact (e.g., the force or pressure of the contact or a substitute for the force or pressure of the contact), determining if there is movement of the contact and tracking the movement across the touch-sensitive surface (e.g., detecting one or more finger-dragging events), and determining if the contact has ceased (e.g., detecting a finger-up event or a break in contact). Contact/motion module **230** receives contact data from the touch-sensitive surface. Determining movement of the point of contact, which is represented by a series of contact data, optionally includes determining speed (magnitude), velocity (magnitude and direction), and/or an acceleration (a change in magnitude and/or direction) of the point of contact. These operations are, optionally, applied to single contacts (e.g., one finger contacts) or to multiple simultaneous contacts (e.g., “multitouch”/multiple finger contacts). In some embodiments, contact/motion module **230** and display controller **256** detect contact on a touchpad.

In some embodiments, contact/motion module **230** uses a set of one or more intensity thresholds to determine whether an operation has been performed by a user (e.g., to determine whether a user has “clicked” on an icon). In some embodiments, at least a subset of the intensity thresholds are determined in accordance with software parameters (e.g., the intensity thresholds are not determined by the activation thresholds of particular physical actuators and can be adjusted without changing the physical hardware of device **200**). For example, a mouse “click” threshold of a trackpad or touch screen display can be set to any of a large range of predefined threshold values without changing the trackpad or touch screen display hardware. Additionally, in some implementations, a user of the device is provided with software settings for adjusting one or more of the set of intensity thresholds (e.g., by adjusting individual intensity thresholds and/or by adjusting a plurality of intensity thresholds at once with a system-level click “intensity” parameter).

Contact/motion module **230** optionally detects a gesture input by a user. Different gestures on the touch-sensitive surface have different contact patterns (e.g., different motions, timings, and/or intensities of detected contacts). Thus, a gesture is, optionally, detected by detecting a particular contact pattern. For example, detecting a finger tap gesture includes detecting a finger-down event followed by detecting a finger-up (liftoff) event at the same position (or substantially the same position) as the finger-down event (e.g., at the position of an icon). As another example, detecting a finger swipe gesture on the touch-sensitive surface includes detecting a finger-down event followed by detecting one or more finger-dragging events, and subsequently followed by detecting a finger-up (liftoff) event.

Graphics module **232** includes various known software components for rendering and displaying graphics on touch screen **212** or other display, including components for

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changing the visual impact (e.g., brightness, transparency, saturation, contrast, or other visual property) of graphics that are displayed. As used herein, the term “graphics” includes any object that can be displayed to a user, including, without limitation, text, web pages, icons (such as user-interface objects including soft keys), digital images, videos, animations, and the like.

In some embodiments, graphics module **232** stores data representing graphics to be used. Each graphic is, optionally, assigned a corresponding code. Graphics module **232** receives, from applications etc., one or more codes specifying graphics to be displayed along with, if necessary, coordinate data and other graphic property data, and then generates screen image data to output to display controller **256**.

Haptic feedback module **233** includes various software components for generating instructions used by tactile output generator(s) **267** to produce tactile outputs at one or more locations on device **200** in response to user interactions with device **200**.

Text input module **234**, which is, in some examples, a component of graphics module **232**, provides soft keyboards for entering text in various applications (e.g., contacts **237**, email **240**, IM **241**, browser **247**, and any other application that needs text input).

GPS module **235** determines the location of the device and provides this information for use in various applications (e.g., to telephone module **238** for use in location-based dialing; to camera module **243** as picture/video metadata; and to applications that provide location-based services such as weather widgets, local yellow page widgets, and map/navigation widgets).

Digital assistant client module **229** includes various client-side digital assistant instructions to provide the client-side functionalities of the digital assistant. For example, digital assistant client module **229** is capable of accepting voice input (e.g., speech input), text input, touch input, and/or gestural input through various user interfaces (e.g., microphone **213**, accelerometer(s) **268**, touch-sensitive display system **212**, optical sensor(s) **264**, other input control devices **216**, etc.) of portable multifunction device **200**. Digital assistant client module **229** is also capable of providing output in audio (e.g., speech output), visual, and/or tactile forms through various output interfaces (e.g., speaker **211**, touch-sensitive display system **212**, tactile output generator(s) **267**, etc.) of portable multifunction device **200**. For example, output is provided as voice, sound, alerts, text messages, menus, graphics, videos, animations, vibrations, and/or combinations of two or more of the above. During operation, digital assistant client module **229** communicates with DA server **106** using RF circuitry **208**.

User data and models **231** include various data associated with the user (e.g., user-specific vocabulary data, user preference data, user-specified name pronunciations, data from the user’s electronic address book, to-do lists, shopping lists, etc.) to provide the client-side functionalities of the digital assistant. Further, user data and models **231** include various models (e.g., speech recognition models, statistical language models, natural language processing models, ontology, task flow models, service models, etc.) for processing user input and determining user intent.

In some examples, digital assistant client module **229** utilizes the various sensors, subsystems, and peripheral devices of portable multifunction device **200** to gather additional information from the surrounding environment of the portable multifunction device **200** to establish a context associated with a user, the current user interaction, and/or

the current user input. In some examples, digital assistant client module 229 provides the contextual information or a subset thereof with the user input to DA server 106 to help infer the user's intent. In some examples, the digital assistant also uses the contextual information to determine how to prepare and deliver outputs to the user. Contextual information is referred to as context data.

In some examples, the contextual information that accompanies the user input includes sensor information, e.g., lighting, ambient noise, ambient temperature, images or videos of the surrounding environment, etc. In some examples, the contextual information can also include the physical state of the device, e.g., device orientation, device location, device temperature, power level, speed, acceleration, motion patterns, cellular signals strength, etc. In some examples, information related to the software state of DA server 106, e.g., running processes, installed programs, past and present network activities, background services, error logs, resources usage, etc., and of portable multifunction device 200 is provided to DA server 106 as contextual information associated with a user input.

In some examples, the digital assistant client module 229 selectively provides information (e.g., user data 231) stored on the portable multifunction device 200 in response to requests from DA server 106. In some examples, digital assistant client module 229 also elicits additional input from the user via a natural language dialogue or other user interfaces upon request by DA server 106. Digital assistant client module 229 passes the additional input to DA server 106 to help DA server 106 in intent deduction and/or fulfillment of the user's intent expressed in the user request.

A more detailed description of a digital assistant is described below with reference to FIGS. 7A-7C. It should be recognized that digital assistant client module 229 can include any number of the sub-modules of digital assistant module 726 described below.

Applications 236 include the following modules (or sets of instructions), or a subset or superset thereof:

- Contacts module 237 (sometimes called an address book or contact list);
- Telephone module 238;
- Video conference module 239;
- E-mail client module 240;
- Instant messaging (IM) module 241;
- Workout support module 242;
- Camera module 243 for still and/or video images;
- Image management module 244;
- Video player module;
- Music player module;
- Browser module 247;
- Calendar module 248;
- Widget modules 249, which includes, in some examples, one or more of: weather widget 249-1, stocks widget 249-2, calculator widget 249-3, alarm clock widget 249-4, dictionary widget 249-5, and other widgets obtained by the user, as well as user-created widgets 249-6;
- Widget creator module 250 for making user-created widgets 249-6;
- Search module 251;
- Video and music player module 252, which merges video player module and music player module;
- Notes module 253;
- Map module 254; and/or
- Online video module 255.

Examples of other applications 236 that are stored in memory 202 include other word processing applications,

other image editing applications, drawing applications, presentation applications, JAVA-enabled applications, encryption, digital rights management, voice recognition, and voice replication.

In conjunction with touch screen 212, display controller 256, contact/motion module 230, graphics module 232, and text input module 234, contacts module 237 are used to manage an address book or contact list (e.g., stored in application internal state 292 of contacts module 237 in memory 202 or memory 470), including: adding name(s) to the address book; deleting name(s) from the address book; associating telephone number(s), e-mail address(es), physical address(es) or other information with a name; associating an image with a name; categorizing and sorting names; providing telephone numbers or e-mail addresses to initiate and/or facilitate communications by telephone 238, video conference module 239, e-mail 240, or IM 241; and so forth.

In conjunction with RF circuitry 208, audio circuitry 210, speaker 211, microphone 213, touch screen 212, display controller 256, contact/motion module 230, graphics module 232, and text input module 234, telephone module 238 are used to enter a sequence of characters corresponding to a telephone number, access one or more telephone numbers in contacts module 237, modify a telephone number that has been entered, dial a respective telephone number, conduct a conversation, and disconnect or hang up when the conversation is completed. As noted above, the wireless communication uses any of a plurality of communications standards, protocols, and technologies.

In conjunction with RF circuitry 208, audio circuitry 210, speaker 211, microphone 213, touch screen 212, display controller 256, optical sensor 264, optical sensor controller 258, contact/motion module 230, graphics module 232, text input module 234, contacts module 237, and telephone module 238, video conference module 239 includes executable instructions to initiate, conduct, and terminate a video conference between a user and one or more other participants in accordance with user instructions.

In conjunction with RF circuitry 208, touch screen 212, display controller 256, contact/motion module 230, graphics module 232, and text input module 234, e-mail client module 240 includes executable instructions to create, send, receive, and manage e-mail in response to user instructions. In conjunction with image management module 244, e-mail client module 240 makes it very easy to create and send e-mails with still or video images taken with camera module 243.

In conjunction with RF circuitry 208, touch screen 212, display controller 256, contact/motion module 230, graphics module 232, and text input module 234, the instant messaging module 241 includes executable instructions to enter a sequence of characters corresponding to an instant message, to modify previously entered characters, to transmit a respective instant message (for example, using a Short Message Service (SMS) or Multimedia Message Service (MMS) protocol for telephony-based instant messages or using XMPP, SIMPLE, or IMPS for Internet-based instant messages), to receive instant messages, and to view received instant messages. In some embodiments, transmitted and/or received instant messages include graphics, photos, audio files, video files and/or other attachments as are supported in an MMS and/or an Enhanced Messaging Service (EMS). As used herein, "instant messaging" refers to both telephony-based messages (e.g., messages sent using SMS or MMS) and Internet-based messages (e.g., messages sent using XMPP, SIMPLE, or IMPS).

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In conjunction with RF circuitry 208, touch screen 212, display controller 256, contact/motion module 230, graphics module 232, text input module 234, GPS module 235, map module 254, and music player module, workout support module 242 includes executable instructions to create workouts (e.g., with time, distance, and/or calorie burning goals); communicate with workout sensors (sports devices); receive workout sensor data; calibrate sensors used to monitor a workout; select and play music for a workout; and display, store, and transmit workout data.

In conjunction with touch screen 212, display controller 256, optical sensor(s) 264, optical sensor controller 258, contact/motion module 230, graphics module 232, and image management module 244, camera module 243 includes executable instructions to capture still images or video (including a video stream) and store them into memory 202, modify characteristics of a still image or video, or delete a still image or video from memory 202.

In conjunction with touch screen 212, display controller 256, contact/motion module 230, graphics module 232, text input module 234, and camera module 243, image management module 244 includes executable instructions to arrange, modify (e.g., edit), or otherwise manipulate, label, delete, present (e.g., in a digital slide show or album), and store still and/or video images.

In conjunction with RF circuitry 208, touch screen 212, display controller 256, contact/motion module 230, graphics module 232, and text input module 234, browser module 247 includes executable instructions to browse the Internet in accordance with user instructions, including searching, linking to, receiving, and displaying web pages or portions thereof, as well as attachments and other files linked to web pages.

In conjunction with RF circuitry 208, touch screen 212, display controller 256, contact/motion module 230, graphics module 232, text input module 234, e-mail client module 240, and browser module 247, calendar module 248 includes executable instructions to create, display, modify, and store calendars and data associated with calendars (e.g., calendar entries, to-do lists, etc.) in accordance with user instructions.

In conjunction with RF circuitry 208, touch screen 212, display controller 256, contact/motion module 230, graphics module 232, text input module 234, and browser module 247, widget modules 249 are mini-applications that can be downloaded and used by a user (e.g., weather widget 249-1, stocks widget 249-2, calculator widget 249-3, alarm clock widget 249-4, and dictionary widget 249-5) or created by the user (e.g., user-created widget 249-6). In some embodiments, a widget includes an HTML (Hypertext Markup Language) file, a CSS (Cascading Style Sheets) file, and a JavaScript file. In some embodiments, a widget includes an XML (Extensible Markup Language) file and a JavaScript file (e.g., Yahoo! Widgets).

In conjunction with RF circuitry 208, touch screen 212, display controller 256, contact/motion module 230, graphics module 232, text input module 234, and browser module 247, the widget creator module 250 are used by a user to create widgets (e.g., turning a user-specified portion of a web page into a widget).

In conjunction with touch screen 212, display controller 256, contact/motion module 230, graphics module 232, and text input module 234, search module 251 includes executable instructions to search for text, music, sound, image, video, and/or other files in memory 202 that match one or more search criteria (e.g., one or more user-specified search terms) in accordance with user instructions.

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In conjunction with touch screen 212, display controller 256, contact/motion module 230, graphics module 232, audio circuitry 210, speaker 211, RF circuitry 208, and browser module 247, video and music player module 252 includes executable instructions that allow the user to download and play back recorded music and other sound files stored in one or more file formats, such as MP3 or AAC files, and executable instructions to display, present, or otherwise play back videos (e.g., on touch screen 212 or on an external, connected display via external port 224). In some embodiments, device 200 optionally includes the functionality of an MP3 player, such as an iPod (trademark of Apple Inc.).

In conjunction with touch screen 212, display controller 256, contact/motion module 230, graphics module 232, and text input module 234, notes module 253 includes executable instructions to create and manage notes, to-do lists, and the like in accordance with user instructions.

In conjunction with RF circuitry 208, touch screen 212, display controller 256, contact/motion module 230, graphics module 232, text input module 234, GPS module 235, and browser module 247, map module 254 are used to receive, display, modify, and store maps and data associated with maps (e.g., driving directions, data on stores and other points of interest at or near a particular location, and other location-based data) in accordance with user instructions.

In conjunction with touch screen 212, display controller 256, contact/motion module 230, graphics module 232, audio circuitry 210, speaker 211, RF circuitry 208, text input module 234, e-mail client module 240, and browser module 247, online video module 255 includes instructions that allow the user to access, browse, receive (e.g., by streaming and/or download), play back (e.g., on the touch screen or on an external, connected display via external port 224), send an e-mail with a link to a particular online video, and otherwise manage online videos in one or more file formats, such as H.264. In some embodiments, instant messaging module 241, rather than e-mail client module 240, is used to send a link to a particular online video. Additional description of the online video application can be found in U.S. Provisional Patent Application No. 60/936,562, "Portable Multifunction Device, Method, and Graphical User Interface for Playing Online Videos," filed Jun. 20, 2007, and U.S. patent application Ser. No. 11/968,067, "Portable Multifunction Device, Method, and Graphical User Interface for Playing Online Videos," filed Dec. 31, 2007, the contents of which are hereby incorporated by reference in their entirety.

Each of the above-identified modules and applications corresponds to a set of executable instructions for performing one or more functions described above and the methods described in this application (e.g., the computer-implemented methods and other information processing methods described herein). These modules (e.g., sets of instructions) need not be implemented as separate software programs, procedures, or modules, and thus various subsets of these modules can be combined or otherwise rearranged in various embodiments. For example, video player module can be combined with music player module into a single module (e.g., video and music player module 252, FIG. 2A). In some embodiments, memory 202 stores a subset of the modules and data structures identified above. Furthermore, memory 202 stores additional modules and data structures not described above.

In some embodiments, device 200 is a device where operation of a predefined set of functions on the device is performed exclusively through a touch screen and/or a touchpad. By using a touch screen and/or a touchpad as the primary input control device for operation of device 200, the

number of physical input control devices (such as push buttons, dials, and the like) on device 200 is reduced.

The predefined set of functions that are performed exclusively through a touch screen and/or a touchpad optionally include navigation between user interfaces. In some embodiments, the touchpad, when touched by the user, navigates device 200 to a main, home, or root menu from any user interface that is displayed on device 200. In such embodiments, a “menu button” is implemented using a touchpad. In some other embodiments, the menu button is a physical push button or other physical input control device instead of a touchpad.

FIG. 2B is a block diagram illustrating exemplary components for event handling in accordance with some embodiments. In some embodiments, memory 202 (FIG. 2A) or 470 (FIG. 4) includes event sorter 270 (e.g., in operating system 226) and a respective application 236-1 (e.g., any of the aforementioned applications 237-251, 255, 480-490).

Event sorter 270 receives event information and determines the application 236-1 and application view 291 of application 236-1 to which to deliver the event information. Event sorter 270 includes event monitor 271 and event dispatcher module 274. In some embodiments, application 236-1 includes application internal state 292, which indicates the current application view(s) displayed on touch-sensitive display 212 when the application is active or executing. In some embodiments, device/global internal state 257 is used by event sorter 270 to determine which application(s) is (are) currently active, and application internal state 292 is used by event sorter 270 to determine application views 291 to which to deliver event information.

In some embodiments, application internal state 292 includes additional information, such as one or more of: resume information to be used when application 236-1 resumes execution, user interface state information that indicates information being displayed or that is ready for display by application 236-1, a state queue for enabling the user to go back to a prior state or view of application 236-1, and a redo/undo queue of previous actions taken by the user.

Event monitor 271 receives event information from peripherals interface 218. Event information includes information about a sub-event (e.g., a user touch on touch-sensitive display 212, as part of a multi-touch gesture). Peripherals interface 218 transmits information it receives from I/O subsystem 206 or a sensor, such as proximity sensor 266, accelerometer(s) 268, and/or microphone 213 (through audio circuitry 210). Information that peripherals interface 218 receives from I/O subsystem 206 includes information from touch-sensitive display 212 or a touch-sensitive surface.

In some embodiments, event monitor 271 sends requests to the peripherals interface 218 at predetermined intervals. In response, peripherals interface 218 transmits event information. In other embodiments, peripherals interface 218 transmits event information only when there is a significant event (e.g., receiving an input above a predetermined noise threshold and/or for more than a predetermined duration).

In some embodiments, event sorter 270 also includes a hit view determination module 272 and/or an active event recognizer determination module 273.

Hit view determination module 272 provides software procedures for determining where a sub-event has taken place within one or more views when touch-sensitive display 212 displays more than one view. Views are made up of controls and other elements that a user can see on the display.

Another aspect of the user interface associated with an application is a set of views, sometimes herein called application views or user interface windows, in which information is displayed and touch-based gestures occur. The application views (of a respective application) in which a touch is detected correspond to programmatic levels within a programmatic or view hierarchy of the application. For example, the lowest level view in which a touch is detected is called the hit view, and the set of events that are recognized as proper inputs is determined based, at least in part, on the hit view of the initial touch that begins a touch-based gesture.

Hit view determination module 272 receives information related to sub events of a touch-based gesture. When an application has multiple views organized in a hierarchy, hit view determination module 272 identifies a hit view as the lowest view in the hierarchy which should handle the sub-event. In most circumstances, the hit view is the lowest level view in which an initiating sub-event occurs (e.g., the first sub-event in the sequence of sub-events that form an event or potential event). Once the hit view is identified by the hit view determination module 272, the hit view typically receives all sub-events related to the same touch or input source for which it was identified as the hit view.

Active event recognizer determination module 273 determines which view or views within a view hierarchy should receive a particular sequence of sub-events. In some embodiments, active event recognizer determination module 273 determines that only the hit view should receive a particular sequence of sub-events. In other embodiments, active event recognizer determination module 273 determines that all views that include the physical location of a sub-event are actively involved views, and therefore determines that all actively involved views should receive a particular sequence of sub-events. In other embodiments, even if touch sub-events were entirely confined to the area associated with one particular view, views higher in the hierarchy would still remain as actively involved views.

Event dispatcher module 274 dispatches the event information to an event recognizer (e.g., event recognizer 280). In embodiments including active event recognizer determination module 273, event dispatcher module 274 delivers the event information to an event recognizer determined by active event recognizer determination module 273. In some embodiments, event dispatcher module 274 stores in an event queue the event information, which is retrieved by a respective event receiver 282.

In some embodiments, operating system 226 includes event sorter 270. Alternatively, application 236-1 includes event sorter 270. In yet other embodiments, event sorter 270 is a stand-alone module, or a part of another module stored in memory 202, such as contact/motion module 230.

In some embodiments, application 236-1 includes a plurality of event handlers 290 and one or more application views 291, each of which includes instructions for handling touch events that occur within a respective view of the application's user interface. Each application view 291 of the application 236-1 includes one or more event recognizers 280. Typically, a respective application view 291 includes a plurality of event recognizers 280. In other embodiments, one or more of event recognizers 280 are part of a separate module, such as a user interface kit (not shown) or a higher level object from which application 236-1 inherits methods and other properties. In some embodiments, a respective event handler 290 includes one or more of: data updater 276, object updater 277, GUI updater 278, and/or event data 279 received from event sorter 270. Event

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handler **290** utilizes or calls data updater **276**, object updater **277**, or GUI updater **278** to update the application internal state **292**. Alternatively, one or more of the application views **291** include one or more respective event handlers **290**. Also, in some embodiments, one or more of data updater **276**, object updater **277**, and GUI updater **278** are included in a respective application view **291**.

A respective event recognizer **280** receives event information (e.g., event data **279**) from event sorter **270** and identifies an event from the event information. Event recognizer **280** includes event receiver **282** and event comparator **284**. In some embodiments, event recognizer **280** also includes at least a subset of: metadata **283**, and event delivery instructions **288** (which include sub-event delivery instructions).

Event receiver **282** receives event information from event sorter **270**. The event information includes information about a sub-event, for example, a touch or a touch movement. Depending on the sub-event, the event information also includes additional information, such as location of the sub-event. When the sub-event concerns motion of a touch, the event information also includes speed and direction of the sub-event. In some embodiments, events include rotation of the device from one orientation to another (e.g., from a portrait orientation to a landscape orientation, or vice versa), and the event information includes corresponding information about the current orientation (also called device attitude) of the device.

Event comparator **284** compares the event information to predefined event or sub-event definitions and, based on the comparison, determines an event or sub event, or determines or updates the state of an event or sub-event. In some embodiments, event comparator **284** includes event definitions **286**. Event definitions **286** contain definitions of events (e.g., predefined sequences of sub-events), for example, event 1 (**287-1**), event 2 (**287-2**), and others. In some embodiments, sub-events in an event (**287**) include, for example, touch begin, touch end, touch movement, touch cancellation, and multiple touching. In one example, the definition for event 1 (**287-1**) is a double tap on a displayed object. The double tap, for example, comprises a first touch (touch begin) on the displayed object for a predetermined phase, a first liftoff (touch end) for a predetermined phase, a second touch (touch begin) on the displayed object for a predetermined phase, and a second liftoff (touch end) for a predetermined phase. In another example, the definition for event 2 (**287-2**) is a dragging on a displayed object. The dragging, for example, comprises a touch (or contact) on the displayed object for a predetermined phase, a movement of the touch across touch-sensitive display **212**, and liftoff of the touch (touch end). In some embodiments, the event also includes information for one or more associated event handlers **290**.

In some embodiments, event definition **287** includes a definition of an event for a respective user-interface object. In some embodiments, event comparator **284** performs a hit test to determine which user-interface object is associated with a sub-event. For example, in an application view in which three user-interface objects are displayed on touch-sensitive display **212**, when a touch is detected on touch-sensitive display **212**, event comparator **284** performs a hit test to determine which of the three user-interface objects is associated with the touch (sub-event). If each displayed object is associated with a respective event handler **290**, the event comparator uses the result of the hit test to determine which event handler **290** should be activated. For example,

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event comparator **284** selects an event handler associated with the sub-event and the object triggering the hit test.

In some embodiments, the definition for a respective event (**287**) also includes delayed actions that delay delivery of the event information until after it has been determined whether the sequence of sub-events does or does not correspond to the event recognizer's event type.

When a respective event recognizer **280** determines that the series of sub-events do not match any of the events in event definitions **286**, the respective event recognizer **280** enters an event impossible, event failed, or event ended state, after which it disregards subsequent sub-events of the touch-based gesture. In this situation, other event recognizers, if any, that remain active for the hit view continue to track and process sub-events of an ongoing touch-based gesture.

In some embodiments, a respective event recognizer **280** includes metadata **283** with configurable properties, flags, and/or lists that indicate how the event delivery system should perform sub-event delivery to actively involved event recognizers. In some embodiments, metadata **283** includes configurable properties, flags, and/or lists that indicate how event recognizers interact, or are enabled to interact, with one another. In some embodiments, metadata **283** includes configurable properties, flags, and/or lists that indicate whether sub-events are delivered to varying levels in the view or programmatic hierarchy.

In some embodiments, a respective event recognizer **280** activates event handler **290** associated with an event when one or more particular sub-events of an event are recognized. In some embodiments, a respective event recognizer **280** delivers event information associated with the event to event handler **290**. Activating an event handler **290** is distinct from sending (and deferred sending) sub-events to a respective hit view. In some embodiments, event recognizer **280** throws a flag associated with the recognized event, and event handler **290** associated with the flag catches the flag and performs a predefined process.

In some embodiments, event delivery instructions **288** include sub-event delivery instructions that deliver event information about a sub-event without activating an event handler. Instead, the sub-event delivery instructions deliver event information to event handlers associated with the series of sub-events or to actively involved views. Event handlers associated with the series of sub-events or with actively involved views receive the event information and perform a predetermined process.

In some embodiments, data updater **276** creates and updates data used in application **236-1**. For example, data updater **276** updates the telephone number used in contacts module **237**, or stores a video file used in video player module. In some embodiments, object updater **277** creates and updates objects used in application **236-1**. For example, object updater **277** creates a new user-interface object or updates the position of a user-interface object. GUI updater **278** updates the GUI. For example, GUI updater **278** prepares display information and sends it to graphics module **232** for display on a touch-sensitive display.

In some embodiments, event handler(s) **290** includes or has access to data updater **276**, object updater **277**, and GUI updater **278**. In some embodiments, data updater **276**, object updater **277**, and GUI updater **278** are included in a single module of a respective application **236-1** or application view **291**. In other embodiments, they are included in two or more software modules.

It shall be understood that the foregoing discussion regarding event handling of user touches on touch-sensitive

displays also applies to other forms of user inputs to operate multifunction devices **200** with input devices, not all of which are initiated on touch screens. For example, mouse movement and mouse button presses, optionally coordinated with single or multiple keyboard presses or holds; contact movements such as taps, drags, scrolls, etc. on touchpads; pen stylus inputs; movement of the device; oral instructions; detected eye movements; biometric inputs; and/or any combination thereof are optionally utilized as inputs corresponding to sub-events which define an event to be recognized.

FIG. **3** illustrates a portable multifunction device **200** having a touch screen **212** in accordance with some embodiments. The touch screen optionally displays one or more graphics within user interface (UI) **300**. In this embodiment, as well as others described below, a user is enabled to select one or more of the graphics by making a gesture on the graphics, for example, with one or more fingers **302** (not drawn to scale in the figure) or one or more styluses **303** (not drawn to scale in the figure). In some embodiments, selection of one or more graphics occurs when the user breaks contact with the one or more graphics. In some embodiments, the gesture optionally includes one or more taps, one or more swipes (from left to right, right to left, upward and/or downward), and/or a rolling of a finger (from right to left, left to right, upward and/or downward) that has made contact with device **200**. In some implementations or circumstances, inadvertent contact with a graphic does not select the graphic. For example, a swipe gesture that sweeps over an application icon optionally does not select the corresponding application when the gesture corresponding to selection is a tap.

Device **200** also includes one or more physical buttons, such as “home” or menu button **304**. As described previously, menu button **304** is used to navigate to any application **236** in a set of applications that is executed on device **200**. Alternatively, in some embodiments, the menu button is implemented as a soft key in a GUI displayed on touch screen **212**.

In one embodiment, device **200** includes touch screen **212**, menu button **304**, push button **306** for powering the device on/off and locking the device, volume adjustment button(s) **308**, subscriber identity module (SIM) card slot **310**, headset jack **312**, and docking/charging external port **224**. Push button **306** is, optionally, used to turn the power on/off on the device by depressing the button and holding the button in the depressed state for a predefined time interval; to lock the device by depressing the button and releasing the button before the predefined time interval has elapsed; and/or to unlock the device or initiate an unlock process. In an alternative embodiment, device **200** also accepts verbal input for activation or deactivation of some functions through microphone **213**. Device **200** also, optionally, includes one or more contact intensity sensors **265** for detecting intensity of contacts on touch screen **212** and/or one or more tactile output generators **267** for generating tactile outputs for a user of device **200**.

FIG. **4** is a block diagram of an exemplary multifunction device with a display and a touch-sensitive surface in accordance with some embodiments. Device **400** need not be portable. In some embodiments, device **400** is a laptop computer, a desktop computer, a tablet computer, a multimedia player device, a navigation device, an educational device (such as a child’s learning toy), a gaming system, or a control device (e.g., a home or industrial controller). Device **400** typically includes one or more processing units (CPUs) **410**, one or more network or other communications interfaces **460**, memory **470**, and one or more communica-

tion buses **420** for interconnecting these components. Communication buses **420** optionally include circuitry (sometimes called a chipset) that interconnects and controls communications between system components. Device **400** includes input/output (I/O) interface **430** comprising display **440**, which is typically a touch screen display. I/O interface **430** also optionally includes a keyboard and/or mouse (or other pointing device) **450** and touchpad **455**, tactile output generator **457** for generating tactile outputs on device **400** (e.g., similar to tactile output generator(s) **267** described above with reference to FIG. **2A**), sensors **459** (e.g., optical, acceleration, proximity, touch-sensitive, and/or contact intensity sensors similar to contact intensity sensor(s) **265** described above with reference to FIG. **2A**). Memory **470** includes high-speed random access memory, such as DRAM, SRAM, DDR RAM, or other random access solid state memory devices; and optionally includes non-volatile memory, such as one or more magnetic disk storage devices, optical disk storage devices, flash memory devices, or other non-volatile solid state storage devices. Memory **470** optionally includes one or more storage devices remotely located from CPU(s) **410**. In some embodiments, memory **470** stores programs, modules, and data structures analogous to the programs, modules, and data structures stored in memory **202** of portable multifunction device **200** (FIG. **2A**), or a subset thereof. Furthermore, memory **470** optionally stores additional programs, modules, and data structures not present in memory **202** of portable multifunction device **200**. For example, memory **470** of device **400** optionally stores drawing module **480**, presentation module **482**, word processing module **484**, website creation module **486**, disk authoring module **488**, and/or spreadsheet module **490**, while memory **202** of portable multifunction device **200** (FIG. **2A**) optionally does not store these modules.

Each of the above-identified elements in FIG. **4** is, in some examples, stored in one or more of the previously mentioned memory devices. Each of the above-identified modules corresponds to a set of instructions for performing a function described above. The above-identified modules or programs (e.g., sets of instructions) need not be implemented as separate software programs, procedures, or modules, and thus various subsets of these modules are combined or otherwise rearranged in various embodiments. In some embodiments, memory **470** stores a subset of the modules and data structures identified above. Furthermore, memory **470** stores additional modules and data structures not described above.

Attention is now directed towards embodiments of user interfaces that can be implemented on, for example, portable multifunction device **200**.

FIG. **5A** illustrates an exemplary user interface for a menu of applications on portable multifunction device **200** in accordance with some embodiments. Similar user interfaces are implemented on device **400**. In some embodiments, user interface **500** includes the following elements, or a subset or superset thereof:

- Signal strength indicator(s) **502** for wireless communication(s), such as cellular and Wi-Fi signals;
- Time **504**;
- Bluetooth indicator **505**;
- Battery status indicator **506**;
- Tray **508** with icons for frequently used applications, such as:
 - Icon **516** for telephone module **238**, labeled “Phone,” which optionally includes an indicator **514** of the number of missed calls or voicemail messages;

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Icon **518** for e-mail client module **240**, labeled “Mail,” which optionally includes an indicator **510** of the number of unread e-mails;

Icon **520** for browser module **247**, labeled “Browser;” and

Icon **522** for video and music player module **252**, also referred to as iPod (trademark of Apple Inc.) module **252**, labeled “iPod;” and

Icons for other applications, such as:

Icon **524** for IM module **241**, labeled “Messages;”

Icon **526** for calendar module **248**, labeled “Calendar;”

Icon **528** for image management module **244**, labeled “Photos;”

Icon **530** for camera module **243**, labeled “Camera;”

Icon **532** for online video module **255**, labeled “Online Video;”

Icon **534** for stocks widget **249-2**, labeled “Stocks;”

Icon **536** for map module **254**, labeled “Maps;”

Icon **538** for weather widget **249-1**, labeled “Weather;”

Icon **540** for alarm clock widget **249-4**, labeled “Clock;”

Icon **542** for workout support module **242**, labeled “Workout Support;”

Icon **544** for notes module **253**, labeled “Notes;” and

Icon **546** for a settings application or module, labeled “Settings,” which provides access to settings for device **200** and its various applications **236**.

It should be noted that the icon labels illustrated in FIG. 5A are merely exemplary. For example, icon **522** for video and music player module **252** is optionally labeled “Music” or “Music Player.” Other labels are, optionally, used for various application icons. In some embodiments, a label for a respective application icon includes a name of an application corresponding to the respective application icon. In some embodiments, a label for a particular application icon is distinct from a name of an application corresponding to the particular application icon.

FIG. 5B illustrates an exemplary user interface on a device (e.g., device **400**, FIG. 4) with a touch-sensitive surface **551** (e.g., a tablet or touchpad **455**, FIG. 4) that is separate from the display **550** (e.g., touch screen display **212**). Device **400** also, optionally, includes one or more contact intensity sensors (e.g., one or more of sensors **459**) for detecting intensity of contacts on touch-sensitive surface **551** and/or one or more tactile output generators **457** for generating tactile outputs for a user of device **400**.

Although some of the examples which follow will be given with reference to inputs on touch screen display **212** (where the touch-sensitive surface and the display are combined), in some embodiments, the device detects inputs on a touch-sensitive surface that is separate from the display, as shown in FIG. 5B. In some embodiments, the touch-sensitive surface (e.g., **551** in FIG. 5B) has a primary axis (e.g., **552** in FIG. 5B) that corresponds to a primary axis (e.g., **553** in FIG. 5B) on the display (e.g., **550**). In accordance with these embodiments, the device detects contacts (e.g., **560** and **562** in FIG. 5B) with the touch-sensitive surface **551** at locations that correspond to respective locations on the display (e.g., in FIG. 5B, **560** corresponds to **568** and **562** corresponds to **570**). In this way, user inputs (e.g., contacts **560** and **562**, and movements thereof) detected by the device on the touch-sensitive surface (e.g., **551** in FIG. 5B) are used by the device to manipulate the user interface on the display (e.g., **550** in FIG. 5B) of the multifunction device when the touch-sensitive surface is separate from the display. It should be understood that similar methods are, optionally, used for other user interfaces described herein.

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Additionally, while the following examples are given primarily with reference to finger inputs (e.g., finger contacts, finger tap gestures, finger swipe gestures), it should be understood that, in some embodiments, one or more of the finger inputs are replaced with input from another input device (e.g., a mouse-based input or stylus input). For example, a swipe gesture is, optionally, replaced with a mouse click (e.g., instead of a contact) followed by movement of the cursor along the path of the swipe (e.g., instead of movement of the contact). As another example, a tap gesture is, optionally, replaced with a mouse click while the cursor is located over the location of the tap gesture (e.g., instead of detection of the contact followed by ceasing to detect the contact). Similarly, when multiple user inputs are simultaneously detected, it should be understood that multiple computer mice are, optionally, used simultaneously, or a mouse and finger contacts are, optionally, used simultaneously.

FIG. 6A illustrates exemplary personal electronic device **600**. Device **600** includes body **602**. In some embodiments, device **600** includes some or all of the features described with respect to devices **200** and **400** (e.g., FIGS. 2A-4). In some embodiments, device **600** has touch-sensitive display screen **604**, hereafter touch screen **604**. Alternatively, or in addition to touch screen **604**, device **600** has a display and a touch-sensitive surface. As with devices **200** and **400**, in some embodiments, touch screen **604** (or the touch-sensitive surface) has one or more intensity sensors for detecting intensity of contacts (e.g., touches) being applied. The one or more intensity sensors of touch screen **604** (or the touch-sensitive surface) provide output data that represents the intensity of touches. The user interface of device **600** responds to touches based on their intensity, meaning that touches of different intensities can invoke different user interface operations on device **600**.

Techniques for detecting and processing touch intensity are found, for example, in related applications: International Patent Application Serial No. PCT/US2013/040061, titled “Device, Method, and Graphical User Interface for Displaying User Interface Objects Corresponding to an Application,” filed May 8, 2013, and International Patent Application Serial No. PCT/US2013/069483, titled “Device, Method, and Graphical User Interface for Transitioning Between Touch Input to Display Output Relationships,” filed Nov. 11, 2013, each of which is hereby incorporated by reference in their entirety.

In some embodiments, device **600** has one or more input mechanisms **606** and **608**. Input mechanisms **606** and **608**, if included, are physical. Examples of physical input mechanisms include push buttons and rotatable mechanisms. In some embodiments, device **600** has one or more attachment mechanisms. Such attachment mechanisms, if included, can permit attachment of device **600** with, for example, hats, eyewear, earrings, necklaces, shirts, jackets, bracelets, watch straps, chains, trousers, belts, shoes, purses, backpacks, and so forth. These attachment mechanisms permit device **600** to be worn by a user.

FIG. 6B depicts exemplary personal electronic device **600**. In some embodiments, device **600** includes some or all of the components described with respect to FIGS. 2A, 2B, and 4. Device **600** has bus **612** that operatively couples I/O section **614** with one or more computer processors **616** and memory **618**. I/O section **614** is connected to display **604**, which can have touch-sensitive component **622** and, optionally, touch-intensity sensitive component **624**. In addition, I/O section **614** is connected with communication unit **630** for receiving application and operating system data, using

Wi-Fi, Bluetooth, near field communication (NFC), cellular, and/or other wireless communication techniques. Device 600 includes input mechanisms 606 and/or 608. Input mechanism 606 is a rotatable input device or a depressible and rotatable input device, for example. Input mechanism 608 is a button, in some examples.

Input mechanism 608 is a microphone, in some examples. Personal electronic device 600 includes, for example, various sensors, such as GPS sensor 632, accelerometer 634, directional sensor 640 (e.g., compass), gyroscope 636, motion sensor 638, and/or a combination thereof, all of which are operatively connected to I/O section 614.

Memory 618 of personal electronic device 600 is a non-transitory computer-readable storage medium, for storing computer-executable instructions, which, when executed by one or more computer processors 616, for example, cause the computer processors to perform the techniques and processes described below. The computer-executable instructions, for example, are also stored and/or transported within any non-transitory computer-readable storage medium for use by or in connection with an instruction execution system, apparatus, or device, such as a computer-based system, processor-containing system, or other system that can fetch the instructions from the instruction execution system, apparatus, or device and execute the instructions. Personal electronic device 600 is not limited to the components and configuration of FIG. 6B, but can include other or additional components in multiple configurations.

As used here, the term “affordance” refers to a user-interactive graphical user interface object that is, for example, displayed on the display screen of devices 200, 400, and/or 600 (FIGS. 2A, 4, and 6A-6B). For example, an image (e.g., icon), a button, and text (e.g., hyperlink) each constitutes an affordance.

As used herein, the term “focus selector” refers to an input element that indicates a current part of a user interface with which a user is interacting. In some implementations that include a cursor or other location marker, the cursor acts as a “focus selector” so that when an input (e.g., a press input) is detected on a touch-sensitive surface (e.g., touchpad 455 in FIG. 4 or touch-sensitive surface 551 in FIG. 5B) while the cursor is over a particular user interface element (e.g., a button, window, slider or other user interface element), the particular user interface element is adjusted in accordance with the detected input. In some implementations that include a touch screen display (e.g., touch-sensitive display system 212 in FIG. 2A or touch screen 212 in FIG. 5A) that enables direct interaction with user interface elements on the touch screen display, a detected contact on the touch screen acts as a “focus selector” so that when an input (e.g., a press input by the contact) is detected on the touch screen display at a location of a particular user interface element (e.g., a button, window, slider, or other user interface element), the particular user interface element is adjusted in accordance with the detected input. In some implementations, focus is moved from one region of a user interface to another region of the user interface without corresponding movement of a cursor or movement of a contact on a touch screen display (e.g., by using a tab key or arrow keys to move focus from one button to another button); in these implementations, the focus selector moves in accordance with movement of focus between different regions of the user interface. Without regard to the specific form taken by the focus selector, the focus selector is generally the user interface element (or contact on a touch screen display) that is controlled by the user so as to communicate the user’s intended interaction with the user interface (e.g., by indicating, to the device, the

element of the user interface with which the user is intending to interact). For example, the location of a focus selector (e.g., a cursor, a contact, or a selection box) over a respective button while a press input is detected on the touch-sensitive surface (e.g., a touchpad or touch screen) will indicate that the user is intending to activate the respective button (as opposed to other user interface elements shown on a display of the device).

As used in the specification and claims, the term “characteristic intensity” of a contact refers to a characteristic of the contact based on one or more intensities of the contact. In some embodiments, the characteristic intensity is based on multiple intensity samples. The characteristic intensity is, optionally, based on a predefined number of intensity samples, or a set of intensity samples collected during a predetermined time period (e.g., 0.05, 0.1, 0.2, 0.5, 1, 2, 5, 10 seconds) relative to a predefined event (e.g., after detecting the contact, prior to detecting liftoff of the contact, before or after detecting a start of movement of the contact, prior to detecting an end of the contact, before or after detecting an increase in intensity of the contact, and/or before or after detecting a decrease in intensity of the contact). A characteristic intensity of a contact is, optionally based on one or more of: a maximum value of the intensities of the contact, a mean value of the intensities of the contact, an average value of the intensities of the contact, a top 10 percentile value of the intensities of the contact, a value at the half maximum of the intensities of the contact, a value at the 90 percent maximum of the intensities of the contact, or the like. In some embodiments, the duration of the contact is used in determining the characteristic intensity (e.g., when the characteristic intensity is an average of the intensity of the contact over time). In some embodiments, the characteristic intensity is compared to a set of one or more intensity thresholds to determine whether an operation has been performed by a user. For example, the set of one or more intensity thresholds includes a first intensity threshold and a second intensity threshold. In this example, a contact with a characteristic intensity that does not exceed the first threshold results in a first operation, a contact with a characteristic intensity that exceeds the first intensity threshold and does not exceed the second intensity threshold results in a second operation, and a contact with a characteristic intensity that exceeds the second threshold results in a third operation. In some embodiments, a comparison between the characteristic intensity and one or more thresholds is used to determine whether or not to perform one or more operations (e.g., whether to perform a respective operation or forgo performing the respective operation) rather than being used to determine whether to perform a first operation or a second operation.

In some embodiments, a portion of a gesture is identified for purposes of determining a characteristic intensity. For example, a touch-sensitive surface receives a continuous swipe contact transitioning from a start location and reaching an end location, at which point the intensity of the contact increases. In this example, the characteristic intensity of the contact at the end location is based on only a portion of the continuous swipe contact, and not the entire swipe contact (e.g., only the portion of the swipe contact at the end location). In some embodiments, a smoothing algorithm is applied to the intensities of the swipe contact prior to determining the characteristic intensity of the contact. For example, the smoothing algorithm optionally includes one or more of: an unweighted sliding-average smoothing algorithm, a triangular smoothing algorithm, a median filter smoothing algorithm, and/or an exponential smoothing algo-

rithm. In some circumstances, these smoothing algorithms eliminate narrow spikes or dips in the intensities of the swipe contact for purposes of determining a characteristic intensity.

The intensity of a contact on the touch-sensitive surface is characterized relative to one or more intensity thresholds, such as a contact-detection intensity threshold, a light press intensity threshold, a deep press intensity threshold, and/or one or more other intensity thresholds. In some embodiments, the light press intensity threshold corresponds to an intensity at which the device will perform operations typically associated with clicking a button of a physical mouse or a trackpad. In some embodiments, the deep press intensity threshold corresponds to an intensity at which the device will perform operations that are different from operations typically associated with clicking a button of a physical mouse or a trackpad. In some embodiments, when a contact is detected with a characteristic intensity below the light press intensity threshold (e.g., and above a nominal contact-detection intensity threshold below which the contact is no longer detected), the device will move a focus selector in accordance with movement of the contact on the touch-sensitive surface without performing an operation associated with the light press intensity threshold or the deep press intensity threshold. Generally, unless otherwise stated, these intensity thresholds are consistent between different sets of user interface figures.

An increase of characteristic intensity of the contact from an intensity below the light press intensity threshold to an intensity between the light press intensity threshold and the deep press intensity threshold is sometimes referred to as a “light press” input. An increase of characteristic intensity of the contact from an intensity below the deep press intensity threshold to an intensity above the deep press intensity threshold is sometimes referred to as a “deep press” input. An increase of characteristic intensity of the contact from an intensity below the contact-detection intensity threshold to an intensity between the contact-detection intensity threshold and the light press intensity threshold is sometimes referred to as detecting the contact on the touch-surface. A decrease of characteristic intensity of the contact from an intensity above the contact-detection intensity threshold to an intensity below the contact-detection intensity threshold is sometimes referred to as detecting liftoff of the contact from the touch-surface. In some embodiments, the contact-detection intensity threshold is zero. In some embodiments, the contact-detection intensity threshold is greater than zero.

In some embodiments described herein, one or more operations are performed in response to detecting a gesture that includes a respective press input or in response to detecting the respective press input performed with a respective contact (or a plurality of contacts), where the respective press input is detected based at least in part on detecting an increase in intensity of the contact (or plurality of contacts) above a press-input intensity threshold. In some embodiments, the respective operation is performed in response to detecting the increase in intensity of the respective contact above the press-input intensity threshold (e.g., a “down stroke” of the respective press input). In some embodiments, the press input includes an increase in intensity of the respective contact above the press-input intensity threshold and a subsequent decrease in intensity of the contact below the press-input intensity threshold, and the respective operation is performed in response to detecting the subsequent decrease in intensity of the respective contact below the press-input threshold (e.g., an “up stroke” of the respective press input).

In some embodiments, the device employs intensity hysteresis to avoid accidental inputs sometimes termed “jitter,” where the device defines or selects a hysteresis intensity threshold with a predefined relationship to the press-input intensity threshold (e.g., the hysteresis intensity threshold is X intensity units lower than the press-input intensity threshold or the hysteresis intensity threshold is 75%, 90%, or some reasonable proportion of the press-input intensity threshold). Thus, in some embodiments, the press input includes an increase in intensity of the respective contact above the press-input intensity threshold and a subsequent decrease in intensity of the contact below the hysteresis intensity threshold that corresponds to the press-input intensity threshold, and the respective operation is performed in response to detecting the subsequent decrease in intensity of the respective contact below the hysteresis intensity threshold (e.g., an “up stroke” of the respective press input). Similarly, in some embodiments, the press input is detected only when the device detects an increase in intensity of the contact from an intensity at or below the hysteresis intensity threshold to an intensity at or above the press-input intensity threshold and, optionally, a subsequent decrease in intensity of the contact to an intensity at or below the hysteresis intensity, and the respective operation is performed in response to detecting the press input (e.g., the increase in intensity of the contact or the decrease in intensity of the contact, depending on the circumstances).

For ease of explanation, the descriptions of operations performed in response to a press input associated with a press-input intensity threshold or in response to a gesture including the press input are, optionally, triggered in response to detecting either: an increase in intensity of a contact above the press-input intensity threshold, an increase in intensity of a contact from an intensity below the hysteresis intensity threshold to an intensity above the press-input intensity threshold, a decrease in intensity of the contact below the press-input intensity threshold, and/or a decrease in intensity of the contact below the hysteresis intensity threshold corresponding to the press-input intensity threshold. Additionally, in examples where an operation is described as being performed in response to detecting a decrease in intensity of a contact below the press-input intensity threshold, the operation is, optionally, performed in response to detecting a decrease in intensity of the contact below a hysteresis intensity threshold corresponding to, and lower than, the press-input intensity threshold.

3. Digital Assistant System

FIG. 7A illustrates a block diagram of digital assistant system 700 in accordance with various examples. In some examples, digital assistant system 700 is implemented on a standalone computer system. In some examples, digital assistant system 700 is distributed across multiple computers. In some examples, some of the modules and functions of the digital assistant are divided into a server portion and a client portion, where the client portion resides on one or more user devices (e.g., devices 104, 122, 200, 400, or 600) and communicates with the server portion (e.g., server system 108) through one or more networks, e.g., as shown in FIG. 1. In some examples, digital assistant system 700 is an implementation of server system 108 (and/or DA server 106) shown in FIG. 1. It should be noted that digital assistant system 700 is only one example of a digital assistant system, and that digital assistant system 700 can have more or fewer components than shown, can combine two or more components, or can have a different configuration or arrangement

of the components. The various components shown in FIG. 7A are implemented in hardware, software instructions for execution by one or more processors, firmware, including one or more signal processing and/or application specific integrated circuits, or a combination thereof.

Digital assistant system 700 includes memory 702, one or more processors 704, input/output (I/O) interface 706, and network communications interface 708. These components can communicate with one another over one or more communication buses or signal lines 710.

In some examples, memory 702 includes a non-transitory computer-readable medium, such as high-speed random access memory and/or a non-volatile computer-readable storage medium (e.g., one or more magnetic disk storage devices, flash memory devices, or other non-volatile solid-state memory devices).

In some examples, I/O interface 706 couples input/output devices 716 of digital assistant system 700, such as displays, keyboards, touch screens, and microphones, to user interface module 722. I/O interface 706, in conjunction with user interface module 722, receives user inputs (e.g., voice input, keyboard inputs, touch inputs, etc.) and processes them accordingly. In some examples, e.g., when the digital assistant is implemented on a standalone user device, digital assistant system 700 includes any of the components and I/O communication interfaces described with respect to devices 200, 400, or 600 in FIGS. 2A, 4, 6A-6B, respectively. In some examples, digital assistant system 700 represents the server portion of a digital assistant implementation, and can interact with the user through a client-side portion residing on a user device (e.g., devices 104, 200, 400, or 600).

In some examples, the network communications interface 708 includes wired communication port(s) 712 and/or wireless transmission and reception circuitry 714. The wired communication port(s) receives and send communication signals via one or more wired interfaces, e.g., Ethernet, Universal Serial Bus (USB), FIREWIRE, etc. The wireless circuitry 714 receives and sends RF signals and/or optical signals from/to communications networks and other communications devices. The wireless communications use any of a plurality of communications standards, protocols, and technologies, such as GSM, EDGE, CDMA, TDMA, Bluetooth, Wi-Fi, VoIP, Wi-MAX, or any other suitable communication protocol. Network communications interface 708 enables communication between digital assistant system 700 with networks, such as the Internet, an intranet, and/or a wireless network, such as a cellular telephone network, a wireless local area network (LAN), and/or a metropolitan area network (MAN), and other devices.

In some examples, memory 702, or the computer-readable storage media of memory 702, stores programs, modules, instructions, and data structures including all or a subset of: operating system 718, communications module 720, user interface module 722, one or more applications 724, and digital assistant module 726. In particular, memory 702, or the computer-readable storage media of memory 702, stores instructions for performing the processes described below. One or more processors 704 execute these programs, modules, and instructions, and reads/writes from/to the data structures.

Operating system 718 (e.g., Darwin, RTXC, LINUX, UNIX, iOS, OS X, WINDOWS, or an embedded operating system such as VxWorks) includes various software components and/or drivers for controlling and managing general system tasks (e.g., memory management, storage device

control, power management, etc.) and facilitates communications between various hardware, firmware, and software components.

Communications module 720 facilitates communications between digital assistant system 700 with other devices over network communications interface 708. For example, communications module 720 communicates with RF circuitry 208 of electronic devices such as devices 200, 400, and 600 shown in FIGS. 2A, 4, 6A-6B, respectively. Communications module 720 also includes various components for handling data received by wireless circuitry 714 and/or wired communications port 712.

User interface module 722 receives commands and/or inputs from a user via I/O interface 706 (e.g., from a keyboard, touch screen, pointing device, controller, and/or microphone), and generate user interface objects on a display. User interface module 722 also prepares and delivers outputs (e.g., speech, sound, animation, text, icons, vibrations, haptic feedback, light, etc.) to the user via the I/O interface 706 (e.g., through displays, audio channels, speakers, touch-pads, etc.).

Applications 724 include programs and/or modules that are configured to be executed by one or more processors 704. For example, if the digital assistant system is implemented on a standalone user device, applications 724 include user applications, such as games, a calendar application, a navigation application, or an email application. If digital assistant system 700 is implemented on a server, applications 724 include resource management applications, diagnostic applications, or scheduling applications, for example.

Memory 702 also stores digital assistant module 726 (or the server portion of a digital assistant). In some examples, digital assistant module 726 includes the following sub-modules, or a subset or superset thereof: input/output processing module 728, speech-to-text (STT) processing module 730, natural language processing module 732, dialogue flow processing module 734, task flow processing module 736, service processing module 738, and speech synthesis processing module 740. Each of these modules has access to one or more of the following systems or data and models of the digital assistant module 726, or a subset or superset thereof: ontology 760, vocabulary index 744, user data 748, task flow models 754, service models 756, and ASR systems 758.

In some examples, using the processing modules, data, and models implemented in digital assistant module 726, the digital assistant can perform at least some of the following: converting speech input into text; identifying a user's intent expressed in a natural language input received from the user; actively eliciting and obtaining information needed to fully infer the user's intent (e.g., by disambiguating words, games, intentions, etc.); determining the task flow for fulfilling the inferred intent; and executing the task flow to fulfill the inferred intent.

In some examples, as shown in FIG. 7B, I/O processing module 728 interacts with the user through I/O devices 716 in FIG. 7A or with a user device (e.g., devices 104, 200, 400, or 600) through network communications interface 708 in FIG. 7A to obtain user input (e.g., a speech input) and to provide responses (e.g., as speech outputs) to the user input. I/O processing module 728 optionally obtains contextual information associated with the user input from the user device, along with or shortly after the receipt of the user input. The contextual information includes user-specific data, vocabulary, and/or preferences relevant to the user input. In some examples, the contextual information also

includes software and hardware states of the user device at the time the user request is received, and/or information related to the surrounding environment of the user at the time that the user request was received. In some examples, I/O processing module 728 also sends follow-up questions to, and receive answers from, the user regarding the user request. When a user request is received by I/O processing module 728 and the user request includes speech input, I/O processing module 728 forwards the speech input to STT processing module 730 (or speech recognizer) for speech-to-text conversions.

STT processing module 730 includes one or more ASR systems 758. The one or more ASR systems 758 can process the speech input that is received through I/O processing module 728 to produce a recognition result. Each ASR system 758 includes a front-end speech pre-processor. The front-end speech pre-processor extracts representative features from the speech input. For example, the front-end speech pre-processor performs a Fourier transform on the speech input to extract spectral features that characterize the speech input as a sequence of representative multi-dimensional vectors. Further, each ASR system 758 includes one or more speech recognition models (e.g., acoustic models and/or language models) and implements one or more speech recognition engines. Examples of speech recognition models include Hidden Markov Models, Gaussian-Mixture Models, Deep Neural Network Models, n-gram language models, and other statistical models. Examples of speech recognition engines include the dynamic time warping based engines and weighted finite-state transducers (WFST) based engines. The one or more speech recognition models and the one or more speech recognition engines are used to process the extracted representative features of the front-end speech pre-processor to produce intermediate recognitions results (e.g., phonemes, phonemic strings, and sub-words), and ultimately, text recognition results (e.g., words, word strings, or sequence of tokens). In some examples, the speech input is processed at least partially by a third-party service or on the user's device (e.g., device 104, 200, 400, or 600) to produce the recognition result. Once STT processing module 730 produces recognition results containing a text string (e.g., words, or sequence of words, or sequence of tokens), the recognition result is passed to natural language processing module 732 for intent deduction. In some examples, STT processing module 730 produces multiple candidate text representations of the speech input. Each candidate text representation is a sequence of words or tokens corresponding to the speech input. In some examples, each candidate text representation is associated with a speech recognition confidence score. Based on the speech recognition confidence scores, STT processing module 730 ranks the candidate text representations and provides the n-best (e.g., n highest ranked) candidate text representation(s) to natural language processing module 732 for intent deduction, where n is a predetermined integer greater than zero. For example, in one example, only the highest ranked (n=1) candidate text representation is passed to natural language processing module 732 for intent deduction. In another example, the five highest ranked (n=5) candidate text representations are passed to natural language processing module 732 for intent deduction.

More details on the speech-to-text processing are described in U.S. Utility application Ser. No. 13/236,942 for "Consolidating Speech Recognition Results," filed on Sep. 20, 2011, the entire disclosure of which is incorporated herein by reference.

In some examples, STT processing module 730 includes and/or accesses a vocabulary of recognizable words via phonetic alphabet conversion module 731. Each vocabulary word is associated with one or more candidate pronunciations of the word represented in a speech recognition phonetic alphabet. In particular, the vocabulary of recognizable words includes a word that is associated with a plurality of candidate pronunciations. For example, the vocabulary includes the word "tomato" that is associated with the candidate pronunciations of /tə'meɪroʊ/ and /tə'matoʊ/. Further, vocabulary words are associated with custom candidate pronunciations that are based on previous speech inputs from the user. Such custom candidate pronunciations are stored in STT processing module 730 and are associated with a particular user via the user's profile on the device. In some examples, the candidate pronunciations for words are determined based on the spelling of the word and one or more linguistic and/or phonetic rules. In some examples, the candidate pronunciations are manually generated, e.g., based on known canonical pronunciations.

In some examples, the candidate pronunciations are ranked based on the commonness of the candidate pronunciation. For example, the candidate pronunciation /tə'meɪroʊ/ is ranked higher than /tə'matoʊ/, because the former is a more commonly used pronunciation (e.g., among all users, for users in a particular geographical region, or for any other appropriate subset of users). In some examples, candidate pronunciations are ranked based on whether the candidate pronunciation is a custom candidate pronunciation associated with the user. For example, custom candidate pronunciations are ranked higher than canonical candidate pronunciations. This can be useful for recognizing proper nouns having a unique pronunciation that deviates from canonical pronunciation. In some examples, candidate pronunciations are associated with one or more speech characteristics, such as geographic origin, nationality, or ethnicity. For example, the candidate pronunciation /tə'meɪroʊ/ is associated with the United States, whereas the candidate pronunciation /tə'matoʊ/ is associated with Great Britain. Further, the rank of the candidate pronunciation is based on one or more characteristics (e.g., geographic origin, nationality, ethnicity, etc.) of the user stored in the user's profile on the device. For example, it can be determined from the user's profile that the user is associated with the United States. Based on the user being associated with the United States, the candidate pronunciation /tə'meɪroʊ/ (associated with the United States) is ranked higher than the candidate pronunciation /tə'matoʊ/ (associated with Great Britain). In some examples, one of the ranked candidate pronunciations is selected as a predicted pronunciation (e.g., the most likely pronunciation).

When a speech input is received, STT processing module 730 is used to determine the phonemes corresponding to the speech input (e.g., using an acoustic model), and then attempt to determine words that match the phonemes (e.g., using a language model). For example, if STT processing module 730 first identifies the sequence of phonemes /tə'meɪroʊ/ corresponding to a portion of the speech input, it can then determine, based on vocabulary index 744, that this sequence corresponds to the word "tomato."

In some examples, STT processing module 730 uses approximate matching techniques to determine words in an utterance. Thus, for example, the STT processing module 730 determines that the sequence of phonemes /tə'meɪroʊ/ corresponds to the word "tomato," even if that particular sequence of phonemes is not one of the candidate sequence of phonemes for that word.

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Natural language processing module 732 (“natural language processor”) of the digital assistant takes the n-best candidate text representation(s) (“word sequence(s)” or “token sequence(s)”) generated by STT processing module 730, and attempts to associate each of the candidate text representations with one or more “actionable intents” recognized by the digital assistant. An “actionable intent” (or “user intent”) represents a task that can be performed by the digital assistant, and can have an associated task flow implemented in task flow models 754. The associated task flow is a series of programmed actions and steps that the digital assistant takes in order to perform the task. The scope of a digital assistant’s capabilities is dependent on the number and variety of task flows that have been implemented and stored in task flow models 754, or in other words, on the number and variety of “actionable intents” that the digital assistant recognizes. The effectiveness of the digital assistant, however, also depends on the assistant’s ability to infer the correct “actionable intent(s)” from the user request expressed in natural language.

In some examples, in addition to the sequence of words or tokens obtained from STT processing module 730, natural language processing module 732 also receives contextual information associated with the user request, e.g., from I/O processing module 728. The natural language processing module 732 optionally uses the contextual information to clarify, supplement, and/or further define the information contained in the candidate text representations received from STT processing module 730. The contextual information includes, for example, user preferences, hardware, and/or software states of the user device, sensor information collected before, during, or shortly after the user request, prior interactions (e.g., dialogue) between the digital assistant and the user, and the like. As described herein, contextual information is, in some examples, dynamic, and changes with time, location, content of the dialogue, and other factors.

In some examples, the natural language processing is based on, e.g., ontology 760. Ontology 760 is a hierarchical structure containing many nodes, each node representing either an “actionable intent” or a “property” relevant to one or more of the “actionable intents” or other “properties.” As noted above, an “actionable intent” represents a task that the digital assistant is capable of performing, i.e., it is “actionable” or can be acted on. A “property” represents a parameter associated with an actionable intent or a sub-aspect of another property. A linkage between an actionable intent node and a property node in ontology 760 defines how a parameter represented by the property node pertains to the task represented by the actionable intent node.

In some examples, ontology 760 is made up of actionable intent nodes and property nodes. Within ontology 760, each actionable intent node is linked to one or more property nodes either directly or through one or more intermediate property nodes. Similarly, each property node is linked to one or more actionable intent nodes either directly or through one or more intermediate property nodes. For example, as shown in FIG. 7C, ontology 760 includes a “restaurant reservation” node (i.e., an actionable intent node). Property nodes “restaurant,” “date/time” (for the reservation), and “party size” are each directly linked to the actionable intent node (i.e., the “restaurant reservation” node).

In addition, property nodes “cuisine,” “price range,” “phone number,” and “location” are sub-nodes of the property node “restaurant,” and are each linked to the “restaurant reservation” node (i.e., the actionable intent node) through

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the intermediate property node “restaurant.” For another example, as shown in FIG. 7C, ontology 760 also includes a “set reminder” node (i.e., another actionable intent node). Property nodes “date/time” (for setting the reminder) and “subject” (for the reminder) are each linked to the “set reminder” node. Since the property “date/time” is relevant to both the task of making a restaurant reservation and the task of setting a reminder, the property node “date/time” is linked to both the “restaurant reservation” node and the “set reminder” node in ontology 760.

An actionable intent node, along with its linked property nodes, is described as a “domain.” In the present discussion, each domain is associated with a respective actionable intent, and refers to the group of nodes (and the relationships there between) associated with the particular actionable intent. For example, ontology 760 shown in FIG. 7C includes an example of restaurant reservation domain 762 and an example of reminder domain 764 within ontology 760. The restaurant reservation domain includes the actionable intent node “restaurant reservation,” property nodes “restaurant,” “date/time,” and “party size,” and sub-property nodes “cuisine,” “price range,” “phone number,” and “location.” Reminder domain 764 includes the actionable intent node “set reminder,” and property nodes “subject” and “date/time.” In some examples, ontology 760 is made up of many domains. Each domain shares one or more property nodes with one or more other domains. For example, the “date/time” property node is associated with many different domains (e.g., a scheduling domain, a travel reservation domain, a movie ticket domain, etc.), in addition to restaurant reservation domain 762 and reminder domain 764.

While FIG. 7C illustrates two example domains within ontology 760, other domains include, for example, “find a movie,” “initiate a phone call,” “find directions,” “schedule a meeting,” “send a message,” and “provide an answer to a question,” “read a list,” “providing navigation instructions,” “provide instructions for a task” and so on. A “send a message” domain is associated with a “send a message” actionable intent node, and further includes property nodes such as “recipient(s),” “message type,” and “message body.” The property node “recipient” is further defined, for example, by the sub-property nodes such as “recipient name” and “message address.”

In some examples, ontology 760 includes all the domains (and hence actionable intents) that the digital assistant is capable of understanding and acting upon. In some examples, ontology 760 is modified, such as by adding or removing entire domains or nodes, or by modifying relationships between the nodes within the ontology 760.

In some examples, nodes associated with multiple related actionable intents are clustered under a “super domain” in ontology 760. For example, a “travel” super-domain includes a cluster of property nodes and actionable intent nodes related to travel. The actionable intent nodes related to travel includes “airline reservation,” “hotel reservation,” “car rental,” “get directions,” “find points of interest,” and so on. The actionable intent nodes under the same super domain (e.g., the “travel” super domain) have many property nodes in common. For example, the actionable intent nodes for “airline reservation,” “hotel reservation,” “car rental,” “get directions,” and “find points of interest” share one or more of the property nodes “start location,” “destination,” “departure date/time,” “arrival date/time,” and “party size.”

In some examples, each node in ontology 760 is associated with a set of words and/or phrases that are relevant to the property or actionable intent represented by the node.

The respective set of words and/or phrases associated with each node are the so-called “vocabulary” associated with the node. The respective set of words and/or phrases associated with each node are stored in vocabulary index **744** in association with the property or actionable intent represented by the node. For example, returning to FIG. 7B, the vocabulary associated with the node for the property of “restaurant” includes words such as “food,” “drinks,” “cuisine,” “hungry,” “eat,” “pizza,” “fast food,” “meal,” and so on. For another example, the vocabulary associated with the node for the actionable intent of “initiate a phone call” includes words and phrases such as “call,” “phone,” “dial,” “ring,” “call this number,” “make a call to,” and so on. The vocabulary index **744** optionally includes words and phrases in different languages.

Natural language processing module **732** receives the candidate text representations (e.g., text string(s) or token sequence(s)) from STT processing module **730**, and for each candidate representation, determines what nodes are implicated by the words in the candidate text representation. In some examples, if a word or phrase in the candidate text representation is found to be associated with one or more nodes in ontology **760** (via vocabulary index **744**), the word or phrase “triggers” or “activates” those nodes. Based on the quantity and/or relative importance of the activated nodes, natural language processing module **732** selects one of the actionable intents as the task that the user intended the digital assistant to perform. In some examples, the domain that has the most “triggered” nodes is selected. In some examples, the domain having the highest confidence value (e.g., based on the relative importance of its various triggered nodes) is selected. In some examples, the domain is selected based on a combination of the number and the importance of the triggered nodes. In some examples, additional factors are considered in selecting the node as well, such as whether the digital assistant has previously correctly interpreted a similar request from a user.

User data **748** includes user-specific information, such as user-specific vocabulary, user preferences, user address, user’s default and secondary languages, user’s contact list, and other short-term or long-term information for each user. In some examples, natural language processing module **732** uses the user-specific information to supplement the information contained in the user input to further define the user intent. For example, for a user request “invite my friends to my birthday party,” natural language processing module **732** is able to access user data **748** to determine who the “friends” are and when and where the “birthday party” would be held, rather than requiring the user to provide such information explicitly in his/her request.

It should be recognized that in some examples, natural language processing module **732** is implemented using one or more machine learning mechanisms (e.g., neural networks). In particular, the one or more machine learning mechanisms are configured to receive a candidate text representation and contextual information associated with the candidate text representation. Based on the candidate text representation and the associated contextual information, the one or more machine learning mechanisms are configured to determine intent confidence scores over a set of candidate actionable intents. Natural language processing module **732** can select one or more candidate actionable intents from the set of candidate actionable intents based on the determined intent confidence scores. In some examples, an ontology (e.g., ontology **760**) is also used to select the one or more candidate actionable intents from the set of candidate actionable intents.

Other details of searching an ontology based on a token string are described in U.S. Utility application Ser. No. 12/341,743 for “Method and Apparatus for Searching Using An Active Ontology,” filed Dec. 22, 2008, the entire disclosure of which is incorporated herein by reference.

In some examples, once natural language processing module **732** identifies an actionable intent (or domain) based on the user request, natural language processing module **732** generates a structured query to represent the identified actionable intent. In some examples, the structured query includes parameters for one or more nodes within the domain for the actionable intent, and at least some of the parameters are populated with the specific information and requirements specified in the user request. For example, the user says “Make me a dinner reservation at a sushi place at 7.” In this case, natural language processing module **732** is able to correctly identify the actionable intent to be “restaurant reservation” based on the user input. According to the ontology, a structured query for a “restaurant reservation” domain includes parameters such as {Cuisine}, {Time}, {Date}, {Party Size}, and the like. In some examples, based on the speech input and the text derived from the speech input using STT processing module **730**, natural language processing module **732** generates a partial structured query for the restaurant reservation domain, where the partial structured query includes the parameters {Cuisine=“Sushi”} and {Time=“7 pm”}. However, in this example, the user’s utterance contains insufficient information to complete the structured query associated with the domain. Therefore, other necessary parameters such as {Party Size} and {Date} are not specified in the structured query based on the information currently available. In some examples, natural language processing module **732** populates some parameters of the structured query with received contextual information. For example, in some examples, if the user requested a sushi restaurant “near me,” natural language processing module **732** populates a {location} parameter in the structured query with GPS coordinates from the user device.

In some examples, natural language processing module **732** identifies multiple candidate actionable intents for each candidate text representation received from STT processing module **730**. Further, in some examples, a respective structured query (partial or complete) is generated for each identified candidate actionable intent. Natural language processing module **732** determines an intent confidence score for each candidate actionable intent and ranks the candidate actionable intents based on the intent confidence scores. In some examples, natural language processing module **732** passes the generated structured query (or queries), including any completed parameters, to task flow processing module **736** (“task flow processor”). In some examples, the structured query (or queries) for the m-best (e.g., m highest ranked) candidate actionable intents are provided to task flow processing module **736**, where m is a predetermined integer greater than zero. In some examples, the structured query (or queries) for the m-best candidate actionable intents are provided to task flow processing module **736** with the corresponding candidate text representation(s).

Other details of inferring a user intent based on multiple candidate actionable intents determined from multiple candidate text representations of a speech input are described in U.S. Utility application Ser. No. 14/298,725 for “System and Method for Inferring User Intent From Speech Inputs,” filed Jun. 6, 2014, the entire disclosure of which is incorporated herein by reference.

Task flow processing module **736** is configured to receive the structured query (or queries) from natural language

processing module 732, complete the structured query, if necessary, and perform the actions required to “complete” the user’s ultimate request. In some examples, the various procedures necessary to complete these tasks are provided in task flow models 754. In some examples, task flow models 754 include procedures for obtaining additional information from the user and task flows for performing actions associated with the actionable intent.

As described above, in order to complete a structured query, task flow processing module 736 needs to initiate additional dialogue with the user in order to obtain additional information, and/or disambiguate potentially ambiguous utterances. When such interactions are necessary, task flow processing module 736 invokes dialogue flow processing module 734 to engage in a dialogue with the user. In some examples, dialogue flow processing module 734 determines how (and/or when) to ask the user for the additional information and receives and processes the user responses. The questions are provided to and answers are received from the users through I/O processing module 728. In some examples, dialogue flow processing module 734 presents dialogue output to the user via audio and/or visual output, and receives input from the user via spoken or physical (e.g., clicking) responses. Continuing with the example above, when task flow processing module 736 invokes dialogue flow processing module 734 to determine the “party size” and “date” information for the structured query associated with the domain “restaurant reservation,” dialogue flow processing module 734 generates questions such as “For how many people?” and “On which day?” to pass to the user. Once answers are received from the user, dialogue flow processing module 734 then populates the structured query with the missing information, or pass the information to task flow processing module 736 to complete the missing information from the structured query.

Once task flow processing module 736 has completed the structured query for an actionable intent, task flow processing module 736 proceeds to perform the ultimate task associated with the actionable intent. Accordingly, task flow processing module 736 executes the steps and instructions in the task flow model according to the specific parameters contained in the structured query. For example, the task flow model for the actionable intent of “restaurant reservation” includes steps and instructions for contacting a restaurant and actually requesting a reservation for a particular party size at a particular time. For example, using a structured query such as: {restaurant reservation, restaurant=ABC Café, date=3/12/2012, time=7 pm, party size=5}, task flow processing module 736 performs the steps of: (1) logging onto a server of the ABC Café or a restaurant reservation system such as OPENTABLE®, (2) entering the date, time, and party size information in a form on the website, (3) submitting the form, and (4) making a calendar entry for the reservation in the user’s calendar.

In some examples, task flow processing module 736 employs the assistance of service processing module 738 (“service processing module”) to complete a task requested in the user input or to provide an informational answer requested in the user input. For example, service processing module 738 acts on behalf of task flow processing module 736 to make a phone call, set a calendar entry, invoke a map search, invoke or interact with other user applications installed on the user device, and invoke or interact with third-party services (e.g., a restaurant reservation portal, a social networking website, a banking portal, etc.). In some examples, the protocols and application programming interfaces (API) required by each service are specified by a

respective service model among service models 756. Service processing module 738 accesses the appropriate service model for a service and generates requests for the service in accordance with the protocols and APIs required by the service according to the service model.

For example, if a restaurant has enabled an online reservation service, the restaurant submits a service model specifying the necessary parameters for making a reservation and the APIs for communicating the values of the necessary parameter to the online reservation service. When requested by task flow processing module 736, service processing module 738 establishes a network connection with the online reservation service using the web address stored in the service model, and sends the necessary parameters of the reservation (e.g., time, date, party size) to the online reservation interface in a format according to the API of the online reservation service.

In some examples, natural language processing module 732, dialogue flow processing module 734, and task flow processing module 736 are used collectively and iteratively to infer and define the user’s intent, obtain information to further clarify and refine the user intent, and finally generate a response (i.e., an output to the user, or the completion of a task) to fulfill the user’s intent. The generated response is a dialogue response to the speech input that at least partially fulfills the user’s intent. Further, in some examples, the generated response is output as a speech output. In these examples, the generated response is sent to speech synthesis processing module 740 (e.g., speech synthesizer) where it can be processed to synthesize the dialogue response in speech form. In yet other examples, the generated response is data content relevant to satisfying a user request in the speech input.

In examples where task flow processing module 736 receives multiple structured queries from natural language processing module 732, task flow processing module 736 initially processes the first structured query of the received structured queries to attempt to complete the first structured query and/or execute one or more tasks or actions represented by the first structured query. In some examples, the first structured query corresponds to the highest ranked actionable intent. In other examples, the first structured query is selected from the received structured queries based on a combination of the corresponding speech recognition confidence scores and the corresponding intent confidence scores. In some examples, if task flow processing module 736 encounters an error during processing of the first structured query (e.g., due to an inability to determine a necessary parameter), the task flow processing module 736 can proceed to select and process a second structured query of the received structured queries that corresponds to a lower ranked actionable intent. The second structured query is selected, for example, based on the speech recognition confidence score of the corresponding candidate text representation, the intent confidence score of the corresponding candidate actionable intent, a missing necessary parameter in the first structured query, or any combination thereof.

Speech synthesis processing module 740 is configured to synthesize speech outputs for presentation to the user. Speech synthesis processing module 740 synthesizes speech outputs based on text provided by the digital assistant. For example, the generated dialogue response is in the form of a text string. Speech synthesis processing module 740 converts the text string to an audible speech output. Speech synthesis processing module 740 uses any appropriate speech synthesis technique in order to generate speech outputs from text, including, but not limited to, concatenation

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tive synthesis, unit selection synthesis, diphone synthesis, domain-specific synthesis, formant synthesis, articulatory synthesis, hidden Markov model (HMM) based synthesis, and sinewave synthesis. In some examples, speech synthesis processing module 740 is configured to synthesize individual words based on phonemic strings corresponding to the words. For example, a phonemic string is associated with a word in the generated dialogue response. The phonemic string is stored in metadata associated with the word. Speech synthesis processing module 740 is configured to directly process the phonemic string in the metadata to synthesize the word in speech form.

In some examples, instead of (or in addition to) using speech synthesis processing module 740, speech synthesis is performed on a remote device (e.g., the server system 108), and the synthesized speech is sent to the user device for output to the user. For example, this can occur in some implementations where outputs for a digital assistant are generated at a server system. And because server systems generally have more processing power or resources than a user device, it is possible to obtain higher quality speech outputs than would be practical with client-side synthesis.

Additional details on digital assistants can be found in the U.S. Utility application Ser. No. 12/987,982, entitled "Intelligent Automated Assistant," filed Jan. 10, 2011, and U.S. Utility application Ser. No. 13/251,088, entitled "Generating and Processing Task Items That Represent Tasks to Perform," filed Sep. 30, 2011, the entire disclosures of which are incorporated herein by reference.

4. Process for Continuous Dialog with a Digital Assistant

FIGS. 8-12 illustrate systems and processes for continuous dialog with a digital assistant. For example, as shown in FIGS. 10A-10B, electronic device 1002 may include any device described herein, including but not limited to devices 104, 200, 400, and 600 (FIGS. 1, 2A, 4, and 6A-6B). Thus, it will be appreciated that the electronic devices depicted in FIGS. 8-12 may correspond to a user device of any type, such as a phone, laptop computer, desktop computer, tablet, wearable device (e.g., smart watch), home speaker, and the like. Furthermore, the processes described herein may be performed by a server with information delivered to and from the device, performed on the device, or a combination thereof.

In general, process 800 depicts various states representing interaction stages between a user of an electronic device and a digital assistant. Specifically, process 800 may begin in an available state 802, which represents a state in which the user has not yet invoked the digital assistant. The user may then invoke the digital assistant in various ways. For example, the user may activate one or more buttons and/or affordances associated with the electronic device (e.g., a physical/mechanical control such as a knob or a button, an object displayed on a touch-sensitive surface, etc.), utter one or more words or phrases (e.g., "Hey Siri," "Siri," "Hi Assistant," etc.), and the like. Once invoked, the digital assistant may enter a listening state 804. In some examples, the user may dismiss the digital assistant while in the listening state, such that the digital assistant returns to the available state 802. While in the listening state 804, the digital assistant may sample audio including speech input from the user. The speech input may include various commands, questions, or other input directed to the digital assistant. Once the user finishes providing a speech input, the digital assistant may begin processing the speech input

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in processing state 806. During processing state 806, the user may interrupt the processing of the speech input by way of a subsequent speech input, activation of one or more buttons/affordances, and the like. If the user interrupts processing state 806, the digital assistant may return to listening state 804 and sample audio including additional speech input from the user.

If the user does not interrupt processing state 806 and digital assistant completes processing of the speech input to obtain one or more results, the digital assistant may enter a response state 808. During response state 808, the digital assistant may provide the one or more results in various ways, such as an audible output, displayed output, a combination of audible and displayed outputs, and the like. During response state 808, the user may interrupt the processing of the speech input by way of a subsequent speech input, activation of one or more buttons/affordances, and the like. If the user interrupts response state 808, the digital assistant may return to listening state 804. If the user does not interrupt response state 808, the digital assistant may enter an attending state 810 once the digital assistant is finished providing the output during response state 808. In particular, during attending state 808, the digital assistant may await one or more forms of re-engagement from the user, as described herein. For example, the user may re-engage the digital assistant by providing additional speech input, looking in a direction of a display of the electronic device and/or a direction of a displayed digital assistant object, providing one or more physical inputs (e.g., via a button and/or affordance), and the like. If a user re-engagement is detected, the digital assistant returns to listening state 804 to sample audio including speech input from the user. While in any of listening state 804, processing state 806, response state 808, or attending state 810, the user may dismiss the digital assistant (e.g., using a speech input, a touch input, and the like). In response to a dismissal, digital assistant may return to available state 802.

With reference to FIG. 9, a process 900 for interacting with a digital assistant is depicted. In general, FIG. 9 illustrates process 900 by which a user may interact with a digital assistant, including various follow-up interactions. Process 900 may be implemented on an electronic device having a display in some examples, including a device such as a smartphone, smart watch, tablet, personal computer, headset, and the like. Initially, the user may invoke the digital assistant at step 902, such that the digital assistant begins listening for audio including user speech input. The user may then provide a first speech input 904 directed to the digital assistant. For example, the user may utter the phrase "Hey Siri," in order to invoke the digital assistant at step 902, and further utter the speech "How old is Barack Obama?" The digital assistant may then process first speech input 904 in order to obtain one or more results responsive to the input. Once the digital assistant has finished processing first speech input 904, the digital assistant may provide a first response 906 (e.g., an audible output and/or displayed output), including the result "Barack Obama is 59 years old." After the digital assistant finishes providing first response 906, a session window 908 may be initiated.

In general, the session window may correspond to a duration of time by which the digital assistant awaits one or more forms of re-engagement from the user. The duration of time may be based, at least in part, upon a variable speech threshold 910. In particular, the duration of time may correspond to the duration of time by which the variable speech threshold 910 exceeds a predetermined speech threshold. The variable speech threshold may also be

adjusted based on a variety of factors, and may generally correspond to a confidence level indicating whether the user still wishes to be engaged with the digital assistant. For instance, the variable speech threshold may decrease over time (i.e., decay) absent factors causing a specific increase or decrease. In some examples, the decay may be linear, exponential, based on a specific function, and the like. For example, the threshold may include **100** possible points, and may decay **20** points per second. The decay may also be based on conversational context, such as the content of user speech input, digital assistant responses, displayed information, and the like.

In some examples, the variable speech threshold may be increased or decreased by a specific amount based on detection of a user gaze. Similarly, absent any such user gaze, the variable speech threshold may not be further adjusted beyond the decay over time. With reference back to FIG. 9, for example, the user may perform action **912**. In particular, after first response **906** has been provided and during session window **908**, the user may perform an action **912** such as looking around an environment of the electronic device. For example, the user may not be looking or gazing at the display of the electronic device when performing action **912**. Accordingly, variable speech threshold **910** is not adjusted in response to action **912**. As a result, variable speech threshold **910** continues to decay at the respective rate.

In some examples, the user may further perform action **914**, which may include the user gazing generally at one or more objects displayed on a display of the electronic device. In response to user action **914**, an increase **916** in the variable speech threshold may occur. For example, the user gaze may be directed at one or more applications displayed on a home screen. In accordance with a determination that the user gaze is directed at the display, the variable speech threshold may be increased by a first amount (e.g., increased by 10 points). The electronic device may alternatively detect a user gaze directed at a displayed digital assistant object (e.g., an “orb” or other object representing that the digital assistant awaiting speech input). In accordance with a determination that the user gaze is directed at the digital assistant object, the variable speech threshold may be increased by a second amount greater than the first amount (e.g., increased by 20 points). The increase **916** in variable speech threshold **910** thus increases the duration of session window **908**.

During session window **908**, one or more additional speech inputs may be received from the user. In response to such speech input, determination is made whether the speech input includes speech directed to the digital assistant. In general, user gaze may be used as context in order to determine whether the speech input includes speech directed to the digital assistant. For instance, at a time prior to or while receiving the additional speech input, the direction of the user gaze (e.g., gaze generally directed at the display, gaze directed at the digital assistant object, etc.) may be used as a factor in determining whether the user is speaking to the digital assistant. As an example, the user may look away from the electronic device **912** in order to speak to another user or entity (e.g., another user within the device environment, another device in the environment, another user through a telephonic device, etc.). While looking away from the electronic device, the user may utter speech **918**, such as “See, I told you Barack Obama was under 60.” After detecting speech **918**, determination is made that speech **918** does not include speech directed to the digital assistant based at least in part on the determination that the user is not looking at the electronic device. The determination may also

be based on utilization of one or more different input models in combination with the content of the user speech, as discussed further herein. In some examples, the digital assistant object may change in appearance when speech not directed to the digital assistant object is detected (e.g., the digital assistant object momentarily shrinking in size, enlarging in size, changing color/animations, etc.).

In some examples, determination is made that the speech input includes speech directed to the digital assistant. For example, the user may utter speech **920** including “How old is Michelle?” Furthermore, user action **914** may precede speech **920**, and may correspond to the user looking at a displayed digital assistant object. Based at least in part of the detection of user action **914**, the subsequent speech **920** may be deemed as directed to the digital assistant. Similarly, as discussed further herein, a relatively high variable speech threshold **910** (at the time of receiving speech input **920**) may also weigh in favor of determining that speech **920** is directed to the digital assistant.

In some examples, various input models are used to determine whether the speech is directed to the digital assistant. In general, the inputs recognized with respect to each model may vary depending on the selectiveness of the model. For example, a permissive model may deem a relatively large number of inputs as potentially indicative that speech is directed to the digital assistant, such as any commands or interrogatories that the digital assistant is capable of processing and responding to (e.g., the phrase “How old is Barack Obama?”). A balanced model may deem a relatively moderate number of inputs as potentially indicative that speech is directed to the digital assistant, such as “global” commands (e.g., “no,” “I said,” “I meant,” “stop,” “wait,” “undo,” “help,” etc.), “contextual” commands (e.g., “next song,” “go back,” etc.), and/or predefined assistant names (e.g., “Siri,” “Assistant,” etc.). A restrictive model may deem a relatively small number of inputs as potentially indicative that speech is directed to the digital assistant (e.g., only predefined assistant names).

Each of the input models may optionally leverage additional context in the determination, such as user gaze, device position/orientation, conversational context, prior word usage, and the like. For example, the user may routinely begin digital assistant commands with the phrase “Can you,” such as “Can you tell me the score of the Packers game,” “Can you read the latest headlines,” or “Can you show the weather for tomorrow.” Usage history corresponding to frequently used words or phrases in prior digital assistant requests may thus be indicative that the same words or phrases are highly likely to be directed to the digital assistant. Thus, in this example, to the extent the user utters “Can you tell me how old Barack Obama is” during a respective session window, determination may be made that the speech input is directed to the digital assistant.

In general, the amount of the increase or decrease in the variable speech threshold may be based on context information associated with the electronic device. For example, the user may be playing a media file such as a song. Based on the currently played media, contextual commands associated with media playback may be indicative that the user’s speech input is directed to the digital assistant and specifically directed to adjusting the playback of the media using the digital assistant. For example, commands such as “turn up the volume,” “next song,” “play that again,” “repeat song,” and the like, may be deemed as directed to the digital assistant to the extent media is currently playing at the electronic device. As another example, the user may be reading a long article or other document on a display of the

device. Here, commands such as “scroll down,” “read out this article,” “next page,” and the like, may be deemed as directed to the digital assistant to the extent these commands are operable on the currently viewed document.

The variable speech threshold may further influence the determination as to whether speech is directed to the digital assistant. Generally, a higher variable speech threshold will weigh in favor of determining that speech is directed to the digital assistant, whereas a lower variable speech threshold will weigh against determining that speech is directed to the digital assistant. The level of variable speech threshold, used as bases for determining whether speech is directed to the digital assistant, may correspond to the level of the threshold at a time the speech input is first detected, at a time during the speech input (e.g., a “middle” portion of the speech input), or at a time corresponding to the end of the speech input. In some examples, an average of the levels of the variable speech threshold across the duration of the speech input may be used.

Various additional factors may influence the amount of increase or decrease in the variable speech threshold. In general, an amount of time between a current speech input (e.g., speech input 918) and a previous interaction, such as user speech (e.g., speech input 904) or a digital assistant response (e.g., first response 906) may influence the adjustment of the variable speech threshold. As an example, speech received immediately after a digital assistant response (e.g., first response 906) may be associated with a high variable speech threshold level 910. Accordingly, such speech may be generally determined as directed to the digital assistant, absent other factors weighing against such a determination (e.g., user looking away from display, irrelevant speech content, etc.). As another example, speech input 918 may be associated with variable speech threshold 910 having a moderate level. Here, the moderate variable speech threshold level may not greatly influence the determination. Thus, combined with the detection of the user looking away from the display via action 912, determination is made that speech 918 is not directed to the digital assistant.

Analysis of the user speech may also influence the adjustment of the variable speech threshold. In particular, prosodic features of user speech may be identified and used as basis for adjusting the variable speech threshold. Prosodic features include attributes such as stress, intonation, rhythm, tempo, word grouping, and the like. As an example, following a digital assistant response including playback of media, the user may utter “Great! Remember this song for later!” Prosodic features of the utterance may indicate an emotional state of the user, in particular, a state of excitement or general satisfaction. Accordingly, the variable speech threshold may be increased given general positive reception of the digital assistant response based on both analysis of the speech content and prosody. In some examples, voice identification using speech recognition may assist with influencing the variable speech threshold adjustment. Here, a speaker identity is determined based on speech recognition. In particular, a first speaker profile and a second speaker profile are obtained, associated with first speech input 904 and subsequent speech input 918, respectfully. The speaker profiles may generally identify various characteristics of the speech, such as pitch, tone, fry, rhythm, tempo, etc. Based on a comparison of the speaker profiles (e.g., comparing the one or more characteristics or a combination of the characteristics), determination is made whether the speaker of each input is the same. To the extent the speaker of each input is the same, the variable speech threshold may be increased.

Application and other digital assistant context may be utilized to influence the variable speech threshold. In general, the user may be interacting with an application on the electronic device, such as a cooking application including various affordances for navigating recipe lists, ingredients, food articles, and the like. The presence of a respective application may be determined, for example, and used as context when interpreting user speech. For example, the user may utter “How much garlic do I need?” The cooking application may be open and may further be displaying a recipe including an ingredient list having “three cloves of garlic.” Accordingly, given the presence of the cooking application and information associated with the cooking application such as the ingredient list, the variable speech threshold is increased based on a determination that the input “How much garlic do I need?” is intended for the digital assistant in the context of using the cooking application. Similarly, additional digital assistants may be utilized, such as a digital assistant associated with a device application. For example, the user may be viewing a personal banking application. The personal banking application may further include a digital assistant representation capable of responding to user requests within the context of the personal banking application. Here, the user may utter “what is my current balance?” Given the context of the personal banking application and content of the user’s speech (e.g., the term “balance” is related to banking), the variable speech threshold is adjusted based on the determination that the user speech is directed to a digital assistant.

In some examples, additional aspects of user gaze are used as basis for adjusting the variable speech threshold. In general, the electronic device may correspond to a headset capable of tracking user gaze within the context of an environment (virtual reality, augmented reality, etc.). A digital assistant object may be represented within the environment, such as a circular object or an “orb” within the user’s perspective. In response to a user gaze towards the digital assistant object, the variable speech threshold may be increased. Similarly, the digital assistant may have recently provided a response to the user. The response may be in the form of a displayed response object within the environment, such as textual representation of the response or similar object. In response to a user gaze towards the response object, the variable speech threshold may be increased. In some examples, additional users may be associated with or otherwise depicted in the environment. In response to a user gaze towards a specific user (or a representation of the user within the environment), between specific users (or between representations of users within the environment), the variable speech threshold may be increased.

In some examples, the user may interrupt a response being provided by the digital assistant. For example, while the first response 906 is being provided, the user may provide an additional speech input (not depicted) such as “Never mind. How old is Michelle Obama?” In response to a determination that the additional speech input is directed to the digital assistant, the providing of first response 906 is ceased. The identification of whether the additional speech input, provided while a digital assistant response is being provided, is directed to the digital assistant may be based on an identification of at least one predefined word associated with a digital assistant response. In particular, words associated with interruptions (e.g., “stop,” “cancel that,” “hold on”), words associated with corrections (e.g., “not that one,” “I said . . .”), and/or user frustration (e.g., “you suck!”, “forget it”). User frustration may further be determined based on prosody information contained within the user speech (e.g.,

volume, stress, intonation, etc.). Follow-up responses may be provided by the digital assistant in response to receiving additional speech inputs during an initial digital assistant response. For example, in response to receiving a speech input including “stop!”, the digital assistant may offer a follow-up response including “Okay, let me know if you have another request.” A session window associated with a variable speech threshold may then be initiated once the follow-up response is provided.

In general, while variable speech threshold **910** exceeds a predetermined speech threshold, the session window is maintained in an active state. While session window **908** is maintained in an active state, audio input is detected in first state. During the first state, speech input is analyzed and determinations are made whether the speech input is directed to the digital assistant as described with respect to FIG. **9**. In some examples, a determination is made that the variable speech threshold does not exceed the predetermined speech threshold. For example, variable speech threshold **910** may be deemed as not exceeding a predetermined minimum threshold (e.g., zero out of 100, five out of 100, etc.) at point **922**. In accordance with a determination that the variable speech threshold **910** does not exceed a predetermined speech threshold, session window **908** is ended. After the session window ends, audio input may be detected in a second state. For example, the second state may be a lower power state than the first state. In the second state, the electronic device may detect audio input and determine whether audio input includes speech directed to invoking the digital assistant. For example, speech including a predefined word or phrase, such as “Hey Siri,” may be utilized in order to invoke the digital assistant.

Referring now to FIG. **10A**, an example interaction between a first user, a digital assistant, and a second user is depicted. In general, first user **1004** may be associated with electronic device **1002**, and may be in proximity to a second user **1008**. Electronic device **1002** may include a digital assistant represented by a digital assistant object **1006**. In some examples, first user **1004** may activate the digital assistant (e.g., via spoken trigger, button press, etc.) such that the display of digital assistant object **1006** indicates that the digital assistant is waiting and capable of responding to spoken question, command, and the like. Once the digital assistant is invoked, first user **1004** may utter first speech input **1010**, including speech “How old is Barack Obama?”. Before and/or while providing first speech input **1010**, first user **1004** may also gaze at digital assistant object **1006**. Determination may be made that first speech input **1010** is directed to the digital assistant, as discussed with respect to FIG. **9**. In response to first speech input **1010**, the digital assistant may provide response **1012**, including the result “Barack Obama is 59 years old.” As discussed with respect to FIG. **9**, a session window may be initiated once response **1012** is provided. Next, the user may utter second speech input **1014**, including “See, I told you he was under 60.” Second speech input **1014** may be provided while the user is not gazing or otherwise looking at the display associated with electronic device **1002**. Specifically, user **1004** may be speaking to user **1008** when uttering speech input **1014**. Here, determination may be made that second speech input **1014** is not directed to the digital assistant, as discussed with respect to FIG. **9**.

Referring now to FIG. **10B**, user **1004** may provide a third speech input **1016** during the session window, including speech “How old is Michelle?”. Determination may be made that third speech input **1016** is directed to the digital assistant, as discussed with respect to FIG. **9**. For example, the

user may no longer be speaking to second user **1008** and also may be gazing at digital assistant object **1006** before and/or while uttering speech input **1016**. In this example, the content of first speech input **1010** and/or response **1012** may also be used as context in order to disambiguate speech input **1016**. In particular, the word “Michelle” may be disambiguated based on the context information “Barack Obama.” In other words, a last name parameter associated with “Michelle” is determined to be the same as the last name parameter identified from first speech input **1010** and/or response **1012**, specifically “Obama.” In response to third speech input **1016**, the digital assistant may provide response **1018**, including the result “Michelle Obama is 56 years old.”

With reference to FIG. **11**, a process **1100** for interacting with a digital assistant is depicted. In general, FIG. **11** illustrates process **1100** by which a user may interact with a digital assistant, including various follow-up interactions. Process **1100** may be implemented on an electronic device that, optionally, does not have a display. For example, process **1100** may be implemented on a home speaker. In some examples, process **1100** is implemented on a first device having a display, such that the first device is further coupled to a second device not having a display, for example, a smartphone, tablet, or personal computer communicatively coupled to headphones (e.g., via wireless or wired connection). In this example, the user may interact with the digital assistant on the first device using the second device, for example, by providing speech input to a smartphone via wireless headphones. Initially, the user may invoke the digital assistant at step **1102**, such that the digital assistant begins listening for audio including user speech input. The user may then provide a first speech input **1104** directed to the digital assistant. For example, the user may utter “Hey Siri,” in order to invoke the digital assistant at **1102**, and followed by the speech “What’s the score of the Warriors Game?” The user may also invoke the digital assistant by tapping on the electronic device (e.g., tapping on a touch sensitive surface of a home speaker), or tapping on a second device communicatively coupled to the electronic device (e.g., tapping on headphones communicatively coupled to a smartphone).

Once the digital assistant has finished processing speech input **1104**, the digital assistant may provide a first response **1106**, such as an audible output with the result “The Warriors are beating the Clippers with a score of 75 to 69.” In an example where a user provides speech input **1104** using a secondary electronic device, response **1106** may also be provided through the secondary electronic device. After the digital assistant finishes providing first response **1106**, a session window **1108** may be initiated. In general, the session window may correspond to a duration of time by which the digital assistant may await one or more forms of re-engagement from the user. The duration of time may be based, at least in part, upon a variable duration. The variable duration of the session window may be based on a variable speech threshold, as discussed with respect to FIG. **9**. For example, in accordance with a determination that a variable speech threshold exceeds a predetermined speech threshold, the session window continues to remain active and the variable duration of the session window is extended. Alternatively, in accordance with a determination that a variable speech threshold does not exceed a predetermined speech threshold, the session window ends.

During session window **1108**, one or more additional speech inputs may be received from the user. In response to such speech input, determination is made whether the

speech input includes speech directed to the digital assistant. For example, the user may provide speech input **1110**, including speech “Wow, looks like a close one.” Upon receiving speech input **1110**, determination is made whether speech input **1110** includes speech directed to the digital assistant. Here, a communicative coupling between a secondary device and the electronic device may be detected and used at least in part as basis for the determination whether speech input **1110** includes speech directed to the digital assistant. For example, the user may provide speech input to the smartphone through the wireless headphones, such that the wireless connection may increase a confidence that the speech input is directed to the digital assistant. In some examples, the content of speech input **1110** is utilized in order to determine whether the speech input includes speech directed to the digital assistant. The content used for the determination may also be based on the type of input model used (permissive, balanced, or restrictive), as discussed with respect to FIG. 9. For example, in response to identifying an interrogatory phrase in the speech input (e.g., “What’s the weather in Cupertino?”), a command in the speech input (e.g., “Play the Lakers game on the Living Room TV”), or a predefined word(s) in the speech input (e.g., “Cancel that”) a determination is made that the second speech input includes speech directed to the digital assistant. Here, upon receiving speech input **1110** including “Wow, looks like a close one,” determination is made that speech input **1110** is not directed to the digital assistant based at least on the content of the speech input. In particular, the speech “Wow, looks like a close one” may be determined to not include an interrogatory phrase, a command, and/or any predefined words or phrases.

In some examples, the user may interrupt a response being provided by the digital assistant. For example, while the first response **1106** is being provided, the user may provide an additional speech input (not depicted) such as “Sorry Siri, I meant the Bucks game.” In response to a determination that the additional speech input is directed to the digital assistant, the providing of first response **1106** is ceased. The identification of whether the additional speech input, provided while a digital assistant response is being provided, is directed to the digital assistant may be based on an identification of at least one predefined word associated with a digital assistant response. In particular, words associated with interruptions, words associated with corrections, and/or user frustration may be utilized in order to determine that the speech is directed to the digital assistant, as discussed with respect to FIG. 9. Additional words may also be utilized, such as names associated with the digital assistant (e.g., “Siri”). A follow-up response may then be provided (not depicted), such as “The Bucks are leading the Bulls by a score of 50-35.” A session window associated with a variable duration may then be initiated once the follow-up response is provided.

In some examples, an additional speech input **1112** is received during session window **1108**, including speech “How much time is left in the game?”. In this example, determination is made that the speech input includes an interrogatory phrase, and thus, that speech input **1112** includes speech directed to the digital assistant. In accordance with a determination that speech input **1112** includes speech directed to the digital assistant, a second session window **1114** associated with a second variable duration is initiated. Generally, an additional session window may begin immediately upon the determination that the respective speech input includes speech directed to the digital assistant. In some examples, the subsequent session window,

such as session window **1114**, may begin once an initial session window, such as session window **1108**, ends. In some examples, instead of initiating a new session window **1114**, the variable duration of session window **1108** is extended based on an increase in the variable speech threshold (e.g., variable speech threshold **1122**) given the determination that speech is directed to the digital assistant. In response to speech input **1112**, a response **1116** corresponding to the input is provided from the digital assistant, such as “There are twelve minutes and twenty seconds left in the game.”

An audible output may be utilized in order to signal an end to a session window, and thus signal to a user that the digital assistant is no longer actively listening for follow-up speech. For instance, in accordance with a determination that speech input **1110** does not include speech directed to the digital assistant (and in the case, not depicted, where additional speech input **1112** is not received), an audible output is provided at a time corresponding to an end of the first variable duration associated with session window **1108**, and specifically at time **1118**. For example, a “chime” or other noise of relatively short duration may be output to the user to signal an end to a respective session window.

In some examples, a determination is made whether audible output criteria is satisfied. In general, the audible output criteria may in part determine whether to output an audible signal indicating the end of a session window. Various scenarios may influence whether such audible output is provided. In particular, the user may be playing media at the electronic device during an interaction with a digital assistant, such as the interaction depicted in process **1100**. In this example, a volume level associated with the playing media may be reduced during times when a session window is active, such as session windows **1108** and **1114**. Specifically, a volume level associated with the playing media may be reduced at a time corresponding to the beginning of the session window and may be raised at a time corresponding to the end of the session window. Here, when media is playing at the electronic device, determination is made that audible output criteria is not satisfied. Other factors may influence whether audible output criteria is satisfied, such as whether a secondary electronic device is being used. For example, audible output criteria may be determined to be satisfied if the user is providing speech to the electronic device through wireless headphones. User preferences, device location, device position and/or orientation, device type, and the like, may also be used to determine whether the audible output criteria is satisfied.

In accordance with a determination that audible output criteria is satisfied, an audible output is provided at a time corresponding to an end of the duration of a respective session window, such as time **1120** associated with session window **1114**. In general, while a session window (such as session windows **1108** and **1114**) are maintained in an active state, audio input is detected in first state. During the first state, speech input is analyzed and determinations made whether the speech input is directed to the digital assistant as described with respect to FIGS. 9 and 11. After the respective session window ends, audio input may be detected in a second state. For example, the second state may be a lower power state than the first state. In the second state, the electronic device may detect audio input and determine whether audio input includes speech directed to invoking the digital assistant. For example, speech including a predefined

word or phrase, such as “Hey Siri,” may be utilized in order to invoke the digital assistant.

5. Digital Assistant Interaction on Lock Screen

With reference to FIG. 12, a process 1200 for interacting with a digital assistant from a device lock screen is depicted. In general, a user may interact with a digital assistant while the respective electronic device, such as device 1202, is locked or unlocked. While the electronic device is locked, a reduced or otherwise limited set of functions may be available relative to when the device is unlocked, such as only providing access to publicly available, non-private information (e.g., weather, sports, etc.). While electronic device 1202 is locked, a status indicator 1204 may be displayed including a lock icon in a “locked” state. A digital assistant object 1206a may also be displayed. The digital assistant object 1206a may be initially displayed with a relatively small size, which may be similar, or the same, to the size of other objects displayed on the device lock screen (e.g., a flashlight affordance and/or a camera affordance). The user may additionally invoke the digital assistant object by touching digital assistant object 1206a through the touch-screen display of electronic device 1202. The touch input to invoke the digital assistant object may be a long press, short press, or force press (e.g., based on one or more threshold touch intensities being exceeded).

A user gaze directed at the electronic device may also be detected in some examples. In particular, one or more cameras and/or sensors on the electronic device may obtain biometric information corresponding to facial features. The biometric information may be compared to reference biometric information in order to authenticate the user as an authorized user of the device. Once the user is authenticated as an authorized user of the device, the device is transitioned from the locked state to an unlocked state as indicated by status indicator 1208. In particular, status indicator 1208 may be displayed including a lock icon in a “unlocked” state.

In some examples, a user gaze is detected as directed at digital assistant object 1206a. One or more outputs may additionally be provided in response to detecting a user gaze at digital assistant object 1206a, such as a haptic output (e.g., a device vibration). One or more animations of digital assistant object 1206a may also change in response to detecting a user gaze directed at digital assistant object. For example, in response to detecting the user gaze, an enlargement of digital assistant object 1206a may be displayed, in particular, such that digital assistant object is displayed as digital assistant object 1206b. The enlarged size of digital assistant object 1206b may be maintained so long as the user gaze remains on digital assistant object 1206b. Accordingly, in response to detecting the user gaze directed away from the digital assistant object, a size reduction in the digital assistant object is displayed by displaying the digital assistant object as having the initial size (e.g., corresponding to digital assistant object 1206a). For example, digital assistant object 1206b may be displayed as shrinking once the user looks away from digital assistant object 1206b. The shrinking of the digital assistant object may also be displayed for a longer period of time than the enlargement of the digital assistant object. In some examples, the digital assistant object size remains the same regardless of the user gaze direction. For example, the digital assistant object size may not change based on user gaze, but may change based on the amplitude of a user’s speech input.

Various animations of the digital assistant object may also be displayed with respect to the user gaze. For example, prior to detecting a user gaze at digital assistant object 1206a, digital assistant object 1206a may be displayed with one or more first colors and/or one or more objects moving at a first speed. In response to detecting the user gaze at digital assistant object 1206a, digital assistant object may be displayed as digital assistant object 1206b. In some examples, digital assistant object 1206b may be displayed with one or more second colors different than the first colors and/or one or more objects moving at a second speed faster than the first speed. Upon detecting the user gaze away from digital assistant object 1206b, digital assistant object may be displayed with the one or more first colors and/or one or more objects moving at the first speed.

In some examples, in response to detecting the gaze, audio input is detected. For example, the user may utter various questions, commands, and other queries once the user gaze at digital assistant object 1206a is detected. In some examples, once a gaze directed away from the digital assistant object is detected, detection of audio input is ceased at a predetermined time after the detected gaze away. For instance, the electronic device may continue to sample audio input to detect user queries after the user gazes away (e.g., for five seconds) from digital assistant object 1206b. If the user resumes a gaze directed at the digital assistant object during the five second duration, the digital assistant may continue to detect audio input until the user gazes away from the digital assistant object for longer than the predetermined duration.

In some examples, in accordance with a determination that audio input includes speech input associated with a task, a corresponding task is performed based on the speech input. In some examples, if the user is not authenticated based on user gaze, the electronic device may be maintained in the locked state while performing the task, based at least in part on a determination that the task requires only non-private information. In some examples, performing the task may require access to personal information. In this regard, the authentication of the user may occur prior to, simultaneously, or after the device detects the user gaze directed at the digital assistant. For example, the user may utter “call Mom”, such that the task requires accessing the user’s contact list to initiate a telephone call to a specific contact titled “Mom.” In such cases, performing the task may include obtaining biometric information corresponding to the user and authenticating the user based on the biometric information, to the extent the user has not already been authenticated. As another example, the user gaze at the digital assistant may also be utilized in order to obtain a biometric signature that is further compared to a reference biometric signature.

A prompt to the user may also be presented requesting the biometric information in some examples. For example, in the scenario where the user is not authenticated at electronic device 1202 prior to gazing at the digital assistant object, biometric information (e.g., facial recognition information, fingerprint information, voiceprint information, etc.) may be used to authenticate the user prior to performing a task that requires personal information.

In some examples, in accordance with a determination that the audio input does not include a speech input, the display size of digital assistant object 1206b is maintained so long as the user gaze remains directed at digital assistant object 1206b. In some examples, the digital assistant object 1206b is displayed as momentarily enlarging (e.g., for one to two seconds), and returning to the size displayed before the

momentary enlargement. The momentary enlargement of digital assistant object **1206** may also be displayed after a predetermined duration associated with audio input not including speech input (e.g., three seconds of silence or non-speech noise). In some examples, alternative animations of digital assistant object **1206b** may be displayed in accordance with a determination that the audio input does not include a speech input. For example, prior to detecting absence of speech input, digital assistant object **1206b** may be displayed with one or more first colors and/or one or more objects moving at a first speed. In accordance with a determination that the audio input does not include a speech input, digital assistant object **1206b** may be displayed with one or more second colors different than the first colors and/or one or more objects moving at a second speed faster than the first speed.

In general, the states of the digital assistant may vary based on displayed size, animation profile, color scheme, and the like. With reference to FIG. **13**, the first digital assistant state may correspond to a digital assistant object having a first size, such as digital assistant object **1302**. For example, once the digital assistant enters the first state and begins listening for user speech directed to the digital assistant (e.g., follow-up speech related to an initial interaction), digital assistant object **1302** is displayed. Digital assistant object **1302** may include various objects moving at a first speed. Once the digital assistant transitions to the second state, the digital assistant object may be displayed as shown via digital assistant object **1304**. In particular, digital assistant object **1304** may be larger than digital assistant object **1302**, the size of digital assistant object **1304** may fluctuate based on received speech amplitude, and the various objects within digital assistant object **1304** may move at a faster speed relative to the objects within digital assistant object **1302**. The animations or color scheme of digital assistant object **1304** may also be dynamically modified based on received speech amplitude.

Once the digital assistant determines that a full user utterance has been detected (e.g., based on endpoint detection), the digital assistant may transition to a third state, such as the state represented by digital assistant object **1306**. The third state may correspond to a processing state. During the processing state, digital assistant object **1306** may shrink in size relative to digital assistant object **1304**, and may include various objects swirling at a respective speed. Here, the objects may be noticeably distinct (e.g., smaller and/or color-less) than the objects within the digital assistant object during other states. Once the digital assistant has finished processing the user speech, the digital assistant may transition to a fourth state, such as the state represented by digital assistant object **1308**. During the fourth state, digital assistant object **1308** may be displayed with a size consistent to that of digital assistant objects **1302** and **1306**. Furthermore, digital assistant object **1308** may include a specific animation profile and specific color scheme (e.g., a more dense collection of swirling objects than depicted in digital assistant object **1302**). Digital assistant object **1308** may represent the digital assistant while digital assistant provides a response based on the user speech input (e.g., an audible and/or displayed response). Once the digital assistant finishes providing the response, digital assistant may return to the first state as depicted by digital assistant object **1302**.

FIGS. **14A-14B** illustrate process **1400** for continuous dialog with a digital assistant according to various examples. Process **1400** is performed, for example, using one or more electronic devices implementing a digital assistant. In some examples, process **1400** is performed using a client-server

system (e.g., system **100**), and the blocks of process **1400** are divided up in any manner between the server (e.g., DA server **106**) and a client device. In other examples, the blocks of process **1400** are divided up between the server and multiple client devices (e.g., a mobile phone and a smart watch). Thus, while portions of process **1400** are described herein as being performed by particular devices of a client-server system, it will be appreciated that process **1400** is not so limited. In other examples, process **1400** is performed using only a client device (e.g., user device **104**) or only multiple client devices. In process **1400**, some blocks are, optionally, combined, the order of some blocks is, optionally, changed, and some blocks are, optionally, omitted. In some examples, additional steps may be performed in combination with the process **1400**.

With reference to FIG. **14A**, at block **1402**, a first speech input directed to a digital assistant is received from a user. At block **1404**, a first response is provided, wherein the first response is based on the first speech input. In some examples, while the first response is being provided, a third speech input is received, and in accordance with a determination that the third speech input includes speech directed to the digital assistant, providing of the first response is ceased. In some examples, identification is made, within the third speech input, of at least one predefined word associated with a digital assistant response. In some examples, based on the identification of the at least one predefined word associated with a digital assistant response, determination is made that the third speech input includes speech directed to the digital assistant. In some examples, the at least one predefined word is associated with a digital assistant response includes at least one of a word associated with an interruption, a word associated with a correction, and a word associated with frustration. By allowing the user to interrupt the digital assistant, the system reduces unnecessary output from the device. Reducing unnecessary output enhances the operability of the device and makes the device more efficient which, additionally, reduces power usage and improves battery life of the device by enabling the user to use the device more quickly and efficiently.

At block **1406**, a session window is initiated, wherein the session window is associated with a variable speech threshold. In some examples, a user gaze directed to a displayed object is detected, and the variable speech threshold is adjusted based on the detected user gaze. In some examples, in accordance with a determination that the user gaze is directed to a displayed object corresponding to a digital assistant object, the variable speech threshold is increased by a first amount. In some examples, in accordance with a determination that the user gaze is directed to a displayed object not corresponding to a digital assistant object, the variable speech threshold is increased by a second amount, wherein the second amount is less than the first amount. At block **1408**, a second speech input is received during the session window. By utilizing a session window associated with a variable speech threshold, the system focuses on a relevant window of time to capture relevant additional speech from the user. Capturing additional speech from the user improves user experience and makes the device more efficient by eliminating the need to submit triggers for re-invoking the digital assistant. Thus, these features reduce power usage and improve battery life of the device by enabling the user to use the device more quickly and efficiently.

With reference to FIG. **14B**, at block **1410**, a determination is made whether the second speech input includes speech directed to the digital assistant. In some examples, a

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user gaze is detected while receiving the second speech input, and in accordance with a determination that the detected user gaze is directed to a display of the electronic device, a determination is made that the second speech input includes speech directed to the digital assistant. In some examples, in accordance with a determination that the detected user gaze is directed to a displayed object corresponding to a digital assistant object, determination is made that the second speech input includes speech directed to the digital assistant. In some examples, a command within the second speech input is identified, and in response to identifying the command, determination is made that the second speech input includes speech directed to the digital assistant. In some examples, at least one predefined word within the second speech input is identified, and in response to identifying the at least one predefined word, determination is made that the second speech input includes speech directed to the digital assistant. In some examples, the at least one predefined word is identified based on at least one of a usage history corresponding to speech input previously provided by the user and a designated trigger word previously stored by the user. By determining whether speech is directed to the digital assistant, the system increases accuracy by ignoring irrelevant speech. Improving accuracy enhances the operability of the device and makes the device more efficient which, additionally, reduces power usage and improves battery life of the device by enabling the user to use the device more quickly and efficiently.

At block **1412**, in accordance with a determination that the second speech input includes speech directed to the digital assistant, a duration associated with the session window is increased. At block **1414**, in accordance with a determination that the second speech input does not include speech directed to the digital assistant, an increase in the duration associated with the session window is forgone. In some examples, increasing a duration associated with the session window includes increasing the variable speech threshold by a respective amount. In some examples, the respective amount is based on context information associated with the electronic device. In some examples, context information associated with the electronic device is retrieved, and at least one word corresponding to the retrieved context information is identified within the second speech input. In some examples, in response to identifying the at least one word corresponding to the retrieved context information, determination is made that the second speech input includes speech directed to the digital assistant. In some examples, context information including an identification of currently playing media is retrieved, and at least one word corresponding to playback of the currently playing media is identified within the second speech input. In some examples, in response to identifying the at least one word corresponding to playback of the currently playing media, determination is made that the second speech input includes speech directed to the digital assistant. By varying the speech threshold, the system improves accuracy by dynamically adjusting the threshold used for recognizing speech based on relevant context. Improving accuracy enhances the operability of the device and makes the device more efficient which, additionally, reduces power usage and improves battery life of the device by enabling the user to use the device more quickly and efficiently.

At block **1416**, a determination is made whether the variable speech threshold does not exceed a predetermined speech threshold. At block **1418**, in accordance with a determination that the variable speech threshold does not exceed a predetermined speech threshold, the session win-

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dow is ended. At block **1420**, in accordance with a determination that the variable speech threshold exceeds a predetermined speech threshold, the session window is maintained in an active state. In some examples, while the session window is active, audio input is detected in a first state. In some examples, after the session window ends, audio input is detected in a second state. In some examples, prosody information associated with the second speech input is identified, and in response to identifying the prosody information, the variable speech threshold is adjusted based on the prosody information. In some examples, a first speaker profile is identified associated with the first speech input and a second speaker profile is identified associated with the second speech input. In some examples, in accordance with a determination that the first speaker profile corresponds to the second speaker profile, the variable speech threshold is increased. In some examples, a user gaze associated with the displayed environment is detected. In some examples, in accordance with detecting a movement of the user gaze towards an object displayed in the environment, the variable speech threshold is adjusted.

By toggling the device between speech recognition states based on speech thresholds the system conserves processing resources by transitioning to low power states when appropriate. Conserving processing resources enhances the operability of the device and makes the device more efficient which, additionally, reduces power usage and improves battery life of the device by enabling the user to use the device more quickly and efficiently.

The operations described above with reference to FIGS. **14A-14B** are optionally implemented by components depicted in FIGS. **1-4**, **6A-6B**, and **7A-7C**. For example, the operations of process **1300** may be implemented by one or more of operating system **718**, applications module **724**, I/O processing module **728**, STT processing module **730**, natural language processing module **732**, vocabulary index **744**, task flow processing module **736**, service processing module **738**, media service(s) **120-1**, or processor(s) **220**, **410**, and **704**. It would be clear to a person having ordinary skill in the art how other processes are implemented based on the components depicted in FIGS. **1-4**, **6A-6B**, and **7A-7C**.

FIGS. **15A-15B** illustrate process **1500** for continuous dialog with a digital assistant according to various examples. Process **1500** is performed, for example, using one or more electronic devices implementing a digital assistant. In some examples, process **1500** is performed using a client-server system (e.g., system **100**), and the blocks of process **1500** are divided up in any manner between the server (e.g., DA server **106**) and a client device. In other examples, the blocks of process **1500** are divided up between the server and multiple client devices (e.g., a mobile phone and a smart watch). Thus, while portions of process **1500** are described herein as being performed by particular devices of a client-server system, it will be appreciated that process **1500** is not so limited. In other examples, process **1500** is performed using only a client device (e.g., user device **104**) or only multiple client devices. In process **1500**, some blocks are, optionally, combined, the order of some blocks is, optionally, changed, and some blocks are, optionally, omitted. In some examples, additional steps may be performed in combination with the process **1500**.

With reference to FIG. **15A**, at block **1502**, a first speech input directed to a digital assistant is received from a user. At block **1504**, a first response is provided based on the first speech input. In some examples, while the first response is being provided, a third speech input is received. In some examples, in accordance with a determination that the third

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speech input includes speech directed to the digital assistant, providing of the first response is ceased. In some examples, identification is made, within the third speech input, of at least one predefined word associated with a digital assistant response. In some examples, based on the identification of the at least one predefined word associated with a digital assistant response, determination is made that the third speech input includes speech directed to the digital assistant. In some examples, the at least one predefined word associated with a digital assistant response includes at least one of a word associated with an interruption, a word associated with a correction, and a word associated with frustration. At block 1506, a first session window is initiated, wherein the first session window is associated with a first variable duration. At block 1508, a second speech input is received during the first session window. By allowing the user to interrupt the digital assistant, the system reduces unnecessary output from the device. Reducing unnecessary output enhances the operability of the device and makes the device more efficient which, additionally, reduces power usage and improves battery life of the device by enabling the user to use the device more quickly and efficiently.

With reference to FIG. 15B, at block 1510, a determination is made whether the second speech input includes speech directed to the digital assistant. In some examples, a coupling between a secondary device and the electronic device is detected, wherein the determination that the second speech input includes speech directed to the digital assistant is made based at least in part on the detection. In some examples, an interrogatory phrase is identified within the second speech input, and in response to identifying the interrogatory phrase, a determination is made that the second speech input includes speech directed to the digital assistant. In some examples, a command is identified within the second speech input, and in response to identifying the command, a determination is made that the second speech input includes speech directed to the digital assistant. In some examples, at least one predefined word is identified within the second speech input, and in response to identifying the at least one predefined word, a determination is made that the second speech input includes speech directed to the digital assistant. In some examples, the at least one predefined word is identified based on a usage history corresponding to speech input previously provided by the user. In some examples, the at least one predefined word is identified based on a designated trigger word previously stored by the user. By determining whether speech is directed to the digital assistant, the system increases accuracy by ignoring irrelevant speech. Improving accuracy enhances the operability of the device and makes the device more efficient which, additionally, reduces power usage and improves battery life of the device by enabling the user to use the device more quickly and efficiently.

At block 1512, in accordance with a determination that the second speech input includes speech directed to the digital assistant, a second session window associated with a second variable duration is initiated. In some examples, context information associated with the electronic device is retrieved, wherein at least one word corresponding to the retrieved context information is identified. In some examples, in response to identifying the at least one word corresponding to the retrieved context information, determination is made that the second speech input includes speech directed to the digital assistant. In some examples, context information including identification of currently playing media is identified, wherein at least one word corresponding to playback of the currently playing media is

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identified within the second speech input. In some examples, in response to identifying the at least one word corresponding to playback of the currently playing media, determination is made that the second speech input includes speech directed to the digital assistant. By utilizing a session window associated with a variable duration, the system focuses on a relevant window of time to capture relevant additional speech from the user. Capturing additional speech from the user improves user experience and makes the device more efficient by eliminating the need to submit triggers for re-invoking the digital assistant. Thus, these features reduce power usage and improve battery life of the device by enabling the user to use the device more quickly and efficiently.

At block 1514, in accordance with a determination that the second speech input does not include speech directed to the digital assistant, an audible output is provided at a time corresponding to an end of the first variable duration. In some examples, in accordance with a determination that the second speech input includes speech not directed to the digital assistant, maintaining the first session window as active until a time corresponding to an end of the first variable duration. In some examples, the first speech input is received from a secondary electronic device, wherein the second speech input is received from the second electronic device. In some examples, the first response is caused to be provided at the secondary electronic device, wherein the audible output is caused to be provided at the secondary electronic device. By determining whether speech is directed to the digital assistant, the system increases accuracy by ignoring irrelevant speech. Improving accuracy enhances the operability of the device and makes the device more efficient which, additionally, reduces power usage and improves battery life of the device by enabling the user to use the device more quickly and efficiently.

At block 1516, a determination is made whether audible output criteria is satisfied. In some examples, in accordance with a determination that the second speech input includes speech directed to the digital assistant and speech corresponding to a request to play a media item, a second response is provided including playback of the media item. In some examples, determination is made that audio output criteria is not satisfied based on the playback of the media item. In some examples, a volume associated with the playback of the media item is raised at a time corresponding to an end of the second variable duration. By adjusting volume based on the playing media, the system improves device operability by interleaving audio cues within playing media. Improving device operability makes the device more efficient which, additionally, reduces power usage and improves battery life of the device by enabling the user to use the device more quickly and efficiently.

At block 1518, in accordance with audible output criteria being satisfied, an audible output is provided at a time corresponding to an end of the second variable duration. At block 1520, in accordance with audible output criteria not being satisfied, an audible output is not provided at a time corresponding to an end of the second variable duration. In some examples, while the second session window is active, audio input is detected in a first state. In some examples, the second session window is ended at the time corresponding to the end of the second variable duration, and after the second session window ends, the audio input is detected in a second state. By providing audible outputs to the user, the system improves device operability by informing the user when the digital assistant is listening for speech queries. Improving device operability makes the device more effi-

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cient which, additionally, reduces power usage and improves battery life of the device by enabling the user to use the device more quickly and efficiently.

FIG. 16 illustrate process 1600 for digital assistant interaction on a lock screen according to various examples. Process 1600 is performed, for example, using one or more electronic devices implementing a digital assistant. In some examples, process 1600 is performed using a client-server system (e.g., system 100), and the blocks of process 1600 are divided up in any manner between the server (e.g., DA server 106) and a client device. In other examples, the blocks of process 1600 are divided up between the server and multiple client devices (e.g., a mobile phone and a smart watch). Thus, while portions of process 1600 are described herein as being performed by particular devices of a client-server system, it will be appreciated that process 1600 is not so limited. In other examples, process 1600 is performed using only a client device (e.g., user device 104) or only multiple client devices. In process 1600, some blocks are, optionally, combined, the order of some blocks is, optionally, changed, and some blocks are, optionally, omitted. In some examples, additional steps may be performed in combination with the process 1600.

At block 1602, while the electronic device is in a locked state, a digital assistant object is displayed as having a first size. At block 1604, a gaze directed at the digital assistant object is detected, wherein the gaze is associated with a user of the electronic device. In some examples, in response to detecting the gaze, a haptic output is provided for a predetermined duration. In some examples, before detecting the gaze directed at the digital assistant object, the digital assistant object is displayed with a first animation, and while detecting the gaze directed at the digital assistant object, the digital assistant object is displayed with a second animation. In some examples, a gaze directed away from the digital assistant object is detected, and in response to detecting the gaze away from the digital assistant object, the digital assistant object is displayed with the first animation. In some examples, while displaying the digital assistant object with the first animation, the digital assistant object is displayed as including one or more objects moving at a first speed. In some examples, while displaying the digital assistant object with the second animation, the digital assistant is displayed as including the one or more objects moving at a second speed, wherein the second speed is faster than the first speed. By displaying various animations of the digital assistant object based on user gaze, the system improves device operability by informing the user when the digital assistant is listening for speech queries. Improving device operability makes the device more efficient which, additionally, reduces power usage and improves battery life of the device by enabling the user to use the device more quickly and efficiently.

At block 1606, in response to detecting the gaze, the digital assistant object is displayed as having a second size larger than the first size. In some examples, displaying the digital assistant object as having a second size larger than the first size includes displaying an enlargement of the digital assistant object. In some examples, while detecting the gaze directed at the digital assistant object, the displayed second size of the digital assistant object is maintained. In some examples, a gaze directed away from the digital assistant object is detected, and in response to detecting the gaze directed away from the digital assistant object, a size reduction in the digital assistant object is displayed by displaying the digital assistant object as having the first size. In some examples, in response to detecting the gaze directed away

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from the digital assistant object, the digital assistant object is displayed as having a first size smaller than the second size, including displaying a shrinking of the digital assistant object. In some examples, displaying an enlargement of the digital assistant object includes displaying the enlargement for a first predetermined duration, and displaying a shrinking of the digital assistant object includes displaying the shrinking for a second predetermined duration longer than the first predetermined duration. By displaying various animations of the digital assistant object based on user gaze, the system improves device operability by informing the user when the digital assistant is listening for speech queries. Improving device operability makes the device more efficient which, additionally, reduces power usage and improves battery life of the device by enabling the user to use the device more quickly and efficiently.

At block 1608, in response to detecting the gaze, audio input is detected. In some examples, a gaze directed away from the digital assistant object is detected, and at a predetermined time after detecting the gaze directed away from the digital assistant object, detecting of the audio input is ceased. By detecting audio for a predetermined time after detecting user gaze away from the digital assistant object, the system improves device operability by listening for relevant speech based on user gaze. Improving device operability makes the device more efficient which, additionally, reduces power usage and improves battery life of the device by enabling the user to use the device more quickly and efficiently.

At block 1610, determination is made whether the audio input includes a speech input. At block 1612, in accordance with a determination that the audio input includes a speech input, a task is performed based on the speech input. In some examples, performing a task based on the speech input includes maintaining the electronic device in the locked state, and performing the task while the electronic device remains in the locked state. In some examples, a determination is made that the task based on the speech input includes a task based on non-private information. In some examples, performing a task based on the speech input includes obtaining biometric information corresponding to the user, and authenticating the user based on the biometric information. In some examples, in response to authenticating the user based on the biometric information, the electronic device is transitioned from the locked state to an unlocked state, and the task is performed while the electronic device remains in the unlocked state. In some examples, in accordance with a determination that the task based on the speech input requires access to private information, the user is prompted to input authentication information. By authenticating the user based on biometric information, the system improves device operability by improving device security. Improving device operability makes the device more efficient which, additionally, reduces power usage and improves battery life of the device by enabling the user to use the device more quickly and efficiently.

At block 1614, in accordance with a determination that the audio input does not include a speech input, the display of the digital assistant object is maintained, while detecting the gaze directed at the digital assistant object, as having the second size larger than the first size. In some examples, in accordance with a determination that the audio input does not include a speech input associated with a user of the electronic device for a predetermined duration, at least one output is provided. In some examples, providing the at least one output includes, while detecting the gaze directed at the

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digital assistant object, displaying the digital assistant object as having a third size larger than the second size, and after displaying the digital assistant object as having a third size larger than the second size, displaying the digital assistant object as having the second size smaller than the third size. In some examples, providing the at least one output includes providing a haptic output for a predetermined duration. In some examples, providing the at least one output includes displaying the digital assistant object as including one or more objects moving at a first speed. By displaying various animations of the digital assistant object based on user gaze, the system improves device operability by informing the user when the digital assistant is listening for speech queries. Improving device operability makes the device more efficient which, additionally, reduces power usage and improves battery life of the device by enabling the user to use the device more quickly and efficiently.

The operations described above with reference to FIG. 16 are optionally implemented by components depicted in FIGS. 1-4, 6A-6B, and 7A-7C. For example, the operations of process 1300 may be implemented by one or more of operating system 718, applications module 724, I/O processing module 728, STT processing module 730, natural language processing module 732, vocabulary index 744, task flow processing module 736, service processing module 738, media service(s) 120-1, or processor(s) 220, 410, and 704. It would be clear to a person having ordinary skill in the art how other processes are implemented based on the components depicted in FIGS. 1-4, 6A-6B, and 7A-7C.

In accordance with some implementations, a computer-readable storage medium (e.g., a non-transitory computer readable storage medium) is provided, the computer-readable storage medium storing one or more programs for execution by one or more processors of an electronic device, the one or more programs including instructions for performing any of the methods or processes described herein.

In accordance with some implementations, an electronic device (e.g., a portable electronic device) is provided that comprises means for performing any of the methods or processes described herein.

In accordance with some implementations, an electronic device (e.g., a portable electronic device) is provided that comprises a processing unit configured to perform any of the methods or processes described herein.

In accordance with some implementations, an electronic device (e.g., a portable electronic device) is provided that comprises one or more processors and memory storing one or more programs for execution by the one or more processors, the one or more programs including instructions for performing any of the methods or processes described herein.

The foregoing description, for purpose of explanation, has been described with reference to specific embodiments. However, the illustrative discussions above are not intended to be exhaustive or to limit the invention to the precise forms disclosed. Many modifications and variations are possible in view of the above teachings. The embodiments were chosen and described in order to best explain the principles of the techniques and their practical applications. Others skilled in the art are thereby enabled to best utilize the techniques and various embodiments with various modifications as are suited to the particular use contemplated.

Although the disclosure and examples have been fully described with reference to the accompanying drawings, it is to be noted that various changes and modifications will become apparent to those skilled in the art. Such changes

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and modifications are to be understood as being included within the scope of the disclosure and examples as defined by the claims.

As described above, one aspect of the present technology is the gathering and use of data available from various sources to improve continuous dialog with a digital assistant. The present disclosure contemplates that in some instances, this gathered data may include personal information data that uniquely identifies or can be used to contact or locate a specific person. Such personal information data can include demographic data, location-based data, telephone numbers, email addresses, twitter IDs, home addresses, data or records relating to a user's health or level of fitness (e.g., vital signs measurements, medication information, exercise information), date of birth, or any other identifying or personal information.

The present disclosure recognizes that the use of such personal information data, in the present technology, can be used to the benefit of users. For example, the personal information such as eye gaze data can be used to determine whether the user is gazing at a digital assistant object. Further, other uses for personal information data that benefit the user are also contemplated by the present disclosure. For instance, health and fitness data may be used to provide insights into a user's general wellness, or may be used as positive feedback to individuals using technology to pursue wellness goals.

The present disclosure contemplates that the entities responsible for the collection, analysis, disclosure, transfer, storage, or other use of such personal information data will comply with well-established privacy policies and/or privacy practices. In particular, such entities should implement and consistently use privacy policies and practices that are generally recognized as meeting or exceeding industry or governmental requirements for maintaining personal information data private and secure. Such policies should be easily accessible by users, and should be updated as the collection and/or use of data changes. Personal information from users should be collected for legitimate and reasonable uses of the entity and not shared or sold outside of those legitimate uses. Further, such collection/sharing should occur after receiving the informed consent of the users. Additionally, such entities should consider taking any needed steps for safeguarding and securing access to such personal information data and ensuring that others with access to the personal information data adhere to their privacy policies and procedures. Further, such entities can subject themselves to evaluation by third parties to certify their adherence to widely accepted privacy policies and practices. In addition, policies and practices should be adapted for the particular types of personal information data being collected and/or accessed and adapted to applicable laws and standards, including jurisdiction-specific considerations. For instance, in the US, collection of or access to certain health data may be governed by federal and/or state laws, such as the Health Insurance Portability and Accountability Act (HIPAA); whereas health data in other countries may be subject to other regulations and policies and should be handled accordingly. Hence different privacy practices should be maintained for different personal data types in each country.

Despite the foregoing, the present disclosure also contemplates embodiments in which users selectively block the use of, or access to, personal information data. That is, the present disclosure contemplates that hardware and/or software elements can be provided to prevent or block access to such personal information data. For example, the present

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technology can be configured to allow users to select to “opt in” or “opt out” of participation in the collection of personal information data during registration for services or anytime thereafter. In another example, users can select not to provide speech input information included in a session window. In yet another example, users can select to limit detail provided with respect to eye gaze data. In addition to providing “opt in” and “opt out” options, the present disclosure contemplates providing notifications relating to the access or use of personal information. For instance, a user may be notified upon downloading an app that their personal information data will be accessed and then reminded again just before personal information data is accessed by the app.

Moreover, it is the intent of the present disclosure that personal information data should be managed and handled in a way to minimize risks of unintentional or unauthorized access or use. Risk can be minimized by limiting the collection of data and deleting data once it is no longer needed. In addition, and when applicable, including in certain health related applications, data de-identification can be used to protect a user’s privacy. De-identification may be facilitated, when appropriate, by removing specific identifiers (e.g., date of birth, etc.), controlling the amount or specificity of data stored (e.g., collecting location data at a city level rather than at an address level), controlling how data is stored (e.g., aggregating data across users), and/or other methods.

Therefore, although the present disclosure broadly covers use of personal information data to implement one or more various disclosed embodiments, the present disclosure also contemplates that the various embodiments can also be implemented without the need for accessing such personal information data. That is, the various embodiments of the present technology are not rendered inoperable due to the lack of all or a portion of such personal information data. For example, continuous dialog can be facilitated by inferring preferences based on non-personal information data or a bare minimum amount of personal information, such as anonymized eye-gaze data, other non-personal information available to the continuous dialog system, or publicly available information.

What is claimed is:

1. An electronic device, comprising:
 - one or more processors;
 - a memory; and
 - one or more programs, wherein the one or more programs are stored in the memory and configured to be executed by the one or more processors, the one or more programs including instructions for:
 - providing a first response based on a first speech input;
 - initiating a first session window;
 - receiving a second speech input during the first session window; and
 - in accordance with a determination that the second speech input includes speech directed to a digital assistant, initiating a second session window associated with a first variable duration.
2. The device of claim 1, the one or more programs including instructions for:
 - receiving the first speech input from a secondary electronic device;
 - receiving the second speech input from the secondary electronic device;
 - causing the first response to be provided at the secondary electronic device; and
 - causing an audible output to be provided at the secondary electronic device.

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3. The device of claim 1, the one or more programs including instructions for:

- detecting a coupling of a secondary device to the electronic device; and
- determining, based at least in part on the detection, that the second speech input includes speech directed to the digital assistant.

4. The device of claim 1, the one or more programs including instructions for:

- identifying an interrogatory phrase within the second speech input; and
- in response to identifying the interrogatory phrase, determining that the second speech input includes speech directed to the digital assistant.

5. The device of claim 1, the one or more programs including instructions for:

- identifying a command within the second speech input; and
- in response to identifying the command, determining that the second speech input includes speech directed to the digital assistant.

6. The device of claim 1, the one or more programs including instructions for:

- identifying at least one predefined word within the second speech input; and
- in response to identifying the at least one predefined word, determining that the second speech input includes speech directed to the digital assistant.

7. The device of claim 1, the one or more programs including instructions for:

- retrieving context information associated with the electronic device;
- identifying, within the second speech input, at least one word corresponding to the retrieved context information; and
- in response to identifying the at least one word corresponding to the retrieved context information, determining that the second speech input includes speech directed to the digital assistant.

8. The device of claim 1, the one or more programs including instructions for:

- retrieving context information including an identification of currently playing media;
- identifying, within the second speech input, at least one word corresponding to playback of the currently playing media; and
- in response to identifying the at least one word corresponding to playback of the currently playing media, determining that the second speech input includes speech directed to the digital assistant.

9. The device of claim 1, the one or more programs including instructions for:

- while the first response is being provided, receiving the second speech input; and
- in accordance with a determination that the second speech input includes speech directed to the digital assistant, ceasing providing the first response.

10. The device of claim 1, the one or more programs including instructions for:

- in accordance with a determination that the second speech input does not include speech directed to the digital assistant, maintaining the first session window as active until a time corresponding to an end of a second variable duration.

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11. The device of claim 1, the one or more programs including instructions for:

in accordance with a determination that the second speech input does not include speech directed to the digital assistant, providing an audible output at a time corresponding to an end of a second variable duration.

12. The device of claim 1, the one or more programs including instructions for:

while the second session window is active, detecting audio input in a first state;

ending the second session window at a time corresponding to the end of the first variable duration; and after the second session window ends, detecting audio input in a second state.

13. A non-transitory computer-readable storage medium storing one or more programs, wherein the one or more programs are stored in memory and configured to be executed by one or more processors of an electronic device, the one or more programs including instructions for:

providing a first response based on a first speech input; initiating a first session window;

receiving a second speech input during the first session window; and

in accordance with a determination that the second speech input includes speech directed to a digital assistant, initiating a second session window associated with a first variable duration.

14. A computer-implemented method, comprising:

at an electronic device with one or more processors and memory:

providing a first response based on a first speech input; initiating a first session window;

receiving a second speech input during the first session window; and

in accordance with a determination that the second speech input includes speech directed to a digital assistant, initiating a second session window associated with a first variable duration.

15. The computer-readable storage medium of claim 13, the one or more programs including instructions for:

receiving the first speech input from a secondary electronic device;

receiving the second speech input from the secondary electronic device;

causing the first response to be provided at the secondary electronic device; and

causing an audible output to be provided at the secondary electronic device.

16. The computer-readable storage medium of claim 13, the one or more programs including instructions for:

detecting a coupling of a secondary device to the electronic device; and

determining, based at least in part on the detection, that the second speech input includes speech directed to the digital assistant.

17. The computer-readable storage medium of claim 13, the one or more programs including instructions for:

identifying an interrogatory phrase within the second speech input; and

in response to identifying the interrogatory phrase, determining that the second speech input includes speech directed to the digital assistant.

18. The computer-readable storage medium of claim 13, the one or more programs including instructions for:

identifying a command within the second speech input; and

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in response to identifying the command, determining that the second speech input includes speech directed to the digital assistant.

19. The computer-readable storage medium of claim 13, the one or more programs including instructions for:

identifying at least one predefined word within the second speech input; and

in response to identifying the at least one predefined word, determining that the second speech input includes speech directed to the digital assistant.

20. The computer-readable storage medium of claim 13, the one or more programs including instructions for:

retrieving context information associated with the electronic device;

identifying, within the second speech input, at least one word corresponding to the retrieved context information; and

in response to identifying the at least one word corresponding to the retrieved context information, determining that the second speech input includes speech directed to the digital assistant.

21. The computer-readable storage medium of claim 13, the one or more programs including instructions for:

retrieving context information including an identification of currently playing media;

identifying, within the second speech input, at least one word corresponding to playback of the currently playing media; and

in response to identifying the at least one word corresponding to playback of the currently playing media, determining that the second speech input includes speech directed to the digital assistant.

22. The computer-readable storage medium of claim 13, the one or more programs including instructions for:

while the first response is being provided, receiving a second speech input; and

in accordance with a determination that the second speech input includes speech directed to the digital assistant, ceasing providing the first response.

23. The computer-readable storage medium of claim 13, the one or more programs including instructions for:

in accordance with a determination that the second speech input does not include speech directed to the digital assistant, maintaining the first session window as active until a time corresponding to an end of a second variable duration.

24. The computer-readable storage medium of claim 13, the one or more programs including instructions for:

in accordance with a determination that the second speech input does not include speech directed to the digital assistant, providing an audible output at a time corresponding to an end of a second variable duration.

25. The computer-readable storage medium of claim 13, the one or more programs including instructions for:

while the second session window is active, detecting audio input in a first state;

ending the second session window at a time corresponding to the end of the first variable duration; and after the second session window ends, detecting audio input in a second state.

26. The method of claim 14, comprising:

receiving the first speech input from a secondary electronic device;

receiving the second speech input from the secondary electronic device;

causing the first response to be provided at the secondary electronic device; and

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causing an audible output to be provided at the secondary electronic device.

27. The method of claim 14, comprising:

detecting a coupling of a secondary device to the electronic device; and

determining, based at least in part on the detection, that the second speech input includes speech directed to the digital assistant.

28. The method of claim 14, comprising:

identifying an interrogatory phrase within the second speech input; and

in response to identifying the interrogatory phrase, determining that the second speech input includes speech directed to the digital assistant.

29. The method of claim 14, comprising:

identifying a command within the second speech input; and

in response to identifying the command, determining that the second speech input includes speech directed to the digital assistant.

30. The method of claim 14, comprising:

identifying at least one predefined word within the second speech input; and

in response to identifying the at least one predefined word, determining that the second speech input includes speech directed to the digital assistant.

31. The method of claim 14, comprising:

retrieving context information associated with the electronic device;

identifying, within the second speech input, at least one word corresponding to the retrieved context information; and

in response to identifying the at least one word corresponding to the retrieved context information, determining that the second speech input includes speech directed to the digital assistant.

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32. The method of claim 14, comprising:

retrieving context information including an identification of currently playing media;

identifying, within the second speech input, at least one word corresponding to playback of the currently playing media; and

in response to identifying the at least one word corresponding to playback of the currently playing media, determining that the second speech input includes speech directed to the digital assistant.

33. The method of claim 14, comprising:

while the first response is being provided, receiving a second speech input; and

in accordance with a determination that the second speech input includes speech directed to the digital assistant, ceasing providing the first response.

34. The method of claim 14, comprising:

in accordance with a determination that the second speech input does not include speech directed to the digital assistant, maintaining the first session window as active until a time corresponding to an end of a second variable duration.

35. The method of claim 14, comprising:

in accordance with a determination that the second speech input does not include speech directed to the digital assistant, providing an audible output at a time corresponding to an end of a second variable duration.

36. The method of claim 14, comprising:

while the second session window is active, detecting audio input in a first state;

ending the second session window at a time corresponding to the end of the first variable duration; and

after the second session window ends, detecting audio input in a second state.

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